

SECTION VIII

CONSEQUENCES OF CHRONIC KIDNEY DISEASE

Mechanisms of Progression in Chronic Kidney Disease

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CHAPTER OUTLINE

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KEY POINTS

- Although the usefulness of the term “chronic kidney disease” has been questioned, the concept of CKD is strongly supported by evidence that in response to nephron loss, a common pathway of mechanisms provokes a vicious cycle of progressive kidney damage that eventuates in further nephron loss and explains the progressive nature of CKD of diverse causes.
- Glomerular hemodynamic adaptations to nephron loss resulting in glomerular capillary hypertension and glomerular hyperfiltration are key factors that drive CKD progression. This hypothesis is strongly supported by trials showing that treatment with both RAAS inhibitors and SGLT2 inhibitors, drugs that act by different mechanisms to reduce glomerular capillary hydraulic pressure, affords kidney protection.
- Nonhemodynamic factors, including proteinuria, tubulointerstitial fibrosis, oxidative stress, acidosis, and renal hypertrophy, act in concert with hemodynamic factors to provoke progressive kidney damage.
- In human CKD, superimposed episodes of acute kidney injury likely play a significant role in accelerating nephron loss in many persons.
- A thorough understanding of the multiple factors that contribute to common pathway mechanisms of CKD progression is essential to inform strategies for achieving kidney protection. Because of the complexity of the interacting mechanisms, it is clear that to achieve optimal protection, the common pathway must be blocked at multiple points.

The introduction of a definition for chronic kidney disease (CKD) by the National Kidney Foundation Kidney Disease Outcomes Quality Initiative (K/DOQI) and its adoption worldwide have made a valuable contribution to raising awareness of the global burden of disease due to CKD.¹ Importantly, the adoption of a classification system for CKD that divides the spectrum of CKD into stages has emphasized the progressive nature of CKD and facilitated the development of stage-specific strategies for slowing

progression, as well as treating the complications of impaired kidney function. These developments highlight the importance of understanding the mechanisms that contribute to CKD progression to inform strategies for slowing such progression. Central to these mechanisms are the adaptations observed in the kidney when nephrons are lost.

The kidney's primary function of maintaining constancy of the extracellular fluid (ECF) volume and composition is

remarkably well preserved until late in the course of CKD. When nephrons are lost through disease or surgical ablation, those remaining undergo remarkable physiologic responses, resulting in hypertrophy and hyperfunction that combine to compensate for the acquired loss of kidney function. Effective kidney function requires close integration of glomerular and tubular functions. Indeed, the preservation of glomerulotubular balance seen until the terminal stages of CKD is fundamental to the intact nephron hypothesis of Bricker, which essentially states that as CKD advances, kidney function is supported by a diminishing pool of functioning (or hyperfunctioning) nephrons, rather than relatively constant numbers of nephrons, each with diminishing function.² This concept has important implications for the mechanisms of disease progression in CKD.

Several decades ago, clinical studies of persons with CKD established that once the glomerular filtration rate (GFR) fell below a critical level, a relentless progression to end-stage kidney failure (ESKF) usually ensued, even when the initial disease activity had abated. The rate of decline of the GFR in a given individual often followed a near-constant linear relationship with time, enabling remarkably accurate predictions of the date at which kidney failure would be reached and kidney replacement therapy (KRT) required. Among persons with diverse kidney diseases, the slope of the GFR-time relationship was found to be a characteristic of individuals rather than typical of their specific kidney diseases. This observation suggested that the progressive nature of CKD could be attributed to a final common pathway of mechanisms, independent of the primary cause of nephropathy.³ Within this framework, Brenner and colleagues⁴ formulated a unifying hypothesis for CKD progression based on the physiologic adaptations observed in experimental models. The central tenets of the common pathway theory state that CKD progression occurs, in general, through focal nephron loss, and that the adaptive responses of surviving nephrons, although initially serving to increase single-nephron GFR and offset the overall loss in clearance, ultimately prove detrimental to the kidney. Over time, glomerulosclerosis and tubular atrophy further reduce nephron number, fueling a self-perpetuating cycle of nephron destruction, culminating in kidney failure. Though recent clinical studies have reported that the progression of CKD is variable and nonlinear in some persons,⁵ these common pathway mechanisms remain relevant.

In this chapter, the functional and structural adaptations observed in remaining nephrons following substantial reductions in functioning kidney mass and the mechanisms thought to be responsible for them are described in detail. How these changes may, in time, prove maladaptive and contribute to the progressive kidney injury described earlier are then considered. Given the growing worldwide burden of CKD that causes substantial morbidity and mortality in individuals and threatens to overburden health care systems, it could be argued that the further elucidation of the mechanisms of CKD progression resulting in more effective interventions to slow its advance should remain among the highest priorities for nephrologists and health care systems today (for more on the global burden of kidney disease see [Chapter 73](#)).

STRUCTURAL AND FUNCTIONAL ADAPTATION OF THE KIDNEY TO NEPHRON LOSS

ALTERATIONS IN GLOMERULAR PHYSIOLOGY

Glomerular hemodynamic responses to nephron loss have been studied largely in animals subjected to surgical ablation of kidney mass. It was recognized several decades ago that unilateral nephrectomy in rats resulted in a rapid increase in function of the remaining kidney, detectable 3 days after nephrectomy, such that the GFR achieved a maximum of 70% to 85% of the previous two-kidney value after 2 to 3 weeks. More recent observations in conscious rats have reported a maximal increase of approximately 50% in the GFR of a single kidney at 8 days after uninephrectomy and a 300% increase in the GFR of the remnant kidney at 16 days after 5/6 nephrectomy ([Fig. 50.1](#)).⁶ Because no new nephrons are formed in mature rodents, the observed rise in the GFR represents an increase in the filtration rate of remaining nephrons.

A detailed study of glomerular hemodynamics was facilitated by the identification of a rat strain, Munich-Wistar, which is unique in regularly bearing glomeruli on the kidney surface. This allowed micropuncture of the glomerulus and direct measurement of intraglomerular pressures, as well as sampling of blood from afferent and efferent arterioles. These techniques made possible the study of mechanisms underlying the compensatory rise in the GFR after kidney mass ablation. Increases in whole-kidney GFR at 2 to 4 weeks after unilateral nephrectomy were attributable to an increase in the single-nephron GFR (SNGFR) averaging 83%, achieved in large part by a rise in the glomerular plasma flow rate (Q_A), which, in turn, resulted from dilation of afferent and, to a lesser extent, efferent arterioles. Although the systemic blood pressure (BP) was not elevated, glomerular capillary hydraulic pressure (P_{GC}) and the glomerular transcapillary pressure difference (ΔP) increased significantly after uninephrectomy, accounting for an estimated 25% of the rise in SNGFR.⁷ The glomerular ultrafiltration coefficient, K_f (the product of glomerular hydraulic permeability and surface area available for filtration), was unaltered at this stage but could become elevated later.⁸

With more extensive nephron loss, even greater compensatory increases in SNGFR were observed. In Munich-Wistar rats studied 7 days after unilateral nephrectomy and infarction of 5/6 of the contralateral kidney, SNGFR in the remnant kidney was more than double that of two-kidney controls. This increment was again attributable to large increases in Q_A and a substantial rise in P_{GC} . Efferent and afferent arteriolar resistances were reduced, but the decrease in afferent arteriolar resistance was again proportionately greater, accounting for the observed rise in P_{GC} .⁹ Comparison of kidney infarction versus surgical excision models of 5/6 nephrectomy subsequently found that changes in arteriolar resistance were similar, but that P_{GC} was significantly more elevated in the infarction model, indicating that glomerular transmission of elevated systemic BP (initially absent in the surgical excision model)

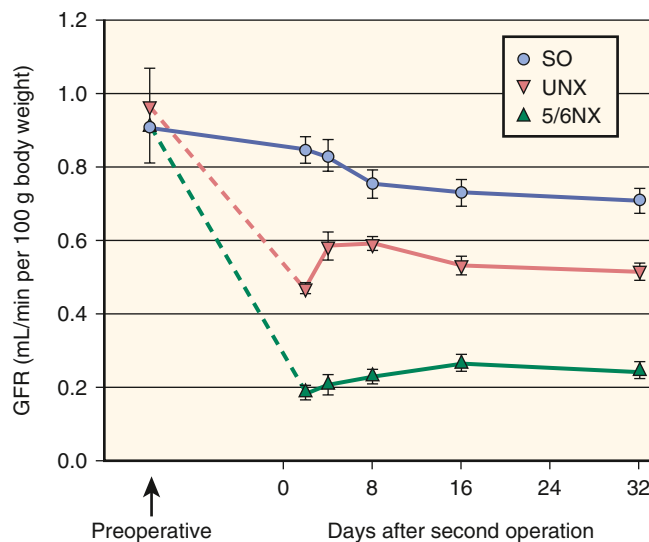


Fig. 50.1 Glomerular filtration rate (GFR) in conscious rats before and after sham operations (SO), uninephrectomy (UNX), or 5/6 nephrectomy (5/6NX). Values are mean $\pm$ standard error of the mean. (From Chamberlain RM, Shirley DG. Time course of the renal functional response to partial nephrectomy: measurements in conscious rats. *Exp Physiol.* 2007;92:251–262.)

also contributes to the increase in P_{GC} .^{10,11} Changes in K_f after extensive renal mass ablation appeared to be time dependent, with a decrease reported at 2 weeks after surgery¹² and an increase at 4 weeks.¹³ Further studies have indicated that glomerular hemodynamic responses to nephron loss seem to be similar between the superficial cortical and juxtamedullary nephrons.¹⁴ The rise in SNGFR associated with kidney mass ablation is often referred to as *glomerular hyperfiltration* and the elevated P_{GC} is termed *glomerular hypertension*. Together, these terms encompass the central concepts underlying the hemodynamic adaptations in the remnant kidney.

Glomerular hemodynamic adaptations to nephron loss may show interspecies variation. In dogs, increases in SNGFR observed 4 weeks after 3/4 or 7/8 nephrectomy were attributable largely to increases in Q_A and K_f . In contrast to the findings in rodents, ΔP in dogs was only modestly elevated, although after ablation of 7/8 of their renal mass, P_{GC} did increase significantly, independent of arterial pressure, again as a result of relatively greater relaxation of afferent versus efferent arterioles.¹⁵

In humans, the effects of nephron loss on the physiology of the remnant kidney have been studied mainly in healthy individuals undergoing donor nephrectomy for kidney transplantation. Inulin clearance studies of the earliest kidney donors have revealed that the total GFR in the donor's remaining kidney had increased from 65% to 70% of the previous two-kidney value by 1 week after nephrectomy. A meta-analysis of data from 48 studies that included 2988 living kidney donors has estimated that the total GFR decreased, on average, by only 17 mL/min/1.73 m² after uninephrectomy.¹⁶ These observations imply that the single-kidney GFR (and therefore also the average SNGFR) increases by 30% to 40% after uninephrectomy in humans. There is currently no method for measuring SNGFR or P_{GC} in humans, but detailed studies in 21 healthy kidney

donors have reported that the observed 40% increase in single-kidney GFR could be accounted for by the observed increase in renal plasma flow and a rise in K_f resulting from glomerular hypertrophy without the need for an increase in P_{GC} .¹⁷

MEDIATORS OF THE GLOMERULAR HEMODYNAMIC RESPONSES TO NEPHRON LOSS

The specific factors that are sensed after kidney mass ablation and serve as signals to initiate the adjustments in glomerular hemodynamics responsible for the increase in remnant kidney GFR remain to be identified. However, the effector mechanisms have been studied extensively; the hemodynamic changes can be attributed to the net effects of complex interactions of several factors, each having specific and sometimes opposing actions on the various determinants of glomerular ultrafiltration. Several vasoactive substances, including angiotensin II (Ang II), aldosterone, natriuretic peptides (NPs), endothelins (ETs), eicosanoids, and bradykinin, have been implicated. Moreover, sustained increases in the SNGFR also require resetting of the autoregulatory mechanisms that normally govern the GFR and renal plasma flow (RPF). For a detailed discussion of vasoactive molecules in the kidney, see Chapter 11.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Ang II appears to play a critical role in the development of glomerular capillary hypertension following kidney mass ablation and may also contribute to changes in K_f . The acute infusion of Ang II in normal rats results in a rise in P_{GC} due to a greater increase in efferent than afferent resistance and reductions in Q_A and K_f .^{18,19} Chronic administration of Ang II for 8 weeks resulted in systemic hypertension, lowered single-kidney GFR and, with the exception of K_f , elicited similar glomerular hemodynamic changes to those observed after acute infusion in both normal and uninephrectomized rats.⁸ The importance of the influence of endogenous Ang II on glomerular hemodynamics in remnant kidneys has been revealed by studies with pharmacologic inhibitors of the renin-angiotensin-aldosterone system (RAAS). Chronic treatment of 5/6 nephrectomized rats with either an angiotensin-converting enzyme (ACE) inhibitor^{20,21} or Ang II (subtype 1) receptor blocker (ARB)^{22,23} results in normalization of P_{GC} through a reduction in systemic BP and dilation of both afferent and efferent arterioles. SNGFR, however, remains elevated due to an increase in K_f . Furthermore, acute infusion of an ACE inhibitor or saralasin, a peptide analog receptor antagonist of Ang II, was found to normalize P_{GC} in 5/6 nephrectomized rats through efferent arteriolar dilation, without affecting the mean arterial pressure (MAP).^{12,24} It is unclear why these findings could not be confirmed with the ARB losartan.²⁵

These effects of RAAS inhibition imply that there is increased local activity of endogenous Ang II, yet plasma renin levels show only a transient increase following 5/6 nephrectomy.^{10,26} This suggests differential regulation of the systemic versus intrarenal RAAS and that Ang II is formed locally. Detailed studies have identified that all components of the RAAS are expressed in the kidney.²⁷ Renin mRNA and protein levels are both increased in glomeruli adjacent to the infarction scar in 5/6 nephrectomized rats.²⁸⁻³⁰

Furthermore, kidney renin mRNA levels are increased at days 3 and 7 after kidney mass ablation by infarction but not when a kidney mass is excised surgically, suggesting that kidney infarction activates the RAAS by creating a margin of ischemic tissue around the organizing infarct and explaining the greater severity of hypertension and glomerulosclerosis associated with the infarction model.^{10,11}

Detailed studies of intrarenal Ang II levels following 5/6 nephrectomy achieved by infarction have confirmed these findings by showing higher Ang II levels in the periinfarct portion of the kidney than the intact portion at all time points.²⁶ On the other hand, the studies also showed that the rise in intrarenal Ang II levels following 5/6 nephrectomy was transient. Whereas Ang II levels in the periinfarct portion were elevated compared with sham-operated controls at 2 weeks after surgery, they were not statistically different at 5 or 7 weeks. In the intact portion of the remnant kidney, Ang II levels were similar to controls at 2 and 5 weeks and were lower at 7 weeks.²⁶ Sustained increases in intrarenal Ang II levels are therefore not required to maintain hypertension and progressive kidney injury characteristic of this model. Nevertheless, subsequent studies have shown that the kidney-protective effects of ACE inhibitor and ARB treatment are associated with a reduction in intrarenal Ang II levels in both the periinfarct and intact portions of the remnant kidney.³¹ In contrast, treatment with the dihydropyridine calcium antagonist, nifedipine, did not reduce proteinuria, despite lowering BP to the same levels as the RAAS antagonists and was associated with an increase in intrarenal Ang II.³¹ Thus intrarenal Ang II appears to play a central role in the pathogenesis of hypertension and kidney injury in this model, even in the absence of sustained increases in Ang II levels. Further research is required to explain these findings fully. It could be argued that apparently normal intrarenal Ang II levels are inappropriately high in the context of hypertension and ECF volume expansion seen in these animals or that the total intrarenal Ang II levels measured may have failed to detect important local elevations of Ang II. Ongoing research has elucidated several novel aspects of the intrarenal RAAS, including regulation by several other factors like prorenin receptor, prostaglandin E₂, and Wnt/ β -catenin (positive regulators), as well as Klotho, vitamin D receptor, and liver X receptor (negative regulators).³²

Attention has focused on the potential role of aldosterone in progressive kidney injury. In addition to evidence that aldosterone may exert profibrotic effects in the kidney (see later), experiments have suggested that it may also have important glomerular hemodynamic effects. Previous observations that the deoxycorticosterone (DOCA)–salt model of hypertension is associated with glomerular capillary hypertension prompted detailed studies of microperfused rabbit afferent and efferent arterioles; these found dose-dependent constriction of both arterioles in response to nanomolar concentrations of aldosterone, with greater sensitivity observed in efferent arterioles.³³ These effects were not inhibited by spironolactone and were still present with albumin-bound aldosterone, indicating that they may be mediated by specific membrane receptors rather than the intracellular receptors responsible for most of the actions of aldosterone. Interestingly, aldosterone may also counteract rabbit-afferent arteriolar vasoconstriction

via a nitric oxide (NO)–dependent pathway, an action that would also be expected to increase P_{GC} .^{34,35}

ENDOTHELINS

Endothelins (ETs) are potent vasoconstrictor peptides that act via at least two receptor subtypes, ET_A and ET_B. ET receptors have been identified throughout the body and are most abundant in the lungs and kidneys. ET_A receptors are primarily located on vascular smooth muscle cells and mediate vasoconstriction, as well as cellular proliferation, inflammation, and podocyte damage. ET_B receptors are expressed on vascular endothelial and kidney epithelial cells and appear to play a role as clearance receptors, as well as mediating endothelium-dependent vasodilation via NO and prostacyclin.^{36–40} Kidney production of endothelins is increased after 5/6 nephrectomy, raising the possibility that they may also contribute to the observed glomerular hemodynamic adaptations.^{41,42} Acute and chronic infusion of endothelins elicits dose-dependent reductions in RPF and GFR in normal rats.^{43–46}

Observations regarding the relative effects of endothelins on afferent and efferent arterioles are to some extent contradictory, possibly reflecting different experimental conditions. Despite some differences, most studies in intact animals have reported greater increases in efferent than afferent arteriolar resistance, resulting in an increase in P_{GC} . The ultrafiltration coefficient (K_f) was significantly reduced, and thus the SNGFR was unchanged or decreased.^{47–50} On the other hand, observations in microperfused arterioles have found that endothelins cause greater constriction of the afferent than efferent arteriole. Studies with selective ET_A and ET_B receptor antagonists and ET_B receptor knockout mice were somewhat contradictory, suggesting a complex interaction between ET_A and ET_B receptors in determining the response.^{51,52} Comparative studies in rat juxtamedullary nephron preparations have shown that endothelin is a more potent vasoconstrictor of both afferent and efferent arterioles than Ang II, arginine vasopressin, norepinephrine, sphingosine-1-phosphate, and adenosine triphosphate (ATP).⁵³ Using data from studies of endothelin infusion experiments in healthy human volunteers, detailed mathematical modeling identified the strongest effect to be afferent arteriolar vasoconstriction via ET_A and weaker vasoconstriction of the efferent arteriole. Systemic vascular resistance and renal vascular resistance both increased.⁵⁴ In short-term studies of human subjects with CKD, endothelin receptor antagonists increased kidney blood flow but had little effect on the GFR due to a decrease in filtration fraction, observations consistent with a greater vasodilatory effect on the efferent arteriole.^{55,56} Interestingly these observations were made in subjects already receiving treatment with an ACE inhibitor or ARB. The potential interaction between endothelins and other vasoactive molecules is further illustrated by observations that chronic infusion of Ang II results in increased production of endothelins⁵⁷ and that ET-1 transgenic mice are not hypertensive but show induction of inducible nitric oxide synthase (iNOS), resulting in increased NO production as a probable counterregulatory mechanism to maintain normal BP.⁵⁸ Furthermore, some of the glomerular hemodynamic effects of ETs appear to be modulated by prostaglandins.⁵⁰

Detailed micropuncture studies to elucidate the role of endothelins in remnant kidney hemodynamics have not yet been published. These studies should be facilitated by the ongoing development of specific ET_A and ET_B receptor antagonists.

NATRIURETIC PEPTIDES

Atrial natriuretic peptide (ANP) and other structurally related NPs mediate, in large part, the functional adaptations in tubular sodium reabsorption that maintain sodium excretion in 5/6 nephrectomized rats⁵⁹ but also exert important hemodynamic effects. Circulating ANP levels are elevated in 5/6 nephrectomized rats, and acute administration of a NP antagonist elicited profound decreases in the GFR and RPF in 5/6 nephrectomized rats on a high-salt (but not low-salt) diet, indicating that NPs play an important role in the observed hemodynamic responses to 5/6 nephrectomy.⁶⁰

In another study, brain natriuretic peptide (BNP) levels were found to be elevated after 3/4 nephrectomy in the absence of cardiac dysfunction or upregulation of myocardial BNP gene expression.⁶¹ Further insights into the kidney hemodynamic effects of NP have been gained from observations in normal rats infused with a synthetic ANP. Whole-kidney and single-nephron GFR increased by approximately 20% due entirely to a rise in P_{GC} , resulting from significant afferent arteriolar dilation and efferent arteriolar constriction.⁶² In the previous experiments, some residual elevation in remnant kidney GFR appeared to persist, even after the NP system was suppressed by sodium restriction or a NP receptor antagonist, suggesting that factors other than NP contribute to glomerular hyperfiltration following kidney mass ablation. The potential interaction between NPs and other vasoactive molecules is illustrated by the observation that ANP infusion in normal rats induces an increase in kidney nitric oxide synthase activity.⁶³ In a human study pro-ANP and pro-BNP concentrations were observed to increase threefold at 24 hours after unilateral total or partial nephrectomy but returned to baseline after 5 days. Pro-ANP and pro-BNP concentration were elevated at 12 months after unilateral nephrectomy but not partial nephrectomy.⁶⁴

EICOSANOIDS

Eicosanoids, another family of potent vasoactive molecules present in abundance in the kidney, may also play a role in mediating glomerular hyperfiltration. Urinary excretion per nephron of both vasodilator and vasoconstrictor prostaglandins (PGs) is increased in rats and rabbits after kidney mass ablation.⁶⁵⁻⁶⁷ Infusion of PGE₂, PGI₂, or 6-keto-PGE₁ into the renal artery elicits significant renal vasodilation.⁶⁸ Whereas acute inhibition of prostaglandin synthesis by infusion of the cyclooxygenase (COX) inhibitor indomethacin has no effect on GFR or glomerular hemodynamics in normal rats, indomethacin lowers both the SNGFR and Q_A after 3/4 or 5/6 nephrectomy.^{65,66} On the other hand, chronic treatment with a selective COX-2 inhibitor attenuates the systemic and glomerular hypertension observed in 5/6 nephrectomized rats but has no effect on the GFR.⁶⁹

The relative effects of prostaglandin synthesis inhibitors on afferent and efferent arterioles may vary with time

post-nephrectomy. Afferent arteriolar constriction was the predominant finding reported at 24 hours post-surgery, whereas constriction of both afferent and efferent arterioles was observed at 3 to 4 weeks.^{65,66} Some contribution of thromboxanes to glomerular hemodynamic adjustments after 5/6 nephrectomy in rats is suggested by the increase in the GFR seen after acute infusion of a selective thromboxane synthesis inhibitor.⁶⁷ Thus different eicosanoids appear to exert opposite effects, but the general impression is that the combined effects of vasodilator prostaglandins outweigh those of the vasoconstrictors. This interaction is illustrated by the observation that perfusion of isolated glomeruli with bradykinin resulted in vasodilation of the efferent arteriole that was completely blocked by indomethacin but that this blockade was reversed by a specific antagonist of 20-hydroxyeicosatetraenoic acid (20-HETE), a vasoconstrictor eicosanoid, indicating that the glomerulus produces both vasodilator and vasoconstrictor eicosanoids.⁷⁰ Thromboxane A₂ (TXA₂) may play a role in the development of hypertension after nephron loss. Plasma concentrations of the TXA₂ metabolite, thromboxane B₂, were elevated in rats after 5/6 nephrectomy and TXA₂ receptor expression was increased in aortic rings.⁷¹

NITRIC OXIDE

The extremely short half-life of nitric oxide (NO) precludes direct measurement of nitric oxide levels or the administration of exogenous nitric oxide in experimental models. The actions of nitric oxide have thus been inferred from experiments with inhibitors of nitric oxide synthase (NOS). Intravenous infusion of NOS inhibitors results in systemic and renal vasoconstriction, as well as a reduction in the GFR, in normal rats.^{72,73} Thus nitric oxide appears to exert a tonic effect on the physiologic maintenance of systemic BP and renal perfusion under resting conditions. It is unclear, however, whether nitric oxide plays a specific role in the adaptive hemodynamic changes that follow kidney mass ablation. Indeed, renal expression of NOS and renal nitric oxide generation is reduced in 5/6 nephrectomized rats, whereas systemic production of nitric oxide is increased.^{74,75} MAP and renal vascular resistance increase, whereas renal blood flow (RBF) and the GFR decrease to a similar extent after acute infusion of an endothelial NOS (eNOS) inhibitor, NG-monomethyl-L-arginine (L-NMMA), irrespective of whether given to normal rats or 3 to 4 weeks after unilateral or 5/6 nephrectomy.⁷³ Chronic NOS inhibition with nitro-L-arginine methyl ester (L-NAME) produces elevations in systemic BP and P_{GC} in 5/6 nephrectomized rats without affecting the GFR,⁷⁶ whereas chronic treatment with aminoguanidine, an inhibitor of inducible NOS, has no effect on the GFR, RPF, or P_{GC} .⁷⁵

Similarly, greater increases in BP and proteinuria were observed after 5/6 nephrectomy in eNOS knockout versus wild-type mice.⁷⁷ On the other hand, renal NOS expression and activity are increased early after unilateral nephrectomy, and pretreatment of rats with a subpressor dose of L-NAME prevents the early increase in RBF and decrease in renal vascular resistance usually observed after unilateral nephrectomy.^{78,79} It therefore appears that nitric oxide plays a role in early hemodynamic adaptations to nephron

loss, resulting in an increase in RBF, but in the longer term, nitric oxide retains a tonic influence on systemic and renal hemodynamics without being a specific determinant of the adaptive changes in glomerular hemodynamics.

BRADYKININ

Bradykinin is a potent vasodilatory peptide that is elevated in the remnant kidney²⁶ and may therefore contribute to hemodynamic adaptations after nephron loss. Acute and chronic infusion of bradykinin result in increased RPF but have no effect on the GFR.^{80,81} Micropuncture studies in intact animals are lacking, but studies of isolated perfused afferent arterioles have shown that bradykinin induces a biphasic response with vasodilation at low concentrations and vasoconstriction at higher concentrations. Both effects appear to be mediated by products of COX.⁸² Similar experiments with efferent arterioles have found dose-dependent vasodilation (no biphasic response) that was dependent on cytochrome P450 metabolites but independent of COX products or nitric oxide.⁸³ When glomeruli were perfused with bradykinin, vasodilation of efferent arterioles was again observed but inhibited by a COX inhibitor, indicating that bradykinin induces glomerular production of COX metabolites (prostaglandins) that also contribute to efferent arteriolar dilation.⁷⁰ Further studies are required to elucidate the role of bradykinin after nephron loss.

UROTENSIN II

Urotensin II is the most potent vasoconstrictor identified to date, but its actions appear to vary in different vascular territories and, in some vessels, it may even produce vasodilation. Infusion of exogenous urotensin II has been reported to increase or decrease the GFR in normal rodents in different experiments (see Chapter 11). The vasoconstrictor effects of urotensin II may be mediated, at least in part, by upregulation of renal renin and aldosterone synthase gene expression.⁸⁴ Urotensin II is produced in the kidney, and urotensin receptors have been localized to glomerular arterioles.⁸⁵ Increased renal expression of mRNA for urotensin-related protein (which also binds to the urotensin receptor) and urotensin receptor has been reported in rats after 5/6 nephrectomy.⁸⁶ Further experiments have observed increased immunostaining for urotensin II in tubules and glomeruli after 5/6 nephrectomy that increased as the kidney damage progressed. Increased urotensin receptor expression was also observed at more advanced stages. Treatment with a urotensin receptor antagonist delayed the onset of hypertension and albuminuria but did not attenuate glomerulosclerosis after 13 weeks.⁸⁷ The role of urotensin II in the hemodynamic adaptations that follow nephron loss has yet to be investigated.

ADJUSTMENTS IN RENAL AUTOREGULATORY MECHANISMS

After extensive kidney mass ablation, there is a marked readjustment of the autoregulatory mechanisms that control RPF and the GFR.^{11,88-90} The role of myogenic mechanisms is uncertain, but detailed studies of afferent arteriolar myogenic responses have suggested that their primary role is to protect the glomerulus from elevations in SBP.⁹¹ This notion has been supported by animal

experiments that have reported that rat strains with impaired myogenic response show greater transmission of perfusion pressure to glomerular capillaries and are more susceptible to glomerular injury than strains with an intact myogenic response.⁹² The tubuloglomerular feedback system is reset after kidney mass ablation to permit and sustain the elevations in SNGFR and P_{GC} described previously.^{93,94} Studies have indicated that connecting tubule glomerular feedback mediated by the epithelial sodium channel (ENaC) plays a key role in this process after uninephrectomy.⁹⁵ Resetting appears to occur as early as 20 minutes after unilateral nephrectomy,⁹⁶ in proportion to the extent of kidney ablation. The adjustments observed after uninephrectomy are of lesser magnitudes than those seen after 5/6 nephrectomy.⁹³ The kidney-protective effects of sodium-glucose cotransporter 2 (SGLT2) inhibitors have been attributed at least in part to their natriuretic effect increasing tubuloglomerular feedback to reduce glomerular hyperfiltration. Treatment with the SGLT2 inhibitor, empagliflozin, after 5/6 nephrectomy resulted in an increase in urinary adenosine excretion, a marker of tubuloglomerular feedback activity, that was inversely related to the severity of tubulointerstitial fibrosis.⁹⁷

INTERACTION OF MULTIPLE FACTORS

As is readily appreciated from the previous discussion, the adjustments in glomerular hemodynamics seen after kidney mass ablation represent the net effect of several endogenous vasoactive factors. NPs and vasodilator prostaglandins dilate the preglomerular vessels, whereas bradykinin dilates both afferent and efferent arterioles. On the other hand, Ang II, vasoconstrictor prostaglandins, and possibly endothelins constrict both afferent and efferent arterioles, with a greater effect on the latter. A net fall in preglomerular vascular resistance is observed, whereas efferent arteriolar resistance decreases to a lesser extent. Together with greater transmission of the raised systemic BP to the glomerular capillary network, these alterations in microvascular resistances result in the observed elevations in Q_A , P_{GC} , ΔP , and SNGFR (Table 50.1). The importance of multiple vasoactive factors is illustrated by the observation that treatment of 5/6 nephrectomized rats with omapatrilat, an inhibitor of both ACE and neutral endopeptidase that results in reduced Ang II production, as well as increased NPs and bradykinin levels, lowers P_{GC} more than ACE inhibition alone.⁹⁸

The complexity of factors involved is further illustrated by observations that other molecules involved in the modulation of progressive kidney injury may exert hemodynamic effects by influencing the mediators discussed previously. Acute infusion of hepatocyte growth factor has been shown to induce a decline in BP and GFR, an effect that is mediated by a short-term increase in ET-1 production.⁹⁹ In isolated perfused preparations, platelet-activating factor at picomolar concentrations has been shown to induce glomerular production of NO, resulting in dilation of precontracted efferent arterioles, whereas at nanomolar concentrations, platelet-activating factor constricts efferent arterioles through local release of COX metabolites.¹⁰⁰ The potential role of other recently identified vasoactive molecules such as urotensin II remains to be elucidated.

Table 50.1 Hemodynamic Effects of Vasoactive Molecules Mediating Glomerular Hemodynamic Adaptations After Partial Renal Mass Ablation

Parameter	R_A	R_E	P_{GC}	Q_A	K_f	SNGFR	RPF	GFR
Angiotensin II	↑	↑↑	↑	↓	↓↔	↓↔	↓	↔
Aldosterone	↑	↑↑	↑	?	?	?	?	?
Endothelins	↑↔	↑	↑↔	↓	↓↔	↓↔	↓	↓↔
Natriuretic peptides	↓	↑(?)	↑	↔	↔	↑	↑↔	↑
Prostaglandins	↓	↓	↔	↑	↑	↑	↑	↑
Bradykinin	↑↑	↓	?	?	?	?	↑	↔
Observed changes after partial renal ablation	↓↓	↓	↑	↑	↑↓	↑	—	↓

GFR, Glomerular filtration rate; K_f , glomerular ultrafiltration coefficient; P_{GC} , glomerular capillary hydraulic pressure; Q_A , glomerular plasma flow rate; R_A , afferent arteriolar resistance; R_E , efferent arteriolar resistance; RPF, renal plasma flow; SNGFR, single-nephron GFR.

KIDNEY HYPERTROPHIC RESPONSES TO NEPHRON LOSS

The notion that a single kidney enlarges to compensate for the loss of its partner has been entertained since antiquity. Aristotle (384–322 bc) noted that a single kidney was able to sustain life in animals and that such kidneys were enlarged. In preparation for the first human nephrectomy in 1869, a German surgeon, Gustav Simon, uninephrectomized dogs and noted a 1.5-fold increase in the size of the remaining kidney at 20 days.¹⁰¹ Compensatory kidney hypertrophy has been studied in a variety of species including toads, mice, rats, guinea pigs, rabbits, cats, dogs, pigs, and baboons. Most experimental work has been conducted in rodents subjected to uninephrectomy, but hypertrophic responses have also been studied in response to unilateral ureteric obstruction (UUO) or after nephrotoxin administration.¹⁰²

WHOLE-KIDNEY HYPERTROPHIC RESPONSES

Among the earliest responses to unilateral nephrectomy are biochemical changes that precede cell growth. Increased incorporation of choline, a precursor of cell membrane phospholipid, has been detected as early as 5 minutes,^{103,104} and increased choline kinase activity has been observed at 2 hours after nephrectomy.¹⁰⁵ Activity of ornithine decarboxylase, the enzyme catalyzing the first step of polyamine synthesis, is elevated at 45 to 120 minutes, and polyamine levels peak at 1 to 2 days post-nephrectomy.^{106–108} Early alterations in mRNA metabolism have also been observed. Although there are no changes in the half-life or cytoplasmic distribution of mRNA, a near-25% increase in the fraction of newly synthesized poly(A)-deficient mRNA occurs within 1 hour of uninephrectomy, and total RNA synthesis in the kidney increases by 25% to 100% relative to that in the liver.^{109–111} Ribosomal RNA synthesis is increased by 40% to 50% at 6 hours.¹¹² The rate of protein synthesis is increased at 2 hours and is nearly doubled at 3 hours.¹¹³ Data on cyclic nucleotide levels, which are thought to affect cell growth and proliferation, are conflicting. Some studies have reported elevated levels of cyclic guanosine monophosphate (cGMP) in the remaining kidney as early as 10 minutes after surgery,^{114–116} whereas others have found no consistent changes in cyclic adenosine monophosphate (cAMP) or cGMP levels.^{102,117} Genome-wide analysis of gene expression using cDNA microarrays in remaining rat

kidneys up to 72 hours after uninephrectomy has revealed the dominant response to be suppression of the genes responsible for inhibition of growth and apoptosis.¹¹⁸

Early biochemical changes are followed by a period of rapid growth. DNA synthesis is increased at 24 hours, and increased numbers of mitotic figures are evident at 28 to 36 hours. Both reach a maximum 5- to 10-fold increase at 40 to 72 hours.^{119–122} In rats, kidney weight is increased at 48 to 72 hours after uninephrectomy and increases by 30% to 40% at 2 to 3 weeks (Fig. 50.2).^{79,102} The nephron number is fixed shortly before birth in most species, so this gain in kidney weight is attributable to increased nephron size. Growth is thought to occur largely through cell hypertrophy, accounting for 80% of the increase in kidney mass seen in adult rats and, to a lesser extent, through hyperplasia.¹¹³ Kidney mass continues to rise for 1 to 2 months until a 40% to 50% increase is achieved. The degree of compensatory growth is a function of the extent of kidney ablation. Uninephrectomy has been shown to provoke an 81% increase of residual kidney mass at 4 weeks compared with an increase of 168% after 70% kidney ablation. Normal controls gained 31% in kidney weight over the same period.¹²³

Age diminishes kidney hypertrophic responses. After uninephrectomy, greater increases in kidney weight and more extensive hyperplasia were observed in 5- versus 55-day-old rats,¹²⁴ and aging rats exhibited gains in kidney weight of only one-third to three-quarters of those seen in younger controls.^{102,125–127}

In humans, assessment of kidney hypertrophy after nephrectomy is dependent on radiologic studies. Ultrasound studies have reported increases of 19% to 100% in kidney volume¹²⁸ and, in computed tomography (CT) studies, have found an increase of 30% to 53% in kidney cross-sectional area after nephrectomy.^{129,130} Contrast-enhanced CT has been used to measure kidney parenchymal volume post unilateral nephrectomy. One study reported increases of 12.1% and 8.9% at 1 week and 6 months, respectively.¹³¹ The degree of hypertrophy correlated positively with the function of the kidney removed and negatively with patient age. One relatively large study of living kidney donors observed a 27.6% ± 9.7% increase in the remaining kidney volume at 6 months post donor nephrectomy,¹³² and a further study has reported an increase in kidney cortical

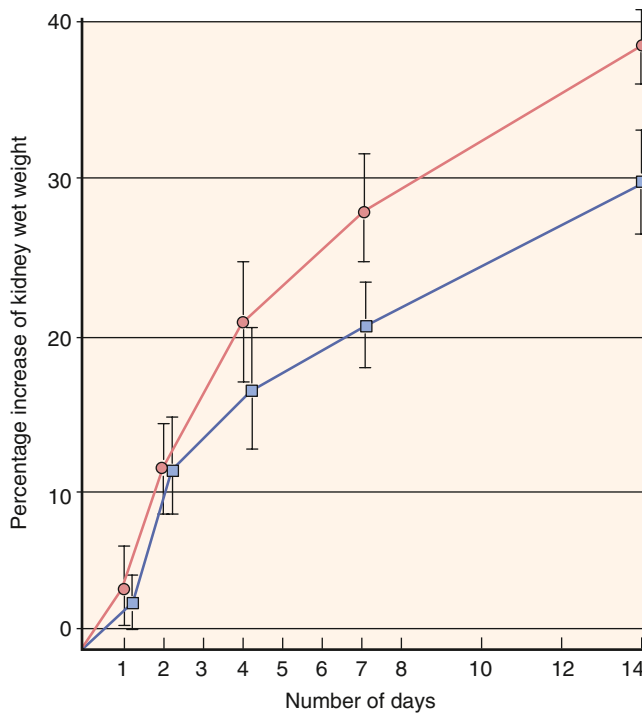


Fig. 50.2 Rate of compensatory renal growth after unilateral nephrectomy (circles) and ureter ligation (squares). (From Dicker SE, Shirley DG. Compensatory hypertrophy of the contralateral kidney after unilateral ureteral ligation. *J Physiol (Lond)*. 1972;220:199–210.)

volume of 27% at a median of 0.8 years post nephrectomy that increased to 35% after a median of 6.1 years.¹⁷ Other studies using CT scans have reported increases in kidney volume of 16.5 to 18.5% at 1 year after nephrectomy for renal cell carcinoma¹³³ and $22.4\% \pm 23.2\%$ at 6 to 12 months after donor nephrectomy.¹³⁴ In a detailed study that used magnetic resonance imaging (MRI) to measure kidney volume, an increase was observed from 198 ± 87 mL preoperatively to 329 ± 175 mL at 3 months post nephrectomy for renal cell carcinoma. The change in kidney volume correlated positively with changes in RBF at 1 week post nephrectomy ($r = 0.6$).¹³⁵ Differences among these studies are likely attributable to the relatively small number of subjects enrolled, wide variation in the time intervals between nephrectomy and assessment of kidney size, and differing indications for nephrectomy.

GLOMERULAR ENLARGEMENT

The principal morphometric change observed in glomeruli after uninephrectomy is an increase in volume. Glomerular enlargement appears to parallel whole-kidney growth and has been detected as early as 4 days after surgery.¹³⁶ The degree of enlargement of superficial and juxtamedullary glomeruli is similar. Proportionally similar increases in number and size of all cell types occur, with preservation of the relative volumes of different glomerular cells.¹⁰² There is consensus that glomerular capillaries increase in length and number (i.e., more branching), but most studies have shown that diameter or cross-sectional surface area of the glomerular capillaries remains constant or increases only minimally.^{137,138} Transplantation of hypertrophied kidneys into uninephrectomized rats has demonstrated regression

of glomerular hypertrophy within 3 weeks, yet the increase in capillary length was maintained.¹³⁸

Glomerular hypertrophy, as evidenced by elevated RNA/DNA and protein/DNA ratios, as well as by increased glomerular volume (V_G) on electron microscopy, has been detected at 2 days after 5/6 nephrectomy in rats.¹³⁹ The initial increase in V_G was due almost entirely to increases in visceral epithelial cell volume, whereas at 14 days the increase in V_G was largely accounted for by mesangial matrix expansion. Although several studies have reported glomerular capillary lengthening after 5/6 nephrectomy, few have detected any increase in cross-sectional area or diameter of the glomerular capillaries.^{140–143} These observations should, however, be considered in the light of important technical considerations. In vitro perfusion of isolated glomeruli has demonstrated that V_G increases as perfusion pressure is raised through physiologic and pathophysiologic ranges. Moreover, glomerular capillary “compliance” in these studies was a function of the baseline V_G , and glomeruli obtained from remnant kidneys post 5/6 nephrectomy had higher compliance than those from control animals.¹⁴⁴ These findings have two important implications. First, although glomerular pressures are only minimally elevated after uninephrectomy, the glomerular capillary hypertension associated with more extensive kidney ablation is likely to contribute significantly to the increase in V_G . Second, estimates of V_G in tissues that have not been perfusion-fixed at the appropriate BP should be interpreted with caution. Direct comparison of V_G in perfusion-fixed versus immersion-fixed kidneys from the same rats yielded estimates of V_G in immersion-fixed samples that were 61% lower than those from perfusion-fixed kidneys.¹⁴⁵

MECHANISMS OF KIDNEY HYPERTROPHY

Despite more than a century of research that identified a large number of mediators or modulators of kidney hypertrophy, the identities of the specific factors that regulate hypertrophy and the stimuli to which these factors respond remained elusive until relatively recently. Kidney innervation does not appear to play a role because kidneys transplanted into bilaterally nephrectomized rats exhibit the same degree of hypertrophy after 3 weeks as kidneys remaining after uninephrectomy.¹⁴⁶ The absence of any reduction in kidney hypertrophy when rats are treated with an ACE inhibitor after uninephrectomy indicates that the renin-angiotensin system also does not play a major role.¹⁴⁷ Several hypotheses were advanced to account for the observed changes associated with kidney hypertrophy and have been discussed in detail in other publications^{101,102} and are summarized here.

SOLUTE LOAD

The notion that hypertrophy after uninephrectomy is stimulated by the need for the remaining kidney to excrete larger amounts of metabolic waste products, necessitating more excretory “work,” was proposed by Sacerdotti in 1896. Subsequently, it became apparent that urea excretion is largely a function of glomerular filtration, whereas the main energy-requiring function of the renal tubules is reabsorption of filtered electrolytes (principally sodium), and water. The hypothesis was therefore modified to view hypertrophy as

a response to the increased demand for water and solute reclamation imposed by an increased SNGFR (solute load hypothesis). Several lines of evidence have supported the concepts underlying the solute load hypothesis. After uninephrectomy, RBF increased by 8% in the remaining kidney and preceded hypertrophy, and treatment with a subpressor dose of the NOS inhibitor L-NAME prevented the rise in RBF and substantially attenuated increases in kidney weight, as well as glomerular and proximal tubule areas, at 7 days postnephrectomy.⁷⁹ In the remnant kidney, proximal tubule sodium absorption increased in parallel with GFR (glomerulotubular balance),¹⁴⁸ and tubules continued to display enhanced fluid reabsorption *in vitro*, implying that the adaptive changes were intrinsic to the tubular epithelial cells (TECs).^{149,150} In chronic glomerulonephritis, a lesion characterized by marked heterogeneity in the SNGFR, there was preservation of the SNGFR to proximal fluid reabsorption ratio and a close correlation between glomerular and proximal tubule hypertrophy. Moreover, sustained increases in the GFR in the absence of kidney mass ablation result in kidney hypertrophy in some conditions including pregnancy (in some but not all studies) and diabetes mellitus.¹⁰¹

On the other hand, experimental maneuvers dissociating kidney solute load from hypertrophy appear to contradict the solute load hypothesis. Total diversion of urine from one kidney into the peritoneum by ureteroperitoneostomy was associated with an increase in the GFR in the contralateral kidney of similar magnitude to that seen after uninephrectomy, but no increase in kidney mass or mitotic activity.¹⁵¹ In another example, potassium depletion resulted in kidney hypertrophy without any increase in GFR.¹⁵² Moreover, the findings that some of the early biochemical changes associated with hypertrophy precede increases in glomerular filtration or sodium reabsorption argue against a causal association of hypertrophy and increased solute load. Despite these conflicting data, there is nevertheless considerable evidence of an association between SNGFR and proximal tubule hypertrophy that may play a role in stimulating kidney growth in the remnant kidney.¹⁰¹

RENOTROPIC FACTORS

Failure of the solute load hypothesis to explain all the experimental data has led others to propose instead that the primary stimulus for kidney hypertrophy is a change in kidney mass and that kidney growth is under the control of specific growth and/or inhibitory factors. Evidence in support of this theory was derived from three types of experiments.¹⁵³

In the first, a stable connection was established between the extracellular space and microcirculation of two animals (parabiosis), and the effects of kidney mass ablation in one animal were assessed in the intact kidneys of its partner.¹⁰¹ Despite some inconsistencies due to variations in methodology, these experiments generally found that uninephrectomy in one animal resulted in hypertrophy of the contralateral kidney and, to a lesser extent, of both kidneys of the parabiotic partner. Bilateral nephrectomy in one partner or a triple nephrectomy produced incremental degrees of hypertrophy in the remaining kidney(s). Furthermore, the hypertrophy was rapidly reversed following cessation of cross circulation.

A second strategy was to inject serum or plasma from uninephrectomized animals into intact subjects and then assess kidney hypertrophy by radiolabeled thymidine uptake or mitotic count.¹⁰¹ Although studies using single small intraperitoneal or subcutaneous doses were negative, the administration of repeated large doses by an intraperitoneal or intravenous route elicited kidney hypertrophy in most subjects studied.

The data that most consistently supported the existence of a renotropic factor were derived from *in vitro* experiments in which kidney tissues were incubated in the presence or absence of plasma or serum from rats subjected to kidney mass ablation. Evidence for hypertrophy was generally assessed by the incorporation of radiolabeled thymidine or uridine into DNA or RNA, respectively. In general, these experiments showed increased uptake of radiolabeled nucleotides after incubation with serum from uninephrectomized animals. This effect appeared to be organ but not species specific. That a tissue factor produced by the kidneys and upregulated after nephrectomy may be required for the activity of a circulating renotropin was suggested by experiments in which kidney extract from rats taken 20 hours after uninephrectomy, in the presence of normal rat serum, was found to stimulate ³H-thymidine incorporation in normal kidney cortex. However, addition of the same extract in the absence of the serum tended to depress ³H-thymidine uptake.¹⁵⁴ Serum taken from bilaterally nephrectomized animals lacked renotropic effects, but these were restored after dialysis of the serum, suggesting the presence of renotropin inhibitory factors that accumulated in the absence of kidney function.¹⁵⁴ Although the specific identity of “renotropin” has remained elusive, several lines of evidence suggested that it was a small protein. Retention of activity after ultrafiltration, dialysis, and removal of albumin from serum implied that renotropin was a molecule of 12 to 25 kDa in size, with no significant binding to albumin.^{101,102}

Several hypotheses were advanced to reconcile the previously observed effects and operation of a putative renotropic system.^{101,102} The following were variously proposed: 1. Renotropin is a circulating substance normally catabolized or excreted by the kidneys; 2. kidney growth is regulated by a specific renotropin-producing tissue inhibited by a factor produced by normal kidneys; and 3. kidney growth is tonically inhibited by a substance produced by normal kidneys, a decrease in the levels of which induce an enzyme in the kidney cortex that cleaves a circulating precursor of renotropin to produce the active molecule.

ENDOCRINE EFFECTS

Several of the major endocrine systems influence kidney growth, but each lacks selective effects on the kidney. There is little evidence that any of these systems represents the specific mediators of compensatory kidney hypertrophy. Whereas early experiments suggested that hypophysectomy inhibits compensatory hypertrophy after uninephrectomy, later studies that controlled for the reduction in kidney mass that usually accompanies hypopituitarism found a degree of hypertrophy comparable with that seen in normal rats.¹⁵⁵

Nevertheless, specific renotropic activity has been identified in a subfraction of ovine pituitary extract associated with a lutropin-like substance.^{101,102} Uninephrectomy is

accompanied by a transient increase in the pulsatile release of growth hormone in male but not female rats, suggesting a role for this hormone in the early phase of hypertrophy in males.¹⁵⁶ When the increase in growth hormone was prevented by the administration of an antagonist to growth hormone-releasing factor or the effects of growth hormone were blocked by a growth hormone receptor blocker, kidney hypertrophy was significantly attenuated.^{157,158} Adrenal hormones appear to play only a small role in kidney hypertrophy. Adrenalectomy does not inhibit compensatory growth after uninephrectomy.¹⁵⁹ Whereas kidney weight relative to body weight is reduced in hypothyroidism and increased by excess thyroid hormone, compensatory hypertrophy still occurs in thyroidectomized rats.¹⁶⁰ Progesterone and estradiol in excess or ovariectomy have little effect on kidney weight, but testosterone appears to play a role, as evidenced by a fall in kidney and body weight after orchidectomy and an increase in kidney weight with excess testosterone.^{161,162} Whereas orchidectomy does not inhibit hypertrophy after uninephrectomy, exogenous testosterone does increase the degree of hypertrophy observed in some but not all studies.^{101,102}

GROWTH FACTORS

Of the numerous growth factors and their receptors that have been localized in the kidney, at least four are associated with kidney hypertrophy.^{163,164} Several lines of evidence suggest a role for insulin-like growth factor-1 (IGF-1). Kidney tissue IGF-1 levels were elevated at 1 to 5 days after uninephrectomy and started to decline within days in some^{165,166} but not all studies.¹⁶⁷ In one study, the level of kidney IGF-1 expression correlated significantly with the extent of kidney mass ablation.¹⁶⁸ On the other hand, Shohat and associates¹⁶⁵ have found an increase in serum IGF-1 levels only at 10 days post nephrectomy, which was still present on day 60. That IGF-1 may be induced independently of GH in the setting of kidney hypertrophy is illustrated by preservation of the increase in kidney IGF-1 in hypophysectomized¹⁶⁹ and GH-deficient rats.¹⁷⁰

Other molecules related to IGF function are also upregulated. Kidney IGF-1 receptor gene expression was increased twofold to fourfold in female rats after uninephrectomy¹⁵⁶; IGF-1 binding protein mRNA was upregulated in the remnant kidney at 2 weeks after 5/6 nephrectomy¹⁷¹; and analysis of the genome-wide transcriptional response to unilateral nephrectomy identified IGF-2 binding protein as one of the activated genes.¹¹⁸ Further evidence has suggested that IGF-1 may in turn promote production of vascular endothelial growth factor (VEGF), implying that VEGF may be a downstream mediator of IGF-1 effects, at least in the pathogenesis of diabetic retinopathy.¹⁷² That VEGF is important for compensatory kidney hypertrophy has been confirmed by the observation that treatment of mice with VEGF antibodies after uninephrectomy completely prevents glomerular hypertrophy and inhibits kidney growth at 7 days.¹⁶⁶ Epidermal growth factor (EGF) in the remaining kidney is increased on day 1 in mice¹⁷³ and by day 5 in rats.¹⁷⁴ In addition, EGF has been shown to induce IGF-I mRNA production in collecting duct cells in vitro, suggesting the existence of a local paracrine system.¹⁷⁵ Increased mRNA levels for both hepatocyte growth factor and its receptor,

c-met, have been demonstrated in the remaining kidney as early as 6 hours after uninephrectomy.^{176,177} In another study, the rise in hepatocyte growth factor message was found to be nonspecific, occurring in both liver and kidney and also in sham-operated rats, whereas the increase in mRNA for c-met was specific for the outer kidney medulla.¹⁷⁸

Despite these associations, the timing of the changes in growth factor levels remains unclear. Whereas some investigators have reported early increases,^{173,179} several others reported changes only at time points when significant hypertrophy is already present, thus failing to provide convincing evidence that they represent the proximal effectors in a renotropic system.^{165,174}

MESANGIAL CELL RESPONSES, A UNIFYING HYPOTHESIS

Mesangial cells play a central role in glomerular function, modulating glomerular capillary blood flow and ultrafiltration surface area. In addition, mesangial cells are both a source of and target for vasoactive molecules, growth factors, cytokines, and extracellular matrix proteins. Studies have suggested that they also play a major role in compensatory kidney hypertrophy. In vitro experiments have found that when mesangial cells from a remaining kidney after uninephrectomy are cultured with serum obtained from rats after uninephrectomy, their conditioned medium induces hypertrophy in tubule cells.¹⁸⁰ Uninephrectomy induces significant transient proliferation of mesangial cells, reaching a peak at 24 hours and ceasing within 72 hours. This proliferation occurs in an environment of increased circulating and kidney growth factors and cytokines, such as growth hormone, IGF-1, and interleukin-10 (IL-10), as well as reduced levels of antiproliferative factors such as transforming growth factor- β (TGF- β) and ANP. A reduction in mesangial cell proliferation occurs in parallel with the onset of tubule cell hypertrophy. IL-10 and TGF- β have been identified as the major mediators of the mesangial regulation of tubule cell hypertrophy. Mesangial cells are the main source of IL-10 in the kidney, where it acts as an autocrine growth factor and induces the expression of TGF- β .¹⁸¹ Mesangial cells are the only resident kidney cells known to produce and activate TGF- β , a process that is regulated by multiple factors, including Ang II, IGF-1, HGF, basic fibroblast growth factor (bFGF), tumor necrosis factor- α (TNF- α), EGF, and platelet-derived growth factor (PDGF), all of which are produced by mesangial cells. IL-10 expression starts to increase in the remaining kidney within hours of uninephrectomy, peaks at 24 hours and returns to normal within several days. In contrast, circulating and kidney TGF- β levels fall in the first 24 hours after nephrectomy and start to rise from 72 hours, reaching a peak at 1 week.¹⁸¹

The importance of IL-10 was confirmed by experiments in which inhibition of IL-10 production resulted in lower TGF- β levels and reduced tubular hypertrophy, resulting in a 20% to 25% reduction in the weight of the remaining kidney.¹⁸¹ IL-10 has no direct effect on tubule cells, whereas TGF- β has been identified as an important mediator of tubule cell hypertrophy.¹⁸² The data presented are therefore consistent with the hypothesis that hyperfiltration after unilateral nephrectomy induces mesangial cell proliferation, as well as the production of IL-10 and other growth factors that induce expression and activation of TGF- β in mesangial

cells. TGF- β in turn stimulates tubule cell hypertrophy¹⁸³ (Fig. 50.3). This goes a long way to unifying the components of the solute load and renotropin hypotheses into a single paradigm to explain the mechanisms of compensatory kidney hypertrophy.

TUBULE CELL RESPONSES

Detailed investigation of cellular responses to kidney mass reduction has begun to elucidate some of the mechanisms involved in compensatory hypertrophy at the cellular level. Kidney hypertrophy is achieved by modulation of the cell cycle, which becomes arrested in the late G1 phase and therefore does not progress to the S phase, resulting in cell hypertrophy instead of hyperplasia. This is achieved through activation of cyclin-dependent kinase (CDK) 4–cyclin D without subsequent engagement of CDK 2–cyclin E, a process thought to be regulated by TGF- β and CDK inhibitor proteins p21^{Waf1}, p27^{kip1}, and p57^{kip2}. Activity of CDK 4–cyclin D complexes increases at 4, 7, and 10 days post nephrectomy, and CDK 2–cyclin E increases at days 2, 4, and 7 to 14, implying that p21^{Waf1}, p27^{kip1}, and p57^{kip2} may play an important role in regulating tubule cell hypertrophy.^{184–186} The required increase in RNA and protein synthesis appears to be mediated at least in part by mammalian target of rapamycin (mTOR), a protein kinase that controls protein synthesis, as well as cell growth and metabolism.

In cells, mTOR exists in two distinct multiprotein complexes, mTORC1 and mTORC2. mTORC1 acts through multiple mediators to regulate protein synthesis and cell size. Two important downstream effectors of mTORC1 are 4E-binding protein 1 and protein kinase p70S6 kinase 1 (S6K1). Transcriptomic analysis after nephrectomy in mice has identified mTORC1 as the most significantly increased gene set at 24 hours, with elevations sustained at 48 and 72 hours.¹⁸⁷ The importance of mTORC1 has been confirmed by the observation that pretreatment of rats with rapamycin, an inhibitor of mTORC1, inhibits kidney hypertrophy after uninephrectomy.¹⁸⁸ Furthermore, experiments in S6K1 knockout mice have found inhibition of 60% to 70% of the hypertrophy observed after uninephrectomy, indicating that S6K1 plays a major role in compensatory hypertrophy.¹⁸⁹

Studies have also identified early upregulation in genes involved in cholesterol biosynthesis after unilateral nephrectomy.¹⁸⁷ A detailed multiomics study has utilized transcriptomics and proteomics to identify the lipid-activated transcription factor, peroxisome proliferator-activated receptor α (PPAR α), as a key mediator involved in hypertrophy of proximal tubule cells in mice after uninephrectomy. Furthermore, treatment with a PPAR α activator, fenofibrate, significantly enhanced hypertrophy, whereas hypertrophy was reduced in mice with deletion of the gene for PPAR α .¹⁹⁰

ADAPTATION OF SPECIFIC TUBULE FUNCTIONS IN RESPONSE TO NEPHRON LOSS

As noted previously, the bulk of the increase in kidney mass following uninephrectomy is due to hypertrophy of the proximal nephron. The more distal nephron segments also

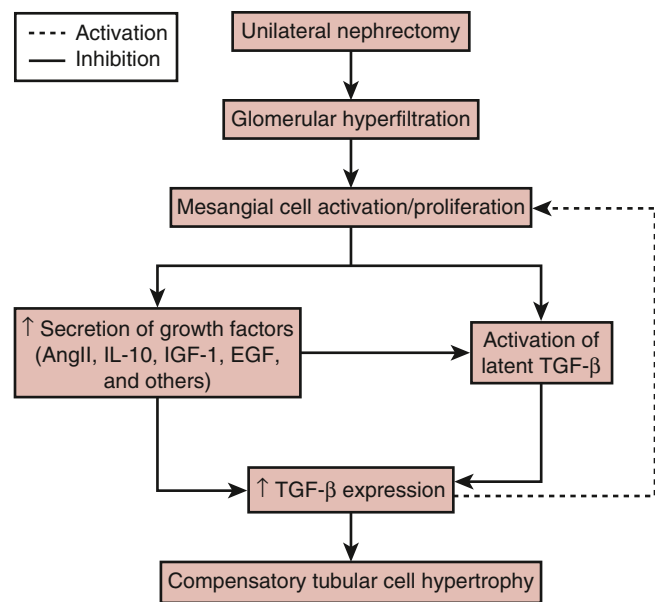


Fig. 50.3 Possible mechanisms involved in compensatory renal growth after unilateral nephrectomy. Ang II, Angiotensin II; EGF, epidermal growth factor; IGF-1, insulin-like growth factor-1; IL-1, interleukin-1; TGF- β , transforming growth factor- β . (From Sinuani I, Beberashvili I, Averbukh Z, et al. Mesangial cells initiate compensatory tubular cell hypertrophy. *Am J Nephrol*. 2010;31:326–331.)

enlarge, but to a lesser extent. In uninephrectomized rats, the proximal convoluted tubule is increased on average by 17% in luminal diameter and 35% in length, yielding a 96% increase in total volume; the distal convoluted tubule is enlarged by 12% in luminal diameter and 17% in length, yielding a 25% increase in total volume.¹⁹¹ Maintenance of homeostasis for various solutes in the presence of a declining GFR requires highly integrated responses from each tubule segment. Whereas some solutes, including creatinine and urea, are chiefly cleared by glomerular filtration and therefore rise gradually in plasma with a declining GFR, for others, the tubule solute handling adapts so that plasma levels remain constant, virtually until kidney failure is reached (Fig. 50.4).

ADAPTATION IN PROXIMAL TUBULE SOLUTE HANDLING

In kidney ablation models, as with the increase in remnant kidney SNGFR, the extent to which the proximal tubule enlarges is inversely proportional to the remnant kidney mass. Proximal tubule enlargement is associated with an increase in proximal fluid reabsorption. In studies of animals and humans with reduced kidney mass, the increase in proximal fluid reabsorption observed was found to be proportional both to the increase in the remnant kidney GFR and the increase in tubular volume.¹⁹² Similarly, in proximal tubules isolated from remnant kidneys, the observed increase in transtubular fluid flux was proportional to the increases in size and protein content of the tubule epithelial cells.^{149,150} Folding of the basolateral membrane of the proximal tubule epithelium was also found to increase, resulting in augmentation of the basolateral surface area, in proportion to the increase in cell volume.¹⁹³ This increase

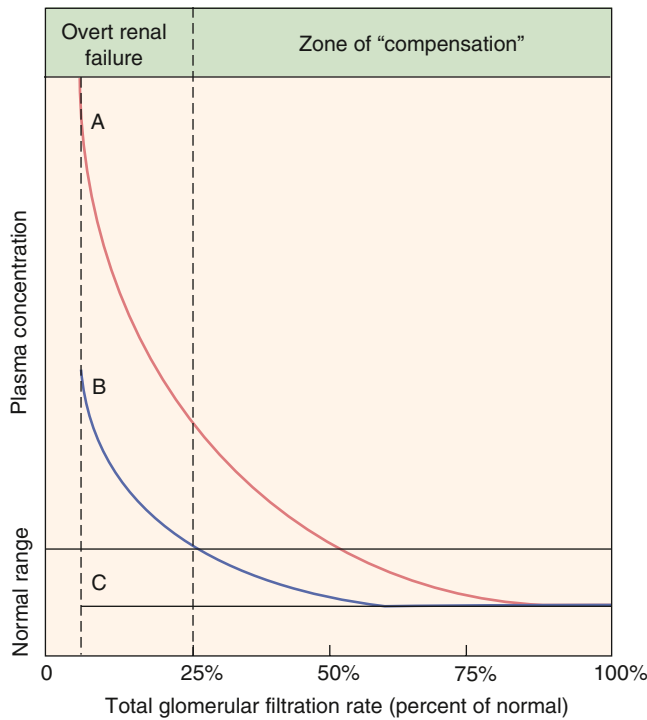


Fig. 50.4 Representative patterns of adaptation for different types of solute in body fluids with declining glomerular filtration rate (GFR). (A) For solutes cleared primarily by glomerular filtration, a rise in serum concentration occurs with each permanent reduction in GFR (e.g., creatinine). (B) For solutes that undergo substantial secretion by the tubules, a rise in serum concentration occurs only after the GFR falls below a critical value due to adaptive increases in tubular secretion (e.g., potassium). (C) For sodium, serum concentration remains normal throughout almost the entire spectrum of GFR decline due to a large capacity to reduce reabsorption by the tubule. (Modified from Bricker NS, et al. in Brenner BM, Rector FC, eds. *The Kidney*, 2nd ed. Philadelphia: WB Saunders; 1981.)

in surface area was accompanied by an increase in activity of $\text{Na}^+\text{-K}^+\text{-ATPase}$, the membrane pump that generates the main driving force for proximal tubule solute and water transport.¹⁹³

Increases in proximal tubule size and surface area are not, however, the only determinants of increased transport activity in this nephron segment. Fluid reabsorption in isolated proximal tubule segments increases within 24 hours of nephrectomy—that is, when the GFR is already increasing but well before significant hypertrophy occurs—implying an intrinsic tubule cell adaptation to nephron loss.¹⁴⁸ This observation also raises the possibility that the increase in proximal fluid reabsorption occurring in response to nephron loss is driven by the increase in the SNGFR.^{101,102} Because solute reclamation is an energy-requiring process, it is not surprising that in uninephrectomized rabbits, the increase in proximal tubule volume is accompanied by a proportional increase in mitochondrial volume.¹⁹⁴ The observation that the increase in kidney mass is outstripped by the rise in the GFR in models of progressive nephron loss implies that kidney energy consumption per unit of remnant kidney mass increases as kidney function declines.¹⁹¹

The rise in SNGFR that occurs in the remnant kidney presents increased loads of glucose, amino acids, and other solutes that should be reabsorbed entirely in the proximal

tubule, provided that the maximal transport capacity was not exceeded. Maximal proximal tubular reabsorptive capacities for glucose and amino acids have been shown to increase in proportion to tubule mass after partial kidney ablation.¹⁹⁵ Some metabolic functions of proximal tubules are also augmented in the remnant kidney to maintain adequate plasma levels of important metabolites including citrulline, arginine, and serine.¹⁹⁶ Other proximal tubule functions, however, are not adjusted in proportion to proximal tubule mass; fractional phosphate reabsorption is decreased, whereas ammoniagenesis increases.^{195,197,198} These adaptations are appropriate homeostatic responses that permit continued excretion of daily phosphate and acid load as the number of functioning nephrons declines.

LOOP OF HENLE AND DISTAL NEPHRON

Although there is little change in cross-sectional area in the thick ascending limb of the loop of Henle, fluid reabsorption in this segment also increases in proportion to the SNGFR.¹⁹¹ In contrast, both the distal tubule and cortical collecting duct enlarge in response to nephron loss.¹⁹¹ Unlike the proximal tubule, however, where the increased reabsorptive capacity is chiefly due to increased tubule dimensions, the increased reabsorptive capacity observed in the distal segments is far greater than would be expected for the corresponding increase in tubule volume, implying a major adaptive increase in active solute transport.¹⁹¹ Levels of mRNA for the sodium-dependent *myo*-inositol cotransporter (SMIT) and Na^+/Cl^- /betaine gamma-aminobutyric acid transporter (BGT-1) are increased in the cortex and outer medulla of remnant kidneys from 5/6 nephrectomized rats.¹⁹⁹ Likewise, potassium secretion by the distal nephron increases in compensation for nephron loss, facilitated by an increase in the basolateral surface area of cortical collecting duct principal cells and an increase in $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity.^{200,201}

GLOMERULOTUBULAR BALANCE

Micropuncture studies have confirmed that proximal fluid reabsorption remains proportional to glomerular filtration over a wide range of SNGFR values in both glomerular and tubulointerstitial diseases.^{202,203} This glomerulotubular balance is critical to the physiologic integrity of remnant nephron function and hence ECF homeostasis. Compensatory increases in the SNGFR in surviving nephrons must be accompanied by similar increases in proximal tubular solute and water reabsorption to avoid overwhelming the distal nephron transport capacity and disrupting its regulation of the volume and composition of the final urine. Conversely, reductions in SNGFR in damaged nephrons must be matched by similar reductions in proximal fluid reabsorption to maintain adequate solute and water delivery to the distal tubule, again permitting excretion of urine of appropriate volume and composition.

Glomerulotubular balance is maintained as follows. The degree of single-nephron hyperfiltration occurring as a consequence of nephron loss determines the passive Starling forces operating in the postglomerular microcirculation, which, in turn, govern net transtubular solute reabsorption.²⁰⁴ Increases in the SNGFR associated

with an increased filtration fraction result in elevated postglomerular capillary protein concentrations, which determine nonlinear increases in oncotic pressure (Π_E), the major determinant of peritubular capillary reabsorptive force (P_r). Reductions in the SNGFR, in contrast, result in a lowered peritubular oncotic pressure, thereby reducing P_r . Thus the SNGFR and proximal fluid reabsorption remain in direct proportion to one another.

Prevention of hyperfiltration by dietary protein restriction has been shown to abrogate the increase in proximal fluid reabsorption in the remnant kidney, underscoring the dependence of proximal tubular function on the level of glomerular filtration.²⁰⁴ In the remnant kidney of rats subjected to extensive kidney mass ablation, absolute fluid reabsorption was found to be markedly increased in proximal portions of superficial and juxtamedullary nephrons, yet fluid delivery to the more distal segments of the nephron was also somewhat increased.²⁰⁵ In the setting of nephron loss, sodium reabsorption by the loop of Henle has been shown to remain proportional to sodium delivery to that segment, indicating preservation of tubulotubular balance, a mechanism that maintains appropriate distal solute and water delivery in the presence of progressive nephron loss. Until the adaptive capacities of these mechanisms are finally exhausted, the operation of glomerulotubular balance and tubulotubular balance ensures that the distal tubule mechanisms that determine final urine volume and composition are not overwhelmed by unregulated distal delivery of water and solute.²⁰⁶ In keeping with these physiologic observations, morphologic studies have shown that within the same kidney, nephrons associated with damaged glomeruli are usually atrophic and presumably hypofunctioning or nonfunctioning, whereas those associated with healthier glomeruli are usually hypertrophic and hyperfunctioning.²⁰⁷

To maintain homeostasis in the presence of continued food and water intake, specific mechanisms that enhance single-nephron water and solute excretion must come into play, in addition to adjustments in the SNGFR and tubular reabsorption that occur in response to nephron loss. These mechanisms are not unique to the setting of impaired GFR, however, and are also engaged when the normal kidney is challenged to excrete extraordinary loads of solute and water. In general, the adaptive physiology of the chronically injured kidney is adequate to preserve homeostasis for many solutes under baseline conditions, but the adaptive capacity may easily become overwhelmed by fluctuations in fluid intake, especially by increases in electrolyte and acid loads. Persons with impaired GFR are therefore susceptible to develop volume overload, volume loss, hyperkalemia, and acidosis when the excretory capacity of the kidney is challenged by relatively modest changes in excretory demands.

SODIUM EXCRETION AND EXTRACELLULAR FLUID VOLUME REGULATION

In CKD, ECF volume is often maintained close to normal until kidney failure is reached.²⁰⁸ This remarkable feat is accomplished by an increase in fractional sodium excretion (FE_{Na}) in inverse proportion to decline in the GFR.²⁰⁹ Many studies have attempted to identify which nephron segments

are responsible for the decrease in sodium reabsorption. In summary:

Micropuncture studies in uninephrectomized rats have shown that tubule fluid transit times, as well as the half-time for reabsorption of a stationary saline droplet in the proximal tubule lumen, were not different from controls.¹⁹¹

In remnant kidneys of rats receiving high normal or low-sodium diets, absolute sodium reabsorption was found to increase, but fractional sodium and fluid reabsorption were found to decrease in all three uremic groups.²¹⁰

Micropuncture studies in dogs and rats have failed to detect significant reductions in fractional proximal tubule fluid and sodium reabsorption.²¹¹

Distal sodium delivery was found to be markedly increased in the rat remnant kidney.²¹⁰

Increased solute transport activity has been demonstrated in the distal tubule of uninephrectomized rats.¹⁹¹

Under conditions of hydropenia and salt loading, sodium reabsorption by the medullary collecting duct of the rat remnant kidney was markedly reduced.²¹²

Investigators have also used a computational model of epithelial transport to investigate changes after nephron loss. Similar to experimental data, the simulations found that after uninephrectomy or 5/6 nephrectomy, fractional reabsorption of sodium and water remains normal from the proximal tubule to distal convoluted tubule but decreases in the collecting tubule and collecting duct to maintain sodium balance.²¹³

Taken together, these data suggest that in the setting of reduced GFR, proximal sodium fractional reabsorption remains largely unchanged and decreases in fractional sodium reabsorption occur predominantly in the loop and distal nephron segments.²¹⁴ These physiologic observations were supported by studies investigating changes in sodium transporters after 5/6 nephrectomy. At 4 weeks after surgery, a substantial increase in abundance of the $Na^+K^+2Cl^-$ and Na^+Cl^- cotransporters (expressed chiefly in the loop of Henle and distal tubule, respectively) was observed, whereas marked decreases were observed in both at 12 weeks.²¹⁵ The expression of ENaC α increased throughout the observation period.²¹⁵

In addition to load-dependent tubular adaptations in sodium handling, sodium excretion is also modulated by hormonal influences. Levels of NPs are elevated when GFR is reduced, as a result of reduced clearance and in response to alterations in sodium and volume status.^{60,216} In rats with extensive kidney mass ablation, plasma ANP levels may be restored toward normal levels by dietary sodium restriction, but in response to increases in sodium intake, they rise progressively along with sodium excretion.²¹⁷ The notion that ANP plays an important role in mediating adaptive changes in sodium excretion in the setting of kidney ablation has been confirmed by observations that the administration of a NP receptor antagonist reduces both FE_{Na} and GFR in 5/6 nephrectomized rats receiving normal or high-salt diets but does not alter these variables in rats fed low-salt diets.⁶⁰ Significantly, NPs not only modulate sodium excretion but may also contribute to the attendant glomerular hyperfiltration and thereby further exacerbate kidney injury (see section earlier).

Systemic hypertension has also been proposed by Guyton and associates²¹⁸ to contribute to the increase in FE_{Na} observed with kidney insufficiency. They hypothesized that a constant sodium intake in the presence of a reduced number of functioning nephrons leads to positive sodium balance as a result of reduced excretory capacity. Positive sodium balance leads to an increase in ECF volume and a rise in systemic BP that in turn leads to an increase in FE_{Na} and reestablishes the steady state. In support of this hypothesis, salt intake has been shown to be critical to the development of hypertension in subtotaly nephrectomized dogs,²¹⁹ and uremic persons have been found to exhibit marked sodium retention when treated with vasodilating antihypertensive agents.²²⁰ On the other hand, a lowered salt intake in 5/6 nephrectomized rats does not prevent the development of systemic hypertension,¹⁴² suggesting that sodium excretion and hypertension are not always interdependent in the setting of extensive kidney mass ablation. Sodium conservation, on the other hand, is also impaired with impaired GFR and, in response to an acute reduction in sodium intake, most persons were unable to reduce sodium excretion below 20 to 30 mEq/day.²²¹ The salt-losing tendency associated with CKD appears to be dependent on the salt load per nephron and may therefore be reversible with adequate dietary sodium restriction. Other factors modulating FE_{Na} in the setting of kidney insufficiency include changes in sympathetic nervous system activity, aldosterone, PGs, and parathyroid hormone (PTH) levels.^{214,222} Sodium homeostasis and volume regulation are discussed in further detail in [chapters 6, 10](#) and [13](#).

URINARY CONCENTRATION AND DILUTION

ECF homeostasis is usually well maintained until GFR is severely reduced, when the ability of the kidney to excrete a volume load becomes significantly reduced.²¹⁴ Normal generation of solute-free water is about 12 mL/100 mL of the GFR and is dependent on dilution of tubule fluid in the thick ascending limb, maintenance of low water permeability in the distal nephron segments in the absence of antidiuretic hormone (ADH), and decreased hypertonicity of the medullary interstitium during water diuresis. Although the single-nephron capacity to excrete free water per milliliter of the GFR is not reduced in persons with advanced CKD,²²³ the absolute reduction in the GFR reduces the overall capacity of the kidney to excrete a water load. Persons with low GFR therefore cannot adequately dilute their urine and are prone to water intoxication and hyponatremia. Hypothetically, in addition to the excretion of the equivalent of 2 L of so-called isotonic urine per day (obligatory excretion of 600 mOsm/day), normal kidneys, with a GFR of 150 L/day, can excrete up to 18 L of free water/day, whereas failing kidneys, with a GFR of 15 L/day, can only excrete about 1.8 L of free water/day. The minimum urinary osmolality achievable by normal kidneys would therefore approach 30 mOsm/L (600 mOsm/20 L), whereas that of diseased kidneys would be 160 mOsm/L (600 mOsm/3.8 L).

The capacity to produce concentrated urine is also impaired with reduced GFR. Normal urinary concentration requires preservation of the countercurrent mechanism to maintain hypertonicity of the medullary interstitium and

normal water transport across the distal nephron segments in response to ADH. The maximal urinary osmolality in normal subjects is about 1200 mOsm/L. As the GFR decreases, however, maximal urinary osmolality falls and, with a GFR of 15 mL/min/1.73 m², is reduced to about 400 mOsm/L.²²⁴ A normal individual can therefore excrete the obligatory daily 600 mOsm in as little as 0.5 L of urine, whereas a person with a GFR of 15 mL/min/1.73 m² requires a minimum of 1.5 L urine to excrete the same load. Part of the defect in urinary concentrating capacity observed with reduced GFR may be attributed to the high solute load imposed per surviving nephron. However, the osmotic effect of urea was shown to be inadequate to account fully for the reduction in maximal urine concentration, indicating that factors other than osmotic diuresis contribute to reduction in urine-concentrating capacity in persons with reduced GFR.²²⁴

Furthermore, in persons with chronic glomerulonephritis, reduction in urine-concentrating capacity was found to correlate significantly with the degree of medullary fibrosis on kidney biopsy,²²⁵ suggesting that disruption of the medullary architecture, with the consequent loss of medullary hypertonicity, may result in disproportionate impairment of urinary concentrating capacity at any given level of the GFR. Consistent with this observation, persons with primary tubulointerstitial injury (e.g., analgesic nephropathy and sickle cell disease) have markedly impaired urine-concentrating capacity, even early in the course of their illness.^{224,226,227} Similarly, in animal experiments, surgical exposure of the kidney papilla in intact hydropenic rats was found to lead to a reduction in urinary osmolality because of the accompanying alterations in vasa recta flow and ensuing washout of medullary solutes.²²⁸ Interestingly, similar exposure of papillae in rats with remnant kidneys did not affect urinary osmolality, presumably because medullary solute washout had already occurred due to the adaptive responses to nephron loss.

Urinary concentration also depends on water reabsorption in the distal nephron segments in the remnant kidney. Reduction in water reabsorption may be the result of several mechanisms in the failing kidney. A defective cAMP-mediated response to ADH may render the cortical collecting duct resistant to the effects of ADH, resulting in increased water delivery to the papillary collecting duct.²²⁹ Urinary osmolality is inversely proportional to fractional water delivery to the papillary collecting duct in 5/6 nephrectomized rats, despite an increase in absolute water reabsorption per functioning collecting tubule when compared with controls.²²⁸ Persons with reduced GFR are therefore prone to volume depletion in the presence of water deprivation or impaired thirst mechanisms. More commonly, the inability to concentrate urine becomes manifest as nocturia, which develops as kidney function deteriorates. Urinary concentrating and diluting mechanisms are discussed in further detail in [Chapter 10](#).

POTASSIUM EXCRETION

To maintain potassium homeostasis in the presence of continued dietary intake and a reduced number of functioning nephrons, potassium excretion per nephron must increase. In both normal and diseased kidneys, almost all the filtered potassium is reabsorbed in the

proximal tubule and loop of Henle. Potassium excretion is therefore determined predominantly by distal secretion,²¹⁴ although a reduction in potassium reabsorption by the loop of Henle has been shown to contribute to increased potassium excretion in rats with reduced kidney mass.²³⁰ In both normal and partially nephrectomized dogs, urinary potassium excretion was found to correlate directly with serum potassium concentration.²³¹ Similarly, in intact and uninephrectomized rats, net potassium secretion in the distal convoluted tubule occurred only during potassium infusion, whereas potassium secretion by cortical collecting tubules (CCTs) occurred under all conditions and was greater after uninephrectomy.²³² Other studies have confirmed that the CCT is an important site of potassium secretion in the remnant kidney.^{200,212} Secretion of potassium by CCTs isolated from remnant kidneys of rabbits fed normal or high-potassium diets was shown to persist *in vitro* and to be directly related to the dietary potassium content,²⁰⁰ indicating an intrinsic tubular adaptation to potassium load. This adaptation was absent in CCTs from rabbits in which dietary potassium had been reduced in proportion to the amount of kidney mass lost. In addition to variations with the dietary potassium load, the increase in potassium secretion by remnant CCTs was also found to correlate with plasma aldosterone levels, but not with intracellular potassium concentration or $\text{Na}^+\text{-K}^+\text{-ATPase}$.²⁰⁰ In contrast, however, others have reported an increase in cortical and outer medullary $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in homogenates from rat remnant kidneys that was abrogated when potassium intake was reduced in proportion to the reduction in the GFR.²³³ These findings have been largely confirmed using a computational model of epithelial transport. In simulations of an acute potassium load and chronic high-potassium diet, potassium secretion increased in the connecting tubule, but the increase was less after uninephrectomy (18% less with acute and 13% less with chronic loads) and 5/6 nephrectomy (42% less with acute and 31% less with chronic loads) than in sham controls. The increase in potassium secretion per tubule failed to compensate fully for the decrease in nephron number.²³⁴ Finally, the frequent occurrence of hyperkalemia in persons with reduced GFR after treatment with a mineralocorticoid antagonist (MRA) or an ACE inhibitor suggests that “normal” aldosterone levels are required to maintain adequate potassium excretion in this population.²³⁵ In general, therefore, the increase in potassium secretion by surviving nephrons appears to be predominantly determined by the rise in plasma potassium levels after potassium ingestion and by intrinsic tubular adaptation to the increased filtered potassium load.^{231,232} In both dogs and persons with reduced GFR, however, the kaliuretic response to an oral potassium load is attenuated compared with normal subjects, despite higher serum potassium levels.^{231,236} The eventual complete excretion of a potassium load therefore occurs at the expense of a sustained increase in serum potassium levels. Control of potassium excretion is discussed further in [Chapters 6 and 16](#).

ACID-BASE REGULATION

Reduction of the GFR in persons with CKD is associated with the development of metabolic acidosis due to a reduction

in the serum bicarbonate concentration. Normal acid-base balance requires reabsorption of filtered bicarbonate, excretion of titratable acid, ammonia generation, and acidification of tubular luminal fluid by the distal nephron.²¹⁴ With decreasing GFR, acidosis develops as a result of varying degrees of impairment in each of these processes.²³⁷

A reduction in kidney ammonia synthesis is the greatest limitation to acid excretion in CKD. Low serum bicarbonate levels result in maintenance of acidic urine, which stimulates proximal tubule ammoniagenesis and also promotes ammonia conversion, resulting in its entrapment as ammonium in the tubule lumen. Net ammonia production per hypertrophied proximal tubule has been shown to increase in response to nephron loss.²²⁹ With decreasing GFR, however, this increase becomes inadequate to compensate for further nephron loss, and absolute ammonia excretion falls.¹⁹⁸ In addition, disruption of the tubulomedullary ammonium concentration gradient because of structural injury may impair ammonia trapping and therefore reduce ammonium excretion.¹⁹⁸ Bicarbonate reclamation by the nephron occurs predominantly in association with sodium reabsorption in the proximal tubule and is dependent on the generation of a proton gradient in the distal nephron.

Conflicting data with respect to bicarbonate reabsorption in remnant kidneys may reflect species differences. In dogs with remnant kidneys, bicarbonate reabsorption was increased at both proximal and distal micropuncture sampling sites compared with intact controls.²³⁸ In contrast, bicarbonate reabsorption per unit GFR is reduced in humans and rats with reduced GFR,²¹⁴ and some persons demonstrate bicarbonate wasting until the serum bicarbonate level drops below 20 mEq/L.²³⁹ Bicarbonate reabsorption is also reduced in the setting of hyperkalemia, increased ECF volume, and hyperparathyroidism, all of which may be present in persons with reduced GFR.²⁴⁰⁻²⁴² Distal urinary acidification tends to be relatively well preserved in persons with reduced GFR, and urinary pH, although higher than in normal individuals with experimental acidosis, is usually around 5.²⁴³ Urinary excretion of titratable acid is also generally well preserved in the setting of nephron loss as a consequence of increased fractional phosphate excretion.^{195,198} As CKD progresses, acid excretion becomes more dependent on the excretion of titratable acid. Renal acidification mechanisms are discussed more comprehensively in [Chapters 9 and 15](#).

CALCIUM AND PHOSPHATE

Derangements of calcium and phosphate metabolism occurring with reduced GFR are the result of not only impaired urinary excretion of these solutes but also associated abnormalities in vitamin D metabolism and PTH secretion. With progressive GFR decline, 1α -hydroxylation of vitamin D by the kidney decreases, calcium absorption from the gut decreases, serum calcium level tends to decrease, serum phosphate level tends to increase, and PTH secretion increases. In response to increased PTH, calcium is mobilized from bone, renal phosphate excretion is enhanced, and the steady state becomes reestablished, with secondary hyperparathyroidism as the trade-off.²⁴⁴ Serum phosphate concentration does not usually increase until the GFR falls below 20 mL/min/1.73 m², and phosphate

balance is maintained predominantly by an increase in fractional phosphate excretion.²⁴⁵

With moderately reduced GFR, filtered phosphate is not greatly increased and the increase in phosphate excretion must be achieved by a reduction in phosphate reabsorption per nephron.²⁴⁶ With more severe reductions in the GFR, phosphate excretion is maintained by an increase in serum phosphate concentration, as well as by reduced reabsorption per nephron. Sodium-dependent phosphate transport measured in proximal tubular brush border membrane vesicles prepared from the remnant kidneys of dogs was shown to be decreased when compared with that in vesicles derived from normal dogs.¹⁹⁵ Interestingly, however, this decrease was abolished if the partially nephrectomized dog had also undergone parathyroidectomy, indicating that PTH plays an important role in proximal tubular adaptation to phosphate excretion. Studies of isolated proximal tubules from euparathyroid uremic rabbits have shown a reduction in net phosphate flux per unit of reabsorptive surface area and an increase in sensitivity to PTH.¹⁹⁷ The authors postulated that the number of PTH receptors per tubule must increase in the remnant kidney, concomitant with tubular hypertrophy. The levels of mRNA encoding the sodium-coupled phosphate transporter, NaPi-2, are reduced by approximately 50% in remnant kidneys from 5/6 nephrectomized rats.²⁴⁷ In contrast, tubules from hyperparathyroid uremic rabbits demonstrated reduced PTH sensitivity, consistent with the downregulation or persistent occupancy of the PTH receptors.

On the other hand, studies in animals with reduced kidney mass subjected to parathyroidectomy have shown that fractional excretion of phosphate remains inversely proportional to the reduction in the GFR,²⁴⁸ indicating that phosphate excretion is not entirely dependent on the presence of PTH. Fibroblast growth factor 23 (FGF23) has been identified as a major mediator of increased phosphaturia after nephron loss.²⁴⁹ First identified as the primary mediator of autosomal dominant hypophosphatemia, increased FGF23 expression has been shown in transgenic models to increase urinary phosphate excretion through the downregulation of NaPi-2a expression in proximal tubules.²⁵⁰ Further experiments using gene deletion models have found that decreased expression of NaPi-2a and NaPi-2c by exogenous FGF23 is mediated predominantly by FGF receptor 1.²⁵¹ Activation of FGF receptors by FGF23 is critically dependent on binding with its coreceptor, Klotho.²⁴⁹ After 5/6 nephrectomy in rats, investigators observed hyperphosphatemia after a high-phosphate diet, despite high levels of FGF23. This was explained by a marked reduction in levels of Klotho that was prevented by treatment with calcitriol. Further experiments in cultured cells have indicated that Klotho is suppressed by phosphaturia through Wnt/ β -catenin signaling.²⁵² Circulating levels of FGF 23 become elevated early in the course of CKD, with levels rising once the GFR falls below 70 mL/min/1.73 m² in humans,²⁴⁹ although this may depend to some extent on vitamin D status. In one study, FGF23 was elevated early in persons with CKD who were vitamin D replete, but PTH was more frequently elevated than FGF23 in those with vitamin D insufficiency.²⁵³ That FGF23 is important in mediating increased phosphaturia after nephron loss has been demonstrated by experiments

in which the administration of neutralizing anti-FGF23 antibodies after kidney mass reduction resulted in decreased fractional excretion of phosphate and increase in serum phosphate levels.²⁵⁴ Whereas most of the reduction in phosphate reabsorption is achieved in the proximal tubule, there is also some evidence of increased fractional phosphate excretion by the distal tubule in uremic dogs and rats.²⁵⁵

As kidney failure advances, kidney 1 α -hydroxylation of vitamin D decreases, and as a result, calcium absorption from the gut is reduced.²⁵⁶ In addition to its effects on renal phosphate excretion, FGF23 inhibits renal 1 α -hydroxylase activity, thereby reducing levels of 1,25-OH vitamin D.^{250,254} In CKD, fractional intestinal calcium absorption is inversely proportional to blood urea nitrogen levels.²⁵⁶ Calcium excretion, on the other hand, varies widely in persons with CKD, probably due to differences in diet, heterogeneity of vitamin D production, and predominance of glomerular versus tubulointerstitial injury.²⁵⁷ In normal individuals, calcium excretion is mediated by the suppression of PTH-induced reabsorption in the distal nephron and by the suppression of PTH-independent mechanisms in the thick ascending limb. In persons with CKD, fractional calcium excretion usually remains unchanged until the GFR falls below 25 mL/min/1.73 m² when fractional excretion increases due to the obligatory solute diuresis.²¹⁴ Absolute calcium excretion, however, remains low. Hypocalciuria has been shown to be due in part to the attendant hyperparathyroidism.²⁵⁸ Similar findings were obtained in rats with reduced kidney mass, in which parathyroidectomy resulted in increased calcium excretion compared with nonparathyroidectomized controls.²⁵⁹ Renal calcium clearance is increased in persons with tubulointerstitial disease and in rats with surgical papillectomy, suggesting that regulation of calcium reabsorption depends on intact medullary structures and that regulation of calcium excretion may be largely modulated by the distal nephron segments.²¹⁴ The potential contributions of calcium and phosphate to kidney disease progression are discussed later. Calcium and phosphate metabolism are also discussed in greater detail in [Chapters 7 and 17](#).

LONG-TERM ADVERSE CONSEQUENCES OF ADAPTATIONS TO NEPHRON LOSS

The functional and structural adaptations to nephron loss described previously may be regarded as a beneficial response that minimizes the resultant loss of the total GFR. It has been appreciated for several decades, however, that rats subjected to subtotal nephrectomy subsequently develop hypertension, albuminuria, and progressive kidney failure. Detailed histopathologic studies in rat remnant kidneys after 5/6 nephrectomy have revealed mesangial accumulation of hyaline material that progressively encroaches on capillary lumina, obliterating the Bowman space and finally resulting in global sclerosis of the glomerulus. These findings, together with the observation that sclerosed glomeruli are a common finding in human CKD of diverse causes, have led to the hypothesis that glomerular hyperfiltration ultimately results in damage to the remaining glomeruli and contributes to a vicious cycle of progressive nephron loss.

The 5/6 nephrectomy model has been extensively studied to elucidate how the physiologic adaptations, which initially permit greatly augmented function per nephron, ultimately provoke a complex series of adverse effects that eventuate in progressive kidney damage and an inexorable decline in function.⁹

HEMODYNAMIC FACTORS CONTRIBUTING TO PROGRESSIVE NEPHRON LOSS

As early as 1 week after extensive kidney mass ablation, glomerular hyperfiltration and glomerular capillary hypertension were found to be associated with morphologic changes in glomerular cells, including podocyte cell membrane blebs, cytoplasmic attenuation, protein reabsorption droplets and foot process fusion, mesangial expansion, focal lifting of endothelial cells from the basement membrane, and endothelial cell membrane blebs (Figs. 50.5 and 50.6).⁹ Evidence that these morphologic changes were a consequence of the glomerular hemodynamic alterations was provided by studies in rats fed a low-protein diet after 5/6 nephrectomy. This intervention prevented the hemodynamic changes, effectively normalizing Q_A , P_{GC} , and the SNGFR, and abrogated the structural lesions observed in rats on standard diet.⁹ Similar findings were subsequently described in a variety of animal models of CKD, including diabetic nephropathy^{260,261} and DOCA-salt hypertension.²⁶²

Together, these observations led Brenner and colleagues to propose that the hemodynamic adaptations following kidney mass ablation ultimately prove injurious to glomeruli and initiate processes that result in glomerulosclerosis. The resulting obliteration of further glomeruli would induce hyperfiltration in remaining, less affected glomeruli, thereby establishing a vicious cycle of progressive nephron loss. These mechanisms constitute a common pathway for kidney damage that could account for the inexorable progression of CKD, independent of the cause of the initial kidney

injury.⁴ This hypothesis also explains the finding of both atrophic and hypertrophic nephrons typically encountered in chronically diseased kidneys. Further evidence supportive of the so-called hyperfiltration hypothesis was gleaned from the study of experimental diabetic nephropathy, in which glomerular hyperfiltration was also found to be a forerunner of glomerular pathology.^{9,261} Maneuvers such as unilateral nephrectomy, which exacerbates hyperfiltration in the

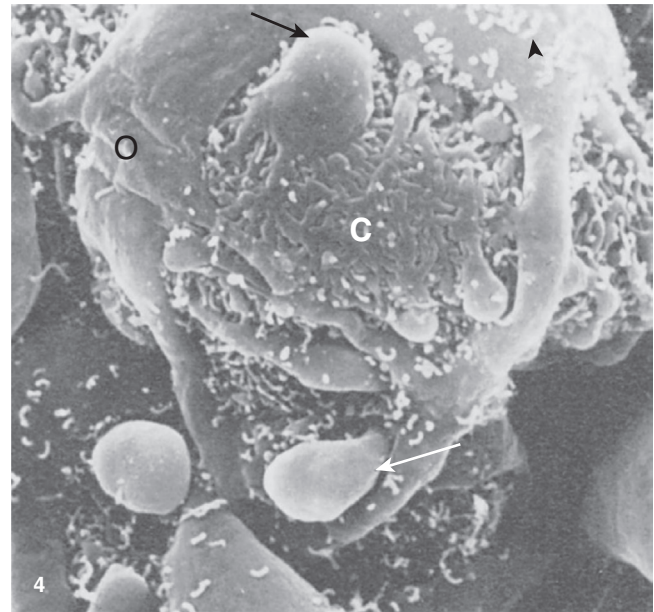


Fig. 50.5 Scanning electron micrograph of a glomerulus from a rat following 5/6 nephrectomy; view from the urinary space. Podocyte cytoplasmic blebs (arrows), numerous microvilli (arrowhead), focal obliteration (O), and coarsening (C) of foot processes are seen. ($\times 3600$.) (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am. J. Physiol.* 1981;241:F85–F93.)

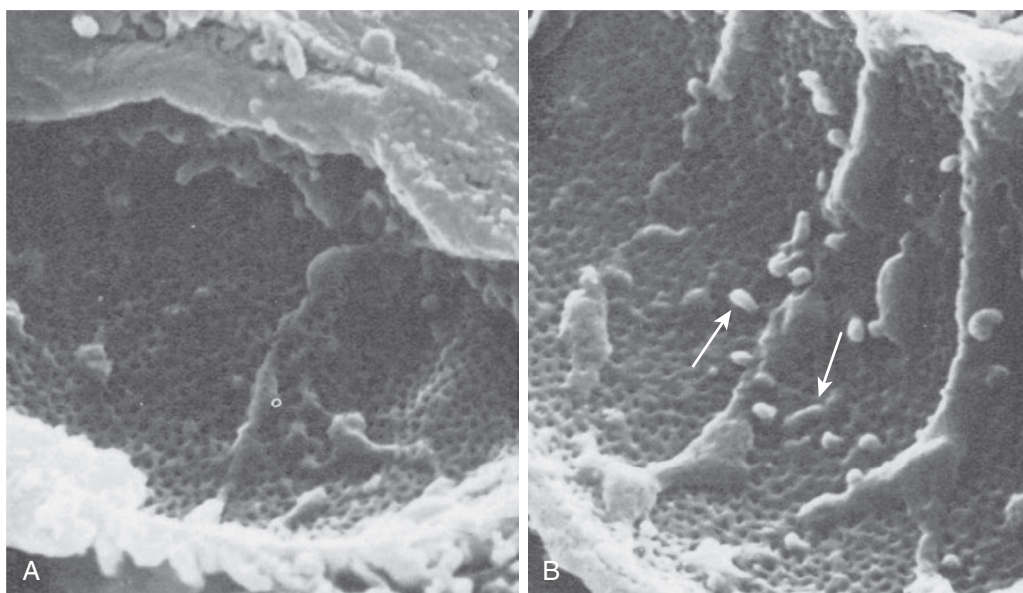


Fig. 50.6 Scanning electron micrographs of glomerular capillaries. (A) Normal endothelial appearance. (B) Rats after 5/6 nephrectomy. Scattered endothelial blebs (arrows) are often present in this group ($\times 18,000$). (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am. J. Physiol.* 1981;241:F85–F93.)

remaining kidney were also found to exacerbate diabetic kidney injury.²⁶³ Furthermore, when the kidney was shielded from elevated perfusion pressure and from glomerular capillary hypertension by creating unilateral renal artery stenosis, the ipsilateral kidney was protected against the development of diabetic injury, which progressed unabated in the contralateral kidney.²⁶⁴ In addition, when glomerular hyperfiltration was reversed in 5/6 nephrectomized rats by transplantation of an isogeneic kidney, hypertension and proteinuria were ameliorated and glomerular injury was limited.²⁶⁵

Similarly, augmenting kidney mass in the Fisher-Lewis rat transplant model normalized P_{GC} and greatly reduced the development of chronic kidney allograft injury.^{266,267} Similarly, it is noteworthy that the phase of transition from an acute, nonhypertensive experimental injury, induced by puromycin aminonucleoside (PAN) administration, to a chronic nephropathy characterized by proteinuria and glomerulosclerosis, is also associated with the development of glomerular capillary hypertension.²⁶⁸

Direct evidence that similar mechanisms may operate in human kidneys has been derived from a study of 14 persons with solitary kidneys who had undergone varying degrees of partial nephrectomy of the remaining kidney for malignancy.²⁶⁹ Before kidney-sparing surgery, proteinuria was absent in all persons. Although serum creatinine levels remained stable after an initial rise of 50% in 12 persons, the 2 persons subjected to the most extensive nephrectomy (75% and 67%, respectively) developed progressive kidney failure and required long-term dialysis. Moreover, among the remaining persons, 7 developed proteinuria, the levels of which were inversely related to the amount of kidney tissue preserved. Kidney biopsy specimens in four persons with moderate to severe proteinuria showed focal segmental glomerulosclerosis (FSGS),²⁶⁹ which later morphometric analysis revealed involvement of virtually all glomeruli examined.²⁷⁰ The importance of kidney mass in humans was further illustrated by an observational study of 749 persons who underwent radical nephrectomy or nephron-sparing surgery for removal of a kidney mass. Those who had nephron-sparing surgery had a significantly lower incidence of reduced GFR (16.0% vs. 44.7%) and proteinuria (13.2% vs. 22.2%) compared with those who underwent radical nephrectomy.²⁷¹ Similarly, a large meta-analysis that included data from 31,729 people who had a radical nephrectomy and 9281 people who had a partial nephrectomy for cancer found a 61% reduction in risk of developing CKD stages 3 to 5 after partial nephrectomy.²⁷²

Finally, whereas unilateral donor nephrectomy is associated with an excellent prognosis in most persons, case-controlled studies have reported a small absolute increase in the long-term risk of ESKF.^{273,274} One detailed study in 51 living donors has reported that hypertension in kidney donors 50 years of age and older was associated with a lower estimated number of functioning nephrons per kidney.²⁷⁵ Several studies have identified hypertension as a risk factor for subsequent GFR decline and albuminuria in living kidney donors.²⁷⁶⁻²⁷⁸ It is therefore likely that living donors with a low nephron number are at increased risk of subsequent kidney damage. Furthermore, detailed assessment of the nontumor portion of kidneys removed during tumor nephrectomy in

936 persons revealed that higher glomerular volume (a marker of likely lower nephron number) and the proportion of sclerosed glomeruli both predicted a higher risk of CKD progression during a median of 6.4 years.²⁷⁹ Thus the risk of progressive CKD after uninephrectomy in humans seems highest in those with lower nephron numbers.

The importance of glomerular hemodynamic factors in the development of progressive kidney damage was further illustrated by studies that reported dramatic protective effects against the development of glomerulosclerosis after chronic inhibition of the RAAS with ACE inhibitor or ARB treatment in 5/6 nephrectomized rats.²⁰⁻²³ Micropuncture studies have shown that like the low-protein diet, the kidney-protective effects of RAAS inhibition were associated with near normalization of the P_{GC} , yet in contrast with the effects of dietary protein restriction, the SNGFR remained elevated.²⁰ This suggested that glomerular capillary hypertension, rather than hyperfiltration per se, was the key factor in the initiation and progression of glomerular injury.

Confirmation of this view has come from an experiment in which rats were treated with a combination of reserpine, hydralazine, and hydrochlorothiazide (triple therapy) to lower arterial pressure to levels similar to those obtained with an ACE inhibitor. In contrast to the glomerular hemodynamic effects of the ACE inhibitor, triple therapy did not alleviate glomerular hypertension or proteinuria, and glomerular injury progressed^{21,22} (Fig. 50.7). Interestingly, within the context of pharmacologic inhibition of the RAAS, the level to which systemic BP is reduced remains a critical determinant of the extent of the kidney protection conferred.²⁸⁰ The effectiveness of both ACE inhibitor and ARB in lowering glomerular pressure and ameliorating glomerular injury has since been observed in several other animal models of chronic kidney disease, including diabetic nephropathy,^{260,281,282} hypertensive kidney disease,^{283,284} experimental chronic renal allograft failure (a model that lacks systemic hypertension but exhibits glomerular capillary hypertension),²⁸⁵⁻²⁸⁷ age-related glomerulosclerosis,^{288,289} and obesity-related glomerulosclerosis.²⁹⁰ That similar mechanisms are relevant in human CKD progression has been strongly suggested by the results of clinical trials showing substantial kidney-protective effects with ACE inhibitor and ARB treatment.²⁹¹⁻²⁹⁵

The importance of glomerular capillary hypertension has been further illustrated by studies of the effects of omapatrilat, a vasopeptidase inhibitor. Micropuncture studies after 5/6 nephrectomy have shown an even greater lowering of P_{GC} with omapatrilat than with ACE inhibitor treatment, despite equivalent effects on systemic BP. In subsequent chronic studies after 5/6 nephrectomy, omapatrilat produced more effective kidney protection than an ACE inhibitor.⁹⁸ Thus among the determinants of glomerular hyperfiltration, glomerular capillary hypertension has been identified as a critical factor in the initiation and progression of glomerular injury.

The kidney-protective effects of sodium-glucose cotransporter 2 (SGLT2) inhibitor treatment further support the importance of glomerular hemodynamic factors in CKD progression.²⁹⁶ These drugs act principally by inhibiting reabsorption of glucose and sodium in the proximal tubule, causing increased delivery of sodium chloride to the macula densa, which provokes tubuloglomerular feedback and

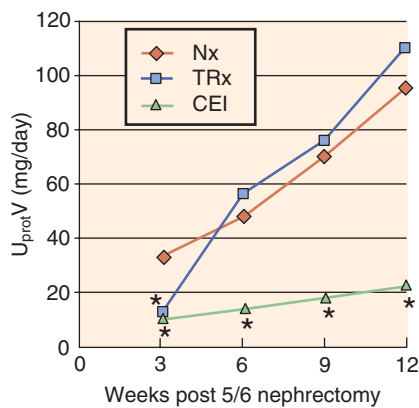


Fig. 50.7 Proteinuria levels following 5/6 nephrectomy in untreated rats (Nx) versus treatment with triple therapy (reserpine, hydralazine, and hydrochlorothiazide [TRx]), Nx + TRx, or enalapril (Nx + CEI). Despite equivalent levels of blood pressure control, enalapril therapy almost completely prevented proteinuria and glomerulosclerosis, whereas triple therapy afforded no renoprotection. $U_{prot}V$, 24-hour urine protein excretion; * $P < .05$ vs. Nx at the same time point. (From Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest*. 1986;77:1993–2000.)

reduces glomerular hyperfiltration by causing constriction of afferent arterioles.²⁹⁷⁻²⁹⁹ In a rodent model of type 1 diabetes, treatment with an SGLT2 inhibitor resulted in a marked reduction in SNGFR that was inversely proportional to the distal tubular chloride concentration, indicating activation of tubuloglomerular feedback.³⁰⁰ Treatment with an SGLT2 inhibitor after 5/6 nephrectomy afforded kidney protection similar to that of ARB treatment and was associated with an increase in urinary adenosine excretion, a marker of tubuloglomerular feedback activity, that was inversely related to the severity of tubulointerstitial fibrosis.⁹⁷

MECHANISMS OF HEMODYNAMICALLY INDUCED INJURY

MECHANICAL STRESS

Several mechanisms have been proposed whereby glomerular hypertension and hyperperfusion may result in glomerular cell injury. Experiments in isolated perfused rat glomeruli have reported significant increases in glomerular volume, with increases in perfusion pressure over the normal and relevant abnormal range.¹⁴⁴ These increases in wall tension and glomerular volume likely result in stretching of glomerular cells. Experimental evidence has suggested that such stretching may have adverse consequences for all three major cell types in the glomerulus. A mathematic model of blood flow and filtration in the glomerular capillary network after 5/6 nephrectomy has estimated that glomerular capillary wall strain is increased approximately threefold versus control, a magnitude sufficient to impact podocyte structure and function.³⁰¹ Furthermore, advances in the study of cellular responses to mechanical stress raised the possibility that glomerular hyperperfusion may also promote the development of glomerulosclerosis through more subtle and complex pathways that induce profibrotic phenotypic alterations in glomerular cells.³⁰²

ENDOTHELIAL CELLS

The vascular endothelium serves a number of complex functions, including acting as a dynamic barrier to leukocytes and plasma proteins, secretion of vasoactive factors (prostacyclin, nitric oxide, and endothelin), conversion of Ang I to Ang II, and expression of cell adhesion molecules. It is also the first cellular structure in the kidney that encounters the mechanical forces imparted by glomerular hyperperfusion. After 5/6 nephrectomy, endothelial cells are activated or injured, resulting in detachment and exposure of the basement membrane (Fig. 50.6). This in turn may induce platelet aggregation, deposition of fibrin, and intracapillary microthrombus formation.^{9,303} It has been recognized for some time that segmental glomerulosclerosis is associated with focal obliteration of capillary loops³⁰⁴ and that interstitial fibrosis is associated with a loss of peritubular capillaries.³⁰⁵ Furthermore, it has been shown that this loss of capillaries in the remnant kidney is associated with a decrease in endothelial cell proliferation and reduced constitutive expression of VEGF by podocytes and renal tubule cells, as well as increased expression of the antiangiogenic factor, thrombospondin-1, by the kidney interstitium.³⁰⁶

Because VEGF is an important endothelial cell angiogenic, survival, and trophic factor, these findings suggest that capillary loss may be caused in part by failure of recovery from hemodynamically mediated endothelial cell injury. Indeed, inhibition of endothelial and mesangial cell proliferation by treatment with everolimus after 5/6 nephrectomy resulted in increased proteinuria, glomerulosclerosis, and interstitial fibrosis associated with reduced glomerular expression of VEGF.³⁰⁷ Furthermore, short-term treatment of rats with VEGF ameliorated both glomerular and peritubular capillary loss after 5/6 nephrectomy.³⁰⁸ This preservation of capillaries was associated with a trend toward less glomerulosclerosis and significantly less interstitial deposition of type III collagen, as well as better preservation of kidney function. Further evidence of the importance of endothelial cells in the preservation of function after nephron loss has been provided by experiments in which bone marrow-derived endothelial progenitor cells (EPCs) were administered to mice after 5/6 nephrectomy. EPC-treated mice showed better preservation of kidney function, less proteinuria, and relatively preserved kidney structure, as well as decreased expression of proinflammatory molecules and restored levels of the angiogenic molecules VEGF, VEGF receptor-2, and thrombospondin 1.³⁰⁹ Long-term studies are required to evaluate further the potential benefits of improving renal angiogenesis in the setting of progressive kidney damage.

Endothelial cells bear numerous receptors that allow them to detect and respond to changes in mechanical forces. Thus exposure of endothelial cells to changes in shear stress, cyclic stretch, or pulsatile barostress that result from glomerular hyperperfusion may induce changes in the expression of genes involved in inflammation, cell-cycle control, apoptosis, thrombosis, and oxidative stress.³¹⁰ The in vitro responses of endothelial cells to mechanical forces have largely been studied in the context of vascular remodeling and atherosclerosis, but it can readily be appreciated that similar responses may affect the development of inflammation and fibrosis in the remnant kidney. Of particular interest are observations that shear

stress can stimulate endothelial expression of adhesion molecules³¹¹ and proinflammatory cytokines.³¹² It is clear therefore why biomechanical activation has emerged as an important paradigm in endothelial cell biology,³¹³ but further studies focusing on glomerular endothelial responses to mechanical stress are required to elucidate the role of these mechanisms in progressive kidney damage.

MESANGIAL CELLS

Mesangial cells are closely associated with the capillaries in the glomerulus and are therefore also exposed to mechanical forces. Evidence from *in vitro* studies has indicated that mesangial cells respond to changes in these mechanical forces in ways that may promote inflammation and fibrosis. Subjecting mesangial cells to cyclical stretch or strain has been shown to induce the proliferation³¹⁴ and synthesis of extracellular matrix constituents.^{315,316} Cyclic stretch also activates the transcription factor nuclear factor- κ B (NF- κ B)³¹⁷ and stimulates monocyte chemoattractant protein 1 (MCP-1),³¹⁸ TGF- β ,³¹⁹ and its receptor,³²⁰ as well as connective tissue growth factor (CTGF) production.³²¹ In cultured mesangial cells, cyclic stretch activates the RAAS³²² and Ang II in turn may induce TGF- β synthesis.³²³

In vitro studies have identified several signaling pathways responsible for stretch-induced signal transduction in mesangial cells. Cyclic stretch activates the serine-threonine kinase Akt through a mechanism requiring phosphatidylinositol-3-kinase and transactivation of EGF receptor and is necessary for the increased synthesis of collagen type 1A1 observed in the mesangial cells. Furthermore, Akt activation was observed in remnant kidney glomeruli, indicating that these mechanisms are also present *in vivo*.³²⁴ The actin cytoskeleton is an important transmitter of mechanical signals, and the GTPase, RhoA, a central regulator of the cytoskeleton, is activated by cyclic stretch. Stretch-induced activation of the mitogen-activated protein kinase Erk, linked to increased matrix production, is dependent on the activation of RhoA.³²⁵ Mesangial cells cultured at ambient pressures of 50 to 60 mm Hg (i.e., levels corresponding to glomerular capillary hypertension) also show enhanced synthesis and secretion of extracellular matrix when compared with cells grown at “normal” pressures of 40 to 50 mm Hg.³²⁶ Exposure of mesangial cells to barostress, achieved by culture under increased barometric pressure, stimulates the expression of cytokines including PDGF-B³²⁷ and MCP-1.³²⁸ Transduction of mechanical forces by mesangial cells has been associated with tyrosine phosphorylation³²⁹ and protein kinase C-induced increases in S-6 kinase activity.³³⁰

PODOCYTES

A growing body of evidence attests to the importance of podocyte injury in a variety of kidney diseases and in CKD progression. Podocytes display morphologic evidence of injury as early as 1 week after 5/6 nephrectomy (see Fig. 50.5)⁹ and 6 months after uninephrectomy.³³¹ Increased numbers of podocytes have been observed in the urine of rats after 5/6 nephrectomy and in human CKD.³³² In 5/6 nephrectomized rats, the number of podocytes in the urine correlated with the severity of proteinuria, as well as mean arterial BP, suggesting that podocyte loss may contribute to CKD progression.³³³ The importance of podocyte injury in

CKD progression is further supported by the observation that amelioration of glomerular damage in 5/6 nephrectomized rats treated with 1,25-dihydroxyvitamin D3 is associated with preservation of podocyte number, as well as prevention of podocyte hypertrophy and injury.³³⁴ Detailed *in vitro* studies have shown that early podocyte injury is associated with dysregulation of calcium homeostasis and disruption of the actomyosin contractile apparatus. The Rho family of small guanosine triphosphatases plays a key role in this process. Calcium influx via transient receptor potential canonical type 5 (TRPC5) channels results in the activation of Rac1 and is associated with increased podocyte migration and proteinuria, whereas calcium influx via TRCP type 6 activates RhoA, resulting in the preservation of stress fibers, prevention of podocyte migration, and maintenance of the filtration barrier.³³⁵ However, subsequent experiments in TRCP6-deficient podocytes have found that TRCP6 is not required for mechanotransduction.³³⁶ Rather, mechanical stretch was found to induce ATP release from podocytes by a process dependent on cholesterol and podocin. ATP release in turn activated the purinergic channel P2X₄, resulting in an influx of calcium and increased disorganization of the actin cytoskeleton.³³⁶

Thus purinergic channels appear to be key mechanotransducers in podocytes. Further research has identified the mechanosensing ion channel, Piezo 1, as an important mechanotransducer in podocytes. Piezo 1 was found to be upregulated in a rodent model of hypertensive nephropathy, and this was associated with increased expression of podocyte injury markers.³³⁷ Because podocytes are attached to the outer aspect of the glomerular basement membrane, it is reasonable to expect that they would be exposed to increased mechanical forces resulting from glomerular hypertension. Confirmation that podocytes respond to such physical forces has been derived from several *in vitro* experiments that examined podocyte responses to stretching. Activation of a voltage-sensitive potassium channels was observed in response to stretching of the podocyte cell membrane,³³⁸ and culture of podocytes under constant stretch-induced activation of the protein kinases Erk1/2 and JNK via the EGF receptor, as well as other changes in cell signaling.³³⁹ Mechanical stretch inhibited podocyte proliferation³⁴⁰ and, in common with TGF- β , reduced α ₃ β ₁-integrin expression and podocyte adhesion *in vitro*.³⁴¹ Exposure to cyclic stretching that mimics pulsatile strain within the glomerulus has been shown to cause reorganization of the actin cytoskeleton,³⁴² upregulation of COX-2, and E prostanoid (EP)-4 receptor expression,³⁴³ as well as podocyte hypertrophy.³⁴⁴ Subsequent experiments using mice with podocyte-specific depletion or overexpression of EP4 receptors have found that prostaglandin E2 acting via EP4 receptors contributes to the development of proteinuria after 5/6 nephrectomy and therefore probably contributes to podocyte injury.³⁴⁵

In another experiment, cyclic stretching of podocytes was associated with increased production of Ang II and TGF- β , as well as upregulation of angiotensin subtype 1 (AT1) receptors, resulting in increased Ang II-dependent apoptosis.³⁴⁶ Cyclic stretch of podocytes also resulted in a 50% reduction of nephrin (a key component of the slit diaphragm) mRNA and protein levels via an

Ang II-dependent mechanism that was inhibited by the peroxisome proliferator-activated receptor- γ (PPAR- γ) agonist, rosiglitazone, which prevented AT1 receptor upregulation.³⁴⁷

The importance of podocyte loss in CKD progression has been demonstrated by experiments in which podocytes were depleted using diphtheria toxin. Loss of >30% of podocytes destabilized glomeruli and provoked progressive podocyte loss even after the initial injury eventuating in glomerulosclerosis. A similar pattern of podocyte loss was observed in the 5/6 nephrectomy model and was ameliorated by treatment with an ACE inhibitor and angiotensin receptor blocker.³⁴⁸ Taken together, these data suggest that stretch-induced podocyte injury is a further mechanism whereby glomerular hypertension contributes to glomerular injury.

CELLULAR INFILTRATION IN REMNANT KIDNEYS

Despite the lack of an obvious immune stimulus, an inflammatory cell infiltrate composed predominantly of macrophages and smaller numbers of lymphocytes is observed in remnant kidneys after 5/6 nephrectomy.³⁴⁹ Interestingly, similar observations have been reported in a rat model of spontaneous kidney agenesis associated with a 60.2% reduction in nephron endowment.³⁵⁰ As discussed earlier, it is possible that the glomerular hemodynamic adaptations to nephron loss may provoke an inflammatory cell response through the effects of mechanical forces on endothelial and mesangial cells. Thus upregulation of renal endothelial adhesion molecules may facilitate the egress of leukocytes from the circulation into the mesangium, where they may participate in further kidney injury. The recruited cellular infiltrate may constitute an abundant source of potent pleiotropic cytokine products, which in turn influence other infiltrating leukocytes, dendritic cells, and kidney cells, stimulating cell proliferation, elaboration of extracellular matrix components, and increased endothelial adhesiveness.³⁵¹

Evidence is emerging that these proposed mechanisms, based largely on *in vitro* observations, are indeed relevant *in vivo*. In the two-kidney, one-clip model of renovascular hypertension, upregulated expression of adhesion molecules and TGF- β , as well as cell infiltration, is observed only in the nonclipped kidney that is exposed to hypertensive perfusion pressure.^{352,353} In the 5/6 nephrectomy model, coordinated upregulation of a variety of cell adhesion molecules, cytokines, and growth factors in association with macrophage infiltration has been observed at time points that precede the development of severe glomerulosclerosis.^{354,355} Furthermore, the kidney protection afforded by an ACE inhibitor or ARB treatment in this model was associated with inhibition of cytokine upregulation and prevention of renal infiltration by macrophages.^{355,356}

Infiltrating macrophages, although present in the glomeruli of remnant kidney, are chiefly distributed in the tubulointerstitial regions,^{349,355} suggesting that they play a role in the development of the TIF that accompanies glomerulosclerosis. Further analysis of the cellular infiltrate has also identified mast cells near areas of TIF.³⁵⁷ It is possible that interstitial infiltrates are recruited as the result of tubulointerstitial cell activation by the downstream effects of cytokines released in the glomeruli. Alternatively, it has

been proposed that excessive uptake of filtered proteins by tubule epithelial cells stimulates expression of cell adhesion and chemoattractant molecules that recruit macrophages and other monocytic cells to tubulointerstitial areas³⁵⁸ (see later for further discussion). The chemokine receptor CCR-1 has been shown to be important in interstitial but not glomerular recruitment of leukocytes. Treatment with a nonpeptide CCR-1 antagonist has been shown to reduce interstitial macrophage infiltration and ameliorate interstitial fibrosis in the UO model, but data are still lacking in the 5/6 nephrectomy model.³⁵⁹ Furthermore, antagonism of MCP-1 signaling through gene therapy—induced production of a mutant form of MCP-1 by skeletal muscle resulted in reduced interstitial macrophage infiltration and amelioration of interstitial fibrosis in mice after UO.³⁶⁰

The identification of renal tubule cells expressing α -smooth muscle actin after 5/6 nephrectomy has raised the possibility that tubule cells may undergo transdifferentiation to a myofibroblast phenotype that contributes to interstitial fibrosis.³⁶¹ Furthermore, the kidney protection observed with mycophenolate treatment in 5/6 nephrectomized rats is associated with reductions in interstitial myofibroblast infiltration and collagen type III deposition.³⁶² There is, however, ongoing debate regarding the origin of interstitial myofibroblasts in CKD. Fate-mapping studies have indicated that mesenchymal cells called *pericytes* are the predominant source of myofibroblasts and that epithelial-to-mesenchymal transdifferentiation is not a source of myofibroblasts or accounts for only a small minority.³⁶³ Other cells that may give rise to interstitial myofibroblasts include resident fibroblasts, endothelial cells, and bone marrow-derived cells.^{364,365}

Several lines of evidence have suggested that this cellular infiltrate contributes to kidney injury and is not merely a consequence of it.³⁶⁶ In one study, multiple linear regression analysis identified glomerular macrophage infiltration in the remnant kidney as a major determinant of mesangial matrix expansion and adhesion formation between the Bowman capsule and glomerular tufts.³⁴⁹ Furthermore, depletion of leukocytes by irradiation in rats delayed the onset of glomerular injury after kidney ablative surgery.³⁶⁷ Several studies have reported amelioration of the cellular infiltrate and kidney injury in the 5/6 nephrectomy model following treatment with the immunosuppressive agent mycophenolate mofetil.³⁶⁸⁻³⁷¹ One study has found that mycophenolate also lowers P_{GC} , which may account for some of its renoprotective effects.³⁷² Several other antiinflammatory interventions have been shown to ameliorate kidney injury after 5/6 nephrectomy. Treatment with the antiinflammatory agent nitroflurbiprofen, also a nitric oxide donor, was associated with moderate kidney protection.³⁷³ Rats treated with a PPAR- γ receptor agonist evidenced significant attenuation of the proteinuria and glomerulosclerosis observed in untreated rats, despite the failure of treatment to lower BP.³⁷⁴ This kidney protection was observed in association with marked reductions in glomerular cell proliferation, glomerular macrophage infiltration, and renal expression of plasminogen activator inhibitor-1 (PAI-1), as well as TGF- β . The authors speculated that some of these effects may have resulted from the known actions of PPAR- γ receptor activation to antagonize the

activities of the transcription factors activator protein-1 (AP-1) and NF- κ B.

Administration of a sphingosine-1-phosphate (S1P) receptor agonist, a novel immunosuppressant that inhibits the egress of T cells and B cells from lymph nodes, attenuated the increase in chemokine receptor expression observed in untreated rats and tended to normalize regulated on activation, normal T cell-expressed and secreted (RANTES) and MCP-1 gene expression.³⁷⁵ Glomerular and interstitial inflammation, as well as fibrosis, were also reduced. Overexpression of the antiinflammatory cytokine interleukin-10 (IL-10) in rats was associated with reduced interstitial inflammation and lower levels of MCP-1, RANTES, interferon- γ (IFN- γ), and IL-2 expression, as well as attenuation of proteinuria, glomerulosclerosis, and TIF.³⁷⁶ Finally, overexpression of the gene for angiotensin, an antiangiogenic factor that also inhibits leukocyte recruitment, as well as neutrophil and macrophage migration, was associated with the inhibition of macrophage and T cell infiltrates in glomeruli, along with the interstitium, reduced MCP-1 expression, and attenuation of glomerulosclerosis and interstitial fibrosis.³⁷⁷ Taken together, these findings strongly support the hypothesis that in addition to direct glomerular cell injury, glomerular hemodynamic adaptations to nephron loss provoke a complex series of proinflammatory and profibrotic responses that further contribute to kidney damage. Treatments that antagonize the mediators of these responses may therefore be of benefit in slowing the rate of CKD progression.

NONHEMODYNAMIC FACTORS CONTRIBUTING TO PROGRESSIVE NEPHRON LOSS

The weight of evidence in support of the hypothesis that glomerular hemodynamic adaptations are central to progressive kidney damage does not exclude that factors not directly attributable to hemodynamic changes are also important. These nonhemodynamic factors have been extensively studied and offer new therapeutic targets for kidney-protective interventions.

PROTEINURIA

Clinical Relevance

Proteinuria

Reduction of proteinuria should be regarded as a key therapeutic goal. Specifically, the dose of RAAS inhibitor treatment should be titrated up to the maximum tolerated dose. Additional treatment with SGLT2 inhibitors (for all persons with proteinuria), nonsteroidal MRA (diabetic kidney disease), and GLP-1RAs (diabetic kidney disease) should be added and blood pressure control optimized to minimize proteinuria.

Abnormal excretion of protein in the urine is the hallmark of experimental and clinical glomerular disease. Whereas immune complex deposition and resulting inflammation account for abnormal permeability of the glomerular filtration barrier to proteins in glomerulonephritis, studies in rats subjected to extensive kidney ablation have shown loss of glomerular barrier function to proteins of similar

molecular size yet in the apparent absence of primary immune-mediated kidney injury or inflammatory response. Sieving studies using dextrans and other macromolecules in rats 7 or 14 days after 5/6 nephrectomy have revealed the loss of both size and charge selectivity of the glomerular filtration barrier. Ultrastructural examination of the remnant kidneys has revealed detachment of glomerular endothelial cells and visceral epithelial cells from the glomerular basement membrane. In addition, protein reabsorption droplets and attenuation of cytoplasm resulting in bleb formation were observed in podocytes. The authors concluded that the altered permselectivity may be caused in part by the separation of endothelial cells from the glomerular basement membrane, allowing access of macromolecules and, in part, to loss of anionic sites in the lamina rara externa, resulting in loss of charge selectivity and detachment of podocytes.³⁷⁸

Studies have identified decreased nephrin expression in podocytes as a further mechanism contributing to proteinuria after 5/6 nephrectomy,³⁷⁹ and in vitro studies have reported a 50% reduction in nephrin expression when podocytes were exposed to cyclic stretching.³⁴⁷ A direct role for Ang II in modulating glomerular capillary permselectivity has been suggested by the observation of marked increases in urinary protein excretion during infusion of Ang II in normal rats. Although some investigators have attributed this to a direct effect of Ang II on the cellular components of the glomerular filtration barrier, resulting in the opening of interendothelial junctions and epithelial cell disruption, others have shown that the increase in proteinuria may be accounted for almost completely by the associated hemodynamic changes, principally a reduction in Q_A and an increase in filtration fraction.³⁸⁰ On the other hand, the notion that Ang II may mediate changes in glomerular permselectivity, independent of its effects on glomerular hemodynamics, has been supported by studies in an isolated perfused rat kidney preparation in which infusion of Ang II augmented urinary protein excretion and enhanced the clearance of tracer macromolecules, independently of any change in filtration fraction.³⁸¹ Furthermore, Ang II and aldosterone have been shown to reduce nephrin expression in podocytes and may therefore directly affect glomerular permselectivity.^{347,382,383}

Proteinuria, long considered simply a marker of glomerular injury, has also been implicated as an effector of injury processes involved in kidney disease progression, especially those resulting in tubulointerstitial fibrosis.³⁸⁴ A large body of research has demonstrated that proximal tubule epithelial cells cultured in media supplemented with high concentrations of albumin, immunoglobulin G (IgG), or transferrin increase secretion of proinflammatory cytokines.³⁸⁵⁻³⁸⁹ Furthermore, the liberation of these molecules was noted to be predominantly from the basolateral aspect of the cells. This would be in keeping with secretion into the kidney interstitium in vivo, thereby contributing to the development of tubulointerstitial inflammation and fibrosis. Other experiments have found apoptosis in tubule cells exposed to high-molecular-weight plasma proteins but not smaller proteins.³⁹⁰ Albumin and transferrin exposure also induced complement activation in tubule cells and reduced binding of factor H, a natural inhibitor of the alternative complement pathway.³⁹¹

Other investigators have proposed that tubulointerstitial inflammation is provoked by misdirection of protein-rich glomerular filtrate into the interstitium due to the formation of adhesions between the glomerular tuft and Bowman capsule or, in the case of crescentic glomerulonephritis, encroachment of the crescent onto the proximal tubule, resulting in occlusion and tubule degeneration.³⁹²

Further studies in the 5/6 nephrectomy model have suggested that tubulointerstitial injury may play an important role in the decline of the GFR, especially in the late stages of progressive kidney injury.³⁹³ By examining serial sections of remnant kidneys, the investigators have shown that in association with a doubling in serum creatinine levels, there was a substantial increase in the proportion of glomeruli no longer connected to tubules (atubular glomeruli) or atrophic tubules. Most of these glomeruli were not globally sclerosed, implying that the tubular injury was responsible for the final loss of function in these nephrons. The authors speculated that the absorption of excess filtered protein may play an important role in this tubular injury.³⁹³

In human studies the magnitude of albuminuria has been identified as an independent risk factor for CKD progression and other adverse outcomes in large meta-analyses, including one with more than 27 million participants.³⁹⁴ Furthermore, a reduction in albuminuria is strongly associated with a reduction in risk of CKD progression. In a large meta-analysis of 29,979 trial participants, each 30% decrease in mean albuminuria by treatment was associated with an average 27% lower hazard for an endpoint of CKD progression (KRT, GFR < 15 mL/min/1.73 m² or doubling of serum creatinine),³⁹⁵ providing strong evidence for the use of change in albuminuria as a surrogate endpoint for kidney-protective efficacy in clinical trials.³⁹⁶

Taken together, the evidence from experimental and clinical studies provides support for the hypothesis that impaired glomerular permselectivity results in excessive filtration of proteins and/or protein-bound molecules that contribute to kidney damage. In addition, the close association between the severity of proteinuria and kidney prognosis implies that reduction of proteinuria should be regarded as an important independent therapeutic goal in clinical strategies seeking to slow the rate of progression of CKD. The mechanisms and consequences of proteinuria are discussed in detail in [Chapter 29](#).

TUBULOINTERSTITIAL FIBROSIS

Together with secondary FSGS, tubulointerstitial fibrosis constitutes a major component of the progressive kidney injury observed in CKD. Tubulointerstitial fibrosis is characterized by inflammatory cell and fibroblast infiltration, accumulation of extracellular matrix, tubule cell loss, and rarefaction of peritubular capillaries. Fibrogenesis starts at small sites of inflammation and then expands if a profibrotic milieu persists. The inflammatory infiltrate is composed of lymphocytes, macrophages, dendritic cells, and mast cells.³⁹⁷ Lymphocytes are recruited early in the process, and their importance is highlighted by the protection from fibrosis observed in Rag-2 null mice, which lack B and T lymphocytes.³⁹⁸ Monocytes are recruited and transdifferentiated into macrophages and fibrocytes. The profibrotic role of macrophages is illustrated by observations that the extent of macrophage accumulation correlates

closely with the severity of fibrosis,³⁹⁹ and macrophage depletion attenuates fibrosis.⁴⁰⁰ Nevertheless, it has been proposed that alternatively activated M2 macrophages exert antiinflammatory actions, and the infusion of cells enriched for M2 macrophages has been shown to reduce kidney fibrosis in mice.⁴⁰¹ Myofibroblasts are the chief source of extracellular matrix production, and the accumulation of interstitial myofibroblasts is central to the pathogenesis of tubulointerstitial fibrosis.

There has been considerable controversy regarding the cellular origins of interstitial myofibroblasts in tubulointerstitial fibrosis. Different investigators have identified resident fibroblasts,⁴⁰² transdifferentiation from tubule epithelial cells and endothelial cells,^{403,404} bone marrow–derived fibrocytes,⁴⁰⁵ and pericytes³⁶³ as possible sources.³⁶⁵ Fate-mapping studies have subsequently indicated that pericytes are the predominant source of myofibroblasts and that epithelial-to-mesenchymal transdifferentiation (EMT) is not a source of myofibroblasts or accounts for only a minority.³⁶³ Further studies have helped resolve the controversy by showing that TECs undergo partial EMT, expressing both epithelial and mesenchymal markers, but remain attached to the tubular basal membrane. Moreover, prevention of EMT substantially ameliorated kidney fibrosis in animal models, confirming that partial EMT of TECs contributes to the pathogenesis of kidney fibrosis.^{406,407}

The excess extracellular matrix is composed largely of collagen types I and II, as well as fibronectin. Investigators have sought to identify the relative contribution of different cell types to the production of collagen I using cell type–specific knockouts. Experiments have revealed that bone marrow–derived cells (fibrocytes) contribute 38% to 50% of collagen deposition in a unilateral ureteral obstruction model. Furthermore, initial collagen I synthesis was attributable to resident mesenchymal fibroblasts and was beneficial in preserving kidney function in this model, whereas bone marrow–derived cells were responsible for later collagen I production that did not affect kidney function.⁴⁰⁸ TECs were found not to contribute directly to collagen I production. Bone marrow–derived cells were similarly shown to contribute substantially to collagen I production in an adenine-induced nephropathy model and, in this case, greater deposition of collagen was associated with lower GFR at all time points.⁴⁰⁸ Targeting of bone marrow–derived cell collagen I production therefore represents an attractive therapeutic option to ameliorate kidney fibrosis.

Experimental evidence suggests that fibrosis starts focally with the development of a fibrogenic microenvironment that triggers fibroblast activation, termed a “fibrogenic niche.”⁴⁰⁹ Matrix accumulation has been proposed to commence with the appearance of collagen nucleators in the interstitial fluid that acts as a scaffold for the deposition of fibrillar collagens. Fibroblasts use collagen fibrils as a scaffold to move through damaged tissue along chemoattractant gradients. Fibrogenesis is promoted by the expression of key growth factors, principally TGF- β .⁴¹⁰ The role of tissue proteases in tubulointerstitial fibrosis is complex and has not been fully characterized. Whereas matrix metalloprotease (MMP) types 2 and 9 degrade collagen type IV (and possibly type I and III) *in vitro*, they do not consistently abrogate TIF

in vivo.^{411,412} Indeed, in some models, MMPs appear to exert profibrotic effects.⁴¹²

Rarefaction of peritubular capillaries is a hallmark of tubulointerstitial fibrosis.⁴¹³ In the early stages, capillaries may be damaged by transient ischemia that promotes apoptosis.⁴¹⁴ Further capillary loss is attributable to an imbalance between proangiogenic and antiangiogenic factors^{415,416} or loss of peritubular endothelial cells through endothelial-mesenchymal transdifferentiation.⁴⁰³ Detailed studies have identified changes in peritubular capillary endothelium in animal models of CKD, including the development of subendothelial spaces, loss of fenestrations, and generation of caveolae and vesicles, as well as increased permeability and degeneration of the microvascular tree.⁴¹⁷ Tissue hypoxia results from the rarefaction of peritubular capillaries and accumulation of extracellular matrix, requiring oxygen to diffuse over greater distances to reach cells. Hypoxia contributes to tubulointerstitial fibrosis by promoting EMT and apoptosis of tubule cells, as well as fibroblast activation and extracellular matrix production.^{418,419} The importance of hypoxia in provoking interstitial (and glomerular) injury has been demonstrated by studies in which induction of hypoxia-inducible factor, a key mediator of protective responses to hypoxia, was associated with amelioration of proteinuria, glomerulosclerosis, and tubulointerstitial fibrosis, as well as decreased macrophage infiltration and expression of type IV collagen and osteopontin.^{420,421} Tubulointerstitial inflammation and fibrosis are discussed in more detail in Chapter 37.

TRANSFORMING GROWTH FACTOR- β AND MOLECULAR MEDIATORS OF KIDNEY FIBROSIS

TGF- β is associated with chronic fibrotic states throughout the body and is the central mediator of kidney fibrosis.⁴²² Three isoforms of TGF- β have been described in mammals, but the focus of most research has been on TGF- β 1. The active form of TGF- β is a dimer that binds to a transmembrane receptor, which is a heterodimer composed of TGF- β receptor types I and II. Intracellular signaling occurs via Smad-dependent and Smad-independent pathways. Smad2 and Smad3 have been identified as the most important intracellular mediators of the actions of TGF- β , with Smad3 having profibrotic effects and Smad2 proposed to have counterregulatory antifibrotic effects. In response to TGF- β receptor signaling, Smad2 and Smad3 are phosphorylated and form a complex with Smad4 that translocates into the nucleus and binds to DNA to modulate the transcription of multiple target genes. Smad3 has been shown to promote transcription of multiple genes involved in fibrosis including PAI-1, proteoglycans, integrins, CTGF, TIMP-1, and collagen types 1, 5, and 6. SMAD 3 also plays a role in EMT.⁴²³ In addition, TGF- β /Smad3 modulates the production of micro RNA molecules that promote fibrosis; levels of profibrotic miR-21, miR-192, and miR-433 are increased, whereas antifibrotic miR-29 and miR-200 are decreased.⁴²⁴ Smad4 is a key modulator of the profibrotic actions of TGF- β and reduces the profibrotic actions of Smad3 by inhibiting its binding to the promoter regions of profibrotic genes. Smad7 is an important negative feedback regulator of TGF- β signaling, which acts by preventing the recruitment and phosphorylation of Smad2 and Smad3. Smad7 also increases

the expression of inhibitor of kappa B (IkB α), an inhibitor of NF- κ B, and may therefore be important in mediating the antiinflammatory actions of TGF- β . Finally, Smad signaling may also be activated by TGF- β -independent mechanisms that are relevant to CKD progression including Ang II and advanced glycation end products (AGEs).⁴²³

In vitro TGF- β elicits overproduction of ECM constituents by mesangial cells, and its expression is increased in several experimental models of kidney disease. These include diabetic nephropathy,⁴²⁵ anti-Thy-1 glomerulonephritis,⁴²⁶ adriamycin-induced nephropathy,⁴²⁷ and chronic allograft nephropathy,⁴²⁸ as well as human glomerulonephritis,^{429,430} HIV nephropathy,⁴³¹ diabetic nephropathy,⁴³² and chronic allograft nephropathy.⁴³³ The role of TGF- β in kidney fibrosis has been further illustrated by experiments in which transfection of the gene for TGF- β into one renal artery produced ipsilateral kidney fibrosis.⁴³⁴ In 5/6 nephrectomized rats, a twofold to threefold increase in remnant kidney mRNA levels for TGF- β was observed, and in situ hybridization revealed elevations in TGF- β mRNA throughout the glomeruli, tubules, and interstitium. Treatment with an ACE inhibitor or ARB resulted in substantial kidney protection and prevented upregulation of TGF- β .^{355,356} Furthermore, in rats treated with an ACE inhibitor or ARB, the extent of glomerulosclerosis correlated closely with remnant kidney TGF- β mRNA levels.²⁸⁰

Several interventions that inhibit the effects of TGF- β have been shown to afford kidney protection in animal models of kidney disease. Transfection of the gene for decorin, a naturally occurring inhibitor of TGF- β , into skeletal muscle limited the progression of kidney injury in anti-Thy-1 glomerulonephritis.⁴³⁵ Administration of anti-TGF- β antibodies to salt-loaded, Dahl salt-sensitive rats ameliorated the hypertension, proteinuria, glomerulosclerosis, and interstitial fibrosis typical of this model.⁴³⁶ Treatment with tranilast (N-[3',4'-dimethoxycinnamoyl]-anthranilic acid; Pharm Chemical, Shanghai Lansheng, Shanghai, China), an inhibitor of TGF- β -induced ECM production, significantly reduced albuminuria, macrophage infiltration, glomerulosclerosis, and interstitial fibrosis in 5/6 nephrectomized rats.⁴³⁷ Transfer of an inducible gene for Smad 7, which blocks TGF- β signaling by inhibiting Smad 2/3 activation, inhibited proteinuria, fibrosis, and myofibroblast accumulation after 5/6 nephrectomy.⁴³⁸ Two weeks of treatment with a polyamide compound designed to suppress transcription of the TGF- β gene significantly reduced proteinuria and prevented upregulation of TGF- β , CTGF, collagen type I α 1, and fibronectin mRNA in the kidney cortex. This also suppressed urinary TGF- β excretion and staining for TGF- β by immunofluorescence in salt-loaded, Dahl salt-sensitive rats.⁴³⁹ Another fibrogenic molecule, CTGF, has also been observed to be overexpressed in kidney biopsies from persons with a variety of kidney diseases.⁴⁴⁰ The specific induction of CTGF expression by exogenous TGF- β in mesangial cells^{321,441} and fibroblasts,⁴⁴² together with the finding that blocking antibodies to TGF- β inhibits increased CTGF expression in mesangial cells exposed to high glucose concentrations,⁴⁴¹ suggests that CTGF may serve as a downstream mediator of the profibrotic effects of TGF- β .⁴⁴³ Further experiments have shown that the fibrotic effects of TGF- β in the remnant kidney are mediated, at least

in part, by the induction of the micro RNAs miR-21, miR-192, and miR-433 via Smad 3.⁴²⁴ In vitro overexpression of miR-192 promotes, but inhibition of miR-192, attenuates, TGF- β -induced production of collagen type I in rat tubule cells.⁴⁴⁴

It is clear that interventions to inhibit the multiple mechanisms of kidney fibrosis show promise as treatments for slowing the progression of CKD. However, despite the promising results of many preclinical studies inhibiting the expression or actions of TGF- β discussed previously, a phase 2 randomized trial of anti-TGF- β_1 monoclonal antibody therapy—added to RAAS inhibitor treatment in persons with diabetic kidney disease, GFR of 20 to 60 mL/min/1.73 m², and urine polymerase chain reaction (PCR) value higher than 800 mg/g—found no benefit with respect to change in serum creatinine levels after a median of 315 days of therapy.⁴⁴⁵ Secondary analyses also found no differences in proteinuria or biomarkers of kidney injury between groups receiving anti-TGF- β_1 antibody versus placebo.

This negative result may have been in part attributed to the relatively short trial duration, inadequate dosing of the antibody, or selection of participants with relatively severe and advanced nephropathy. Other possible explanations include redundancy in fibrotic mechanisms and the fact that TGF- β has potentially beneficial antiinflammatory actions. Nevertheless, it is likely that the development of interventions to inhibit kidney fibrosis will remain an area of active research and result in multiple potential novel therapies.^{446,447}

MICRO RNAs

Micro RNAs (miRNAs) are small noncoding RNAs that have been shown to have important posttranscriptional gene regulatory functions. Attention has been focused on the potential role of miRs to modulate kidney fibrosis. miR-21 and miR-214 have been shown to be upregulated in several experimental models of CKD,^{448,449} and miR-21 has been reported to be upregulated in human transplant kidneys with nephropathy.⁴⁴⁸ Deletion of miR-21 or treatment with anti-miR-21 oligonucleotides was associated with amelioration of interstitial fibrosis in animal models.⁴⁴⁸ Similarly, deletion of miR-214 and treatment with anti-miR-214 were each associated with attenuation of interstitial fibrosis in the UUO model. Importantly, the effects of miR-214 appear to be independent of TGF- β signaling, and TGF- β blockade had an additive antifibrotic effect with miR-214 deletion.⁴⁵⁰ miR-214 has further been demonstrated to suppress autophagy, an essential process that cells use to degrade damaged organelles and protein aggregates in lysosomes. In mouse models of type 1 diabetes, impaired autophagy in the kidneys was associated with elevated miR-214 expression, as well as increased hypertrophy and inflammation. Treatment with anti-miR-214 and proximal tubule-specific miR-214 gene knock out both improved autophagy and reduced diabetes-related kidney hypertrophy. Furthermore, miR-214 gene knock out ameliorated albuminuria.⁴⁵¹ As discussed previously, TGF- β /Smad3 signaling increases the production of profibrotic miRs and decreases the production of antifibrotic miRs. Multiple other miRs have been implicated in the pathogenesis of different forms of AKI and CKD.^{452,453}

On the other hand, some miRs may mediate kidney-protective effects. Expression of miR-204-5p was observed to be decreased in the kidneys of persons with hypertensive and diabetic kidney disease. Furthermore, in a mouse model of hypertensive CKD, *Mir204* gene knock out exacerbated albuminuria, renal interstitial fibrosis, and arterial disease and in a model of diabetes, treatment with anti-miR-204-5p exacerbated albuminuria and cortical fibrosis.⁴⁵⁴

Studies have identified that miRs can be transferred between cells in extracellular vesicles to exert autocrine or paracrine effects.⁴⁵² MicroRNAs may prove to be useful biomarkers to characterize or monitor CKD progression mechanisms and the potential to administer beneficial miRs or antagonize the effects of pathogenic miRs with antagomirs represents a further potential therapeutic strategy for CKD.⁴⁵³

CELL SENEESCENCE

Senescence describes a cell phenotype characterized by permanent cell-cycle arrest. It may be induced in many cell types by a range of stressors including DNA damage, mitochondrial damage, epigenetic factors, oncogene activation, oxidative stress, mechanical stress, and factors secreted by other senescent cells. Importantly, senescent cells remain viable and produce a characteristic secretome termed senescence-associated secretory phenotype (SASP), composed of proinflammatory cytokines, chemokines, growth factors, matrix metalloproteinases, and microRNAs that promote inflammation and fibrosis and are proposed to contribute to the pathogenesis of a wide range of chronic diseases. All types of kidney cells may become senescent, but kidney damage and fibrosis are most closely associated with senescent tubule epithelial cells.⁴⁵⁵ Senescent tubule cells have been identified in several experimental models of CKD. In the 5/6 nephrectomy and adriamycin nephropathy models, overexpression of C-X-C motif chemokine receptor 4 (CXCR4) in these models was associated with increased cell senescence and interstitial fibrosis, whereas knock-down of CXCR4 ameliorated both.⁴⁵⁶ The UUO model is also associated with increased tubule epithelial cell senescence but treatment with the senolytic drug, ABT-263 (also known as navitoclax), which induces apoptosis of senescent cells, resulted in an increase in kidney fibrosis, suggesting that cell senescence may be important for limiting kidney damage during ongoing injury. In contrast, administration of ABT-263 after relief of the ureteric obstruction resulted in decreased fibrosis, implying that cell senescence impaired repair after the injury phase. Similarly, administration of ABT-263 after ischemia-reperfusion injury was associated with reduced tubule cell senescence and less interstitial fibrosis.⁴⁵⁷ In humans, senescent cells have been identified in kidneys from persons with ureteric obstruction,⁴⁵⁷ glomerulonephritis,⁴⁵⁸ CKD, and failing transplants.⁴⁵⁹ In persons with primary focal and segmental glomerulosclerosis, the proportion of tubule cells expressing markers of cell senescence was associated with subsequent rate of GFR decline.⁴⁶⁰ Multiple senolytic drugs and others that antagonize SASP have shown kidney-protective benefits in experimental models, and several are being tested in clinical trials.⁴⁶¹

OXIDATIVE STRESS

CKD is associated with increased oxidative stress that likely contributes to the progression of kidney damage, as well as the pathogenesis of the associated cardiovascular disease.⁴⁶² Superoxide is the primary reactive oxygen species (ROS) that accounts for oxidative stress. Its major source is production by nicotinamide adenine dinucleotide phosphate (NADPH) oxidase, and it is removed (by conversion to hydrogen peroxide by superoxide dismutase (SOD)). Following 5/6 nephrectomy, significant upregulation of NADPH oxidase and downregulation of SOD were observed in the liver and kidneys, indicating that the increase in superoxide is due to a combination of increased production and decreased removal.⁴⁶³ Similarly, in a mouse model of proteinuric CKD induced by the administration of doxorubicin (ADR), NADPH oxidase was upregulated, and extracellular (EC-SOD) was downregulated. Furthermore, EC-SOD knockout mice showed greater albuminuria and more kidney fibrosis after doxorubicin than wild-type mice, confirming that the antioxidant effects of EC-SOD protect against kidney damage.⁴⁶⁴ Examination of human kidney biopsies confirmed that EC-SOD is similarly downregulated in human CKD.⁴⁶⁴ BP was elevated, and nitrotyrosine levels were increased, whereas urine nitric oxide metabolites were decreased, observations consistent with increased nitric oxide inactivation by superoxide. The effects of increased levels of ROS are further compounded by the reduced abundance and activity of antioxidant enzymes (e.g., catalase, glutathione peroxidase, and glutathione),⁴⁶⁵ as well as reduced HDL, apolipoprotein A-1, and thiols.⁴⁶⁶

Adverse consequences of oxidative stress that may contribute to CKD progression include hypertension (due to inactivation of nitric oxide and oxidation of arachidonic acid to generate vasoconstrictive isoprostanes),⁴⁶⁷ inflammation (due to activation of NF- κ B),⁴⁶⁵ fibrosis, and apoptosis,⁴⁶⁶ as well as glomerular filtration barrier damage.⁴⁶⁸ Inflammation may in turn increase oxidative stress due to the generation of ROS by activated leukocytes, thus establishing a vicious cycle of oxidative stress and inflammation.⁴⁶²

Other factors that may contribute to increased ROS production in CKD include Ang II,⁴⁶⁹ reduced production of NO,⁷⁷ and hypertension.⁴⁷⁰ The contribution of cellular injury due to oxidative stress in kidney damage is further illustrated by studies investigating the role of aldehyde dehydrogenase 2 (ALDH2), a mitochondrial enzyme that protects against ROS-induced accumulation of toxic aldehydes. ALDH2 expression was found to be decreased in human kidneys with CKD and in mouse kidneys after damage induced by aristolochic acid (AA) exposure. Furthermore, mutation-induced reduction in ALDH2 expression resulted in increased fibrosis after AA exposure, whereas overexpression of ALDH2 in proximal tubule cells protected them against developing a profibrotic phenotype after exposure to TGF- β . These findings may be particularly relevant in East Asia, where 35% to 45% of populations carry a deactivating mutation in the *ALDH2* gene.⁴⁷¹

The importance of ROS in the progression of CKD has been shown by experiments in which antioxidant therapies including melatonin,⁴⁷² niacin,⁴⁷³ omega-3 fatty acids,⁴⁷⁴ and n-acetylcysteine⁴⁷⁵ have reduced oxidative stress and ameliorated kidney damage in the 5/6 nephrectomy and

other animal models. On the other hand, treatment with tempol, a SOD mimetic, reduced plasma malondialdehyde levels and the number of superoxide-positive cells but did not reduce overall renal oxidative stress, inflammation, or kidney damage.⁴⁷⁶ Specific antioxidant therapies have not yet been shown to be effective in slowing human CKD progression, though several effective kidney-protective drugs are postulated to have antioxidant effects (see discussion of kidney-protective drugs, later).

ACIDOSIS

As the GFR declines, the ability of kidneys to excrete hydrogen ions becomes impaired and a new steady state is achieved that allows excretion of acid at the cost of a persistent metabolic acidosis (see earlier, "Acid-Base Regulation"). Acidosis is present in most persons when the GFR falls below 20% to 25% of normal.⁴⁷⁷ Chronic metabolic acidosis has multiple adverse consequences including increased protein catabolism, increased bone turnover, induction of inflammatory mediators, insulin resistance, and increased production of corticosteroids and PTH.^{477,478} These observations make it reasonable to consider whether acidosis may also contribute to progressive kidney damage in CKD. Early experiments found no persisting kidney damage after dietary acid loading in rats with normal kidney function and no kidney protection associated with sodium bicarbonate treatment in the 5/6 nephrectomy model, suggesting that acidosis does not initiate or exacerbate kidney damage.⁴⁷⁹

Later experiments, however, have found that an acid-generating diet of casein protein induces tubulointerstitial injury in rats with normal kidney function, whereas a non-acid-inducing diet of soy protein does not.⁴⁸⁰ Furthermore, dietary acid supplementation with $(\text{NH}_4)_2\text{SO}_4$ in soy protein-fed rats was associated with tubulointerstitial injury. Similarly, in the 5/6 nephrectomy model, a casein-rich diet was associated with metabolic acidosis, a progressive decline in the GFR, and increasing albuminuria, whereas a soy protein diet was not.⁴⁸¹

Dietary acid supplementation with $(\text{NH}_4)_2\text{SO}_4$ in soy protein-fed rats provoked a decline in the GFR and albuminuria. Treatment with sodium bicarbonate or $\text{Ca}(\text{HCO}_3)_2$, but not sodium chloride, was kidney protective in the casein-fed rats but only if the resultant hypertension—in sodium bicarbonate but not $\text{Ca}(\text{HCO}_3)_2$ -treated rats—was adequately treated. Furthermore, when rats were subject to 2/3 rather than 5/6 nephrectomy, a level of kidney mass reduction not associated with metabolic acidosis, microdialysis identified tissue acid accumulation in muscle and kidney that correlated with the subsequent GFR decline.⁴⁸² Amelioration of tissue acid accumulation by a low acid-generating diet or alkali supplementation was associated with abrogation of the GFR decline, whereas dietary acid supplementation was associated with exacerbation of the GFR decline. Treatment with calcium citrate has also been shown to improve acidosis and reduce glomerular, as well as interstitial injury in the 5/6 nephrectomy model.⁴⁸³ Mechanisms whereby acidosis may contribute to kidney damage after nephron loss include activation of the alternative complement pathway by increased ammoniogenesis,⁴⁸⁴ increased levels of plasma and kidney Ang II,⁴⁸⁵ and induction of ET, along with aldosterone production.⁴⁸⁶

Observational clinical studies have identified acidosis as an independent risk factor for CKD progression,^{487,488} but to date only relatively small studies have investigated the kidney-protective potential of alkali supplementation in human subjects. In the first randomized study in adults with a creatinine clearance of 15 to 30 mL/min, randomization to treatment of acidosis (serum bicarbonate, 16–20 mmol/L) with sodium bicarbonate was associated with a lower decline in creatinine clearance (1.88 vs. 5.93 mL/min) and lower incidence of end-stage kidney disease (ESKD) (6.5% vs. 33%).⁴⁸⁹ The authors conceded, however, that the study was not blinded or placebo controlled. In a second nonrandomized study, treatment of 30 subjects with sodium citrate was associated with reduced urinary excretion of ET-1 and N-acetyl- β -D-glucosaminidase (a marker of tubulointerstitial injury), as well as a lower rate of estimated GFR decline than that observed in 29 untreated controls who were unwilling or unable to take sodium citrate.⁴⁹⁰ One study has reported kidney-protective effects following sodium bicarbonate treatment in early CKD. In a randomized, placebo-controlled trial in subjects with a mean estimated GFR of 75 mL/min/1.73 m², treatment with sodium bicarbonate for 5 years was associated with a slower reduction in the estimated GFR (derived from plasma cystatin C measurements) than placebo or treatment with sodium chloride.⁴⁹¹

Further studies have reported that correction of acidosis with a diet rich in fruit and vegetables is as effective in ameliorating kidney damage in early (CKD stage 1 or 2)⁴⁹² and more advanced (CKD stage 4) disease.⁴⁹³ In a further trial, people with CKD stage 3 and a serum bicarbonate level of 22 to 24 mmol/L were randomized to oral bicarbonate supplementation, a diet rich in fruit and vegetables, or usual care. All participants received treatment with RAAS inhibition, and SBP was controlled to <130 mm Hg. Both interventions achieved an increase in serum bicarbonate

levels and were associated with a decrease in urinary angiotensinogen. After 3 years, both interventions were associated with less albuminuria and GFR decline than the usual care group.⁴⁹⁴ A meta-analysis of 6 randomized studies of dietary interventions for improving acidosis that included 644 participants with eGFR 15 to 40 mL/min/1.73 m² and baseline serum bicarbonate 14 to 24 mmol/L reported an increase in serum bicarbonate concentration (mean difference 2.98, 95% CI 0.77–5.10 mmol/L) difference and eGFR (mean difference 3.16, 95% CI 0.24–6.08 mL/min/1.73 m²) versus controls.⁴⁹⁵

On the other hand, a randomized trial of treatment with a hydrochloric acid binder, verveimer, in persons with eGFR 20 to 40 mL/min/1.73 m² and serum bicarbonate concentration of 12 to 20 mEq/L at baseline found no difference in the combined primary endpoint of KRT, eGFR decline $\geq 40\%$, or death due to kidney failure, though the achieved difference in serum bicarbonate concentration was only approximately 1 mEq/L, which may have been inadequate to achieve a difference in outcomes.⁴⁹⁶

Bicarbonate supplementation and/or dietary intervention is recommended to prevent bicarbonate levels lower than 18 mEq/L in persons with CKD, but further studies are required to investigate whether it is beneficial in the setting of less severe acidosis.^{497,498}

ANGIOTENSIN II

As noted, Ang II plays a central role in the glomerular hemodynamic adaptations observed after kidney mass ablation. Ang subtype 1 receptors are, however, distributed on many cell types in the kidney including mesangial, glomerular epithelial, endothelial, tubule epithelial, and vascular smooth muscle, cells suggesting multiple potential actions of Ang II in the kidney.⁴⁹⁹ Experimental studies have revealed several nonhemodynamic effects of Ang II that may be important in CKD progression (Fig. 50.8). In

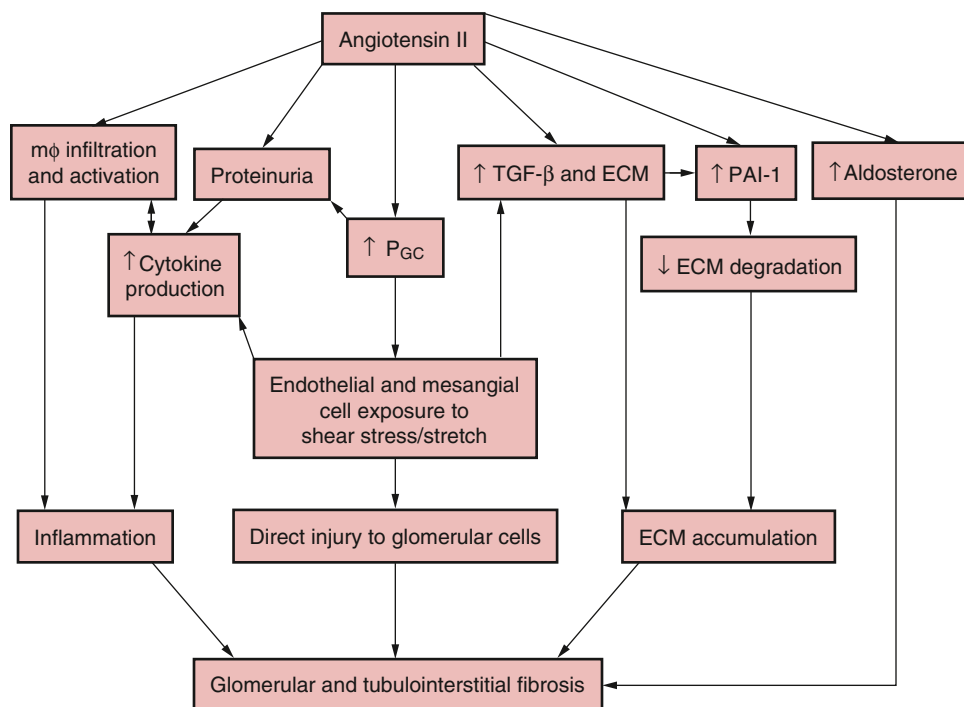


Fig. 50.8 Schematic depicting the central role of angiotensin II through hemodynamic and nonhemodynamic effects in the pathogenesis of progressive renal injury and fibrosis following nephron loss. ECM, Extracellular matrix; mφ, macrophage; PAI-1, plasminogen activator inhibitor-1; P_{GC}, glomerular capillary hydraulic pressure; TGF-β, transforming growth factor-β. (From Taal MW, Brenner BM. Renoprotective benefits of RAS inhibition: from ACEI to angiotensin II antagonists. *Kidney Int.* 2000;57:1803–1817.)

isolated perfused kidneys, the infusion of Ang II results in loss of glomerular size permselectivity and proteinuria, an effect that has been attributed to both the hemodynamic effects of Ang II, resulting in elevations in P_{GC} , and a direct effect of Ang II on glomerular permselectivity.³⁸¹ Furthermore, overexpression of angiotensin subtype 1 receptors on podocytes resulted in albuminuria and FSGS in the absence of hypertension in transgenic rats.⁵⁰⁰ These effects may be explained in part by observations that Ang II-induced apoptosis⁵⁰¹ and decreased nephrin expression³⁸² in podocytes.

In vitro, Ang II has been shown to stimulate mesangial cell proliferation and induce expression of TGF- β , resulting in increased synthesis of ECM.³²³ In vivo, transfection of rat kidneys with human genes for renin and angiotensinogen resulted in glomerular ECM expansion within 7 days.⁵⁰² Ang II also stimulates the production of PAI-1 by endothelial cells and vascular smooth muscle cells⁵⁰³⁻⁵⁰⁵ and may therefore further increase the accumulation of ECM through inhibition of ECM breakdown by MMPs that require conversion to an active form by plasmin. Other reports have indicated that Ang II may directly induce the transcription of a variety of cell adhesion molecules and cytokines, as well as activating the transcription factor NF- κ B⁵⁰⁶⁻⁵⁰⁸ and directly stimulating monocyte activation.⁵⁰⁹ Ang II infusion provoked upregulation of COX-2 expression in rats that was not dependent on BP elevation,⁵¹⁰ and 5/6 nephrectomized rats evidenced Ang II-dependent upregulation of interstitial COX-2 expression.⁵¹¹

In other experiments, Ang II infusion has been shown to induce interstitial macrophage infiltration and increased expression of MCP-1 and TGF- β , effects that were sustained for up to 6 days after cessation of the infusion.⁵¹² In cell culture and animal experiments, Ang II has been shown to promote tubule cell EMT by suppressing expression of miR-429, which inhibits EMT.⁵¹³ Finally, Ang II may have fibrogenic effects via mineralocorticoids (see later). Interestingly, Ang II may also have antifibrotic effects via the angiotensin subtype 2 receptor (AT₂). Ang II appears to upregulate AT₂ receptor expression via an AT₂ receptor-dependent mechanism after 5/6 nephrectomy, and treatment with an AT₂ receptor antagonist exacerbates kidney damage⁵¹⁴ and increased renal PAI-1 expression.⁵¹⁵ Furthermore, overexpression of AT₂ receptors in transgenic mice was associated with reduced albuminuria, as well as decreased glomerular expression of PDGF-BB chain and TGF- β after 5/6 nephrectomy.⁵¹⁶

ALDOSTERONE

Observations that aldosterone stimulates collagen synthesis in the myocardium and that spironolactone treatment affords survival benefits in addition to that achieved with ACE inhibitor alone in persons with heart failure⁵¹⁷ provided initial impetus to studies investigating the potential role of aldosterone in kidney fibrosis. In the remnant kidney model, adrenal hypertrophy and markedly elevated plasma aldosterone levels have been reported. Furthermore, the administration of exogenous aldosterone during inhibition of the RAAS with combination of ACE inhibitor-ARB therapy in the 5/6 nephrectomy model negates the kidney-protective effects of the latter.⁵¹⁸ Further evidence of the role of aldosterone

has been provided by experiments in which rats subjected to adrenalectomy after 5/6 nephrectomy received replacement glucocorticoid but not mineralocorticoid therapy, resulting in less severe kidney injury than rats with intact adrenal glands.⁵¹⁹ In persons with CKD, higher plasma aldosterone concentration was associated with lower eGFR and more severe proteinuria. Furthermore, each doubling of baseline serum aldosterone concentration was associated with an 11% increased risk (HR 1.11, 95% CI, 1.04–1.18) of reaching the composite primary endpoint of ESKD or a $\geq 50\%$ decline in eGFR.⁵²⁰

Further exploration of the physiology of aldosterone has revealed that Ang II may not be the main stimulus for aldosterone secretion and that the mineralocorticoid receptor may undergo nonligand activation by regulatory protein Rac family small guanosine triphosphatase (Rac-1), high glucose, and high sodium concentration.⁵²¹

Mechanisms whereby aldosterone and/or mineralocorticoid receptor activation may contribute to kidney damage include hemodynamic effects (see previously), mesangial cell proliferation,⁵²² apoptosis,⁵²³ hypertrophy and transdifferentiation,⁵²⁴ podocyte injury, and apoptosis associated with reduced expression of nephrin and podocin, resulting in proteinuria,^{383,525,526} proximal tubule cell transdifferentiation and increased production of collagen types III and IV,⁵²⁷ and increased kidney production of ROS,^{383,525} PAI-1,^{528,529} TGF- β ,⁵³⁰ and CTGF.^{531,532}

Early experimental use of aldosterone receptor blockers in 5/6 nephrectomized rats yielded only modest renoprotective effects,^{518,533} but other studies have found significant amelioration of glomerulosclerosis in 5/6 nephrectomized rats treated with spironolactone, alone or in combination with triple antihypertensive therapy or an ARB.^{528,534,535} In some rats, spironolactone was associated with the apparent regression of glomerulosclerosis. Furthermore, the observed kidney protection was associated with inhibition of PAI-1 mRNA expression in the kidney cortex.⁵²⁸ In another experiment, kidney protection, achieved with combination ACE inhibitor and spironolactone treatment, was associated with abrogation of the increase in mesangial cells and decrease in podocytes observed in untreated rats. Combination treatment also attenuated the increase in expression of type IV collagen, TGF- β , and desmin.⁵³⁴ Spironolactone has also been shown to ameliorate kidney damage in other experimental models including diabetic nephropathy,^{531,536} Ren2 transgenic rats,⁵³⁷ radiation nephritis,⁵³⁸ and stroke-prone hypertension.⁵³⁹

HYPERTROPHY

The consistent observation of kidney, and in particular glomerular, hypertrophy after kidney mass reduction has prompted the hypothesis that processes involved in, or resulting from, hypertrophy may contribute to progressive kidney injury in CKD.⁵⁴⁰ The well-documented observation that kidney and glomerular hypertrophy precede the development of diabetic nephropathy and the finding of a positive association between glomerular size and early sclerosis in rats subjected to kidney mass ablation⁵⁴¹ suggests that hypertrophy may play a direct role in the pathogenesis of glomerulosclerosis. Several clinical observations have also supported an association

between glomerular hypertrophy and kidney injury. Oligomeganephronia, a rare congenital condition with a nephron number 25% of normal or less, is characterized by marked hypertrophy of the remaining glomeruli and the development of proteinuria and kidney failure in adolescence, with FSGS as the typical kidney biopsy finding.⁵⁴² In children with minimal change disease, a glomerulopathy generally associated with spontaneous remission and lack of progression to kidney failure, investigators have noted an association between glomerular size and the risk of developing FSGS and kidney failure.⁵⁴³

Several interventions have been used in experiments to interrupt the development of glomerular hypertrophy after kidney mass reduction and thereby assess its role in kidney disease progression, but they have produced contradictory results. Rats subjected to 5/6 nephrectomy were compared with rats in which 2/3 of the left kidney was infarcted and the right ureter drained into the peritoneal cavity, an intervention that apparently results in decreased renal clearance without compensatory kidney hypertrophy.⁵⁴⁰ Micropuncture studies confirmed similar degrees of elevation of P_{GC} and SNGFR in both models. At 4 weeks, however, the maximal planar area of glomeruli was significantly less, and glomerular injury, as assessed by sclerosis index, was significantly reduced in ureteroperitoneostomized rats versus 5/6 nephrectomized controls. Accordingly, the authors concluded that glomerular hypertrophy is more important than glomerular capillary hypertension in the progression of glomerular injury in this model.

Dietary sodium restriction has also been used to inhibit kidney hypertrophy after 5/6 nephrectomy. Although sodium restriction had no effect on glomerular hemodynamics, glomerular volume was significantly reduced in 5/6 nephrectomized rats fed low versus normal sodium diets.¹⁴² Moreover, urinary protein excretion was lower, and glomerulosclerosis was less severe, in rats on restricted sodium intake. These findings were extended by another study in which the effect of sodium restriction in preventing glomerular hypertrophy and ameliorating glomerular injury was confirmed, but which also found that these benefits were overcome by the administration of an androgen that stimulated glomerular hypertrophy, despite sodium restriction. Glomerular hemodynamics were similar among the groups.⁵⁴⁴ On the other hand, treatment with seliciclib, a CDK inhibitor, reduced kidney hypertrophy by 45% after 5/6 nephrectomy but had no effect on kidney damage.⁵⁴⁵

Glomerular hypertrophy may contribute to glomerulosclerosis through a number of different mechanisms. According to the law of Laplace, the increase in glomerular volume could result in an increase in capillary wall tension only if the capillary wall diameter were also increased (Fig. 50.9). Cyclic stretch would then exert stress capable of damaging epithelial, mesangial, and endothelial cells, as described previously. Alternatively, glomerulosclerosis may be viewed as a maladaptive growth response following loss of kidney mass and resulting in excessive mesangial proliferation and extracellular matrix production.⁵⁴⁰ The identification of TGF- β as a key mediator of kidney hypertrophy, as well as an important promoter of

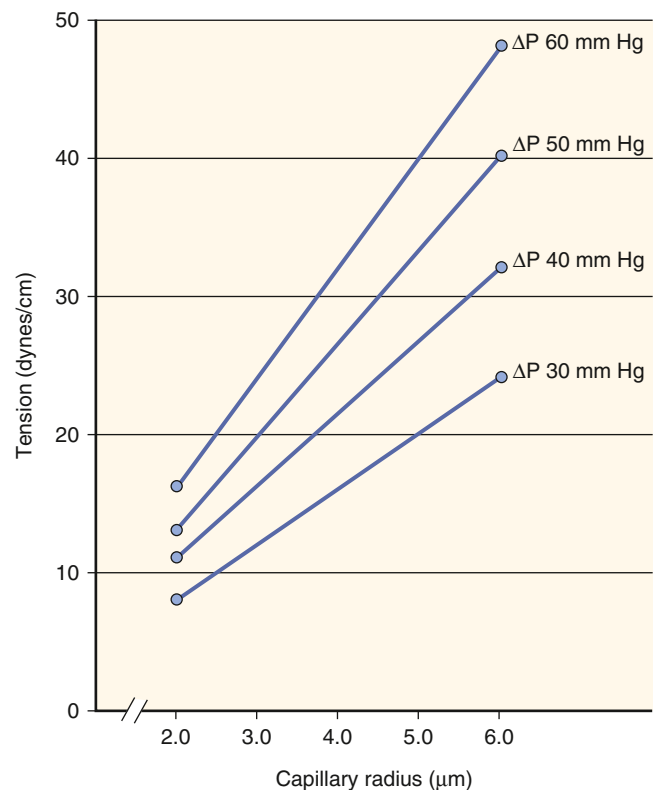


Fig. 50.9 Illustration of the synergistic effects of changes in transcapillary hydrostatic pressure difference (ΔP , mm Hg) and mean glomerular capillary radius (μm) on the calculated capillary wall tension (dynes/cm). (From Bidani AK, Mitchell KD, Schwartz MM, et al. Absence of progressive glomerular injury in a normotensive rat remnant kidney model. *Kidney Int.* 1990;38:28–38.)

kidney fibrosis provides an obvious link between kidney hypertrophy and fibrosis.¹⁸³

Detailed studies of podocyte hypertrophy have found that impaired podocyte hypertrophy in response to glomerular enlargement is a further mechanism that may contribute to the development of proteinuria and FSGS. Using a transgenic rat model (dominant negative AA-4E-BP1 transgene) that resulted in impaired podocyte hypertrophy, the investigators observed a remarkable linear relationship among gain in body weight, proteinuria, and glomerulosclerosis that was accelerated after uninephrectomy. Dietary caloric restriction that prevented weight gain and glomerular enlargement also prevented proteinuria. Analysis of the kidneys from rats with proteinuria demonstrated a mismatch between glomerular tuft volume and total podocyte volume, and electron microscopy revealed a pulling apart of podocyte foot process, resulting in exposed areas of glomerular basement membrane and adhesions to the Bowman capsule.⁵⁴⁶ These findings were extended in the same model by showing that the prevention of glomerular hypertrophy after uninephrectomy by dietary caloric restriction, mTORC1 kinase pathway inhibition (with rapamycin), or treatment with an ACE inhibitor each prevented proteinuria and FSGS. Proliferation of glomerular cells other than podocytes was identified and has been proposed to contribute to the development of

FSGS. Moreover, transcriptomic analysis of gene expression in isolated glomeruli from rats that developed FSGS evidenced a similar pattern of gene expression to that identified in glomeruli from human biopsies of persons with progressive CKD, suggesting that similar factors play a role in human CKD progression.⁵⁴⁷

A UNIFIED HYPOTHESIS OF CHRONIC KIDNEY DISEASE PROGRESSION

In the past, there has tended to be a dichotomy of viewpoints regarding the relative importance of hemodynamic and nonhemodynamic factors in the pathogenesis of glomerulosclerosis and TIF after nephron loss.^{540,548} Proponents of the so-called hypertrophy hypothesis have pointed out that in some experiments, a dissociation between glomerular hemodynamic changes and glomerulosclerosis has been observed, and that in one study, antihypertensive therapy was kidney protective without lowering P_{GC} .⁵⁴⁰ On the other hand, those favoring the hemodynamic hypothesis have noted that treatment with an ACE inhibitor²¹ or ARB²³ in 5/6 nephrectomized rats results in kidney protection without preventing kidney or glomerular hypertrophy and that many of the studies purporting to show a positive association between glomerular hypertrophy and sclerosis failed to report glomerular hemodynamic data. Furthermore, rats subjected to ureteropertoneostomy developed significantly more glomerulosclerosis than sham-operated controls, despite a lack of increase in glomerular size.⁵⁴⁸

Several other observations have suggested that hemodynamic factors override the potential role of hypertrophy in progressive kidney damage. The kidney protection achieved after 5/6 nephrectomy by a low-protein diet (associated with prevention of glomerular hypertrophy) can be reversed by treatment with calcium channel blockers that inhibit renal autoregulation but have no effect on glomerular size.⁵⁴⁹ Comparison of rats subjected to 5/6 nephrectomy by excision versus infarction of two-thirds of the remaining kidney has shown similar increases in glomerular volume, but the infarction model is associated with more severe systemic and glomerular hypertension and glomerulosclerosis.^{10,11}

Despite these apparently conflicting views, it is clear from the earlier discussion that hemodynamic and nonhemodynamic mechanisms often overlap. For example, Ang II, a key mediator of glomerular hemodynamic adaptations after nephron loss, also exerts multiple nonhemodynamic deleterious effects. Furthermore, the inflammatory and profibrotic mechanisms that eventuate in glomerulosclerosis and tubulointerstitial fibrosis may be provoked by both hemodynamic and nonhemodynamic stimuli. A growing appreciation of the complexity of the multiple adaptations that follow nephron loss has facilitated the development of a consensus view that continues to regard raised glomerular capillary pressure as a central factor in initiating glomerulosclerosis, but it also acknowledges that other nonhemodynamic pathogenic mechanisms may act in concert with hemodynamic factors in a complex interplay that results in a vicious cycle of progressive kidney damage (Fig. 50.10). An appreciation

of the many mechanisms involved in CKD progression is essential to inform strategies for achieving optimal kidney protection.

INSIGHTS FROM MODIFIERS OF CHRONIC KIDNEY DISEASE PROGRESSION

Clinical Relevance

Optimal Kidney Protection

The treatment of hypertension with an optimal dose of a RAAS inhibitor is the mainstay of the approach to achieve kidney protection in proteinuric CKD. BP should be reduced to <130/80 mm Hg, and even lower targets should be considered in persons with persistent proteinuria, judged to be at low risk of adverse effects from lower BP. Treatment with an SGLT2 inhibitor should be added for persons with and without diabetes who meet the guideline criteria. For those with type 2 diabetes, the addition of finerenone and/or a GLP-1 receptor agonist should be considered.

KIDNEY-PROTECTIVE DRUG TREATMENTS

Translation of the growing body of knowledge regarding mechanisms of CKD progression has resulted in the development of multiple kidney-protective therapies. Evidence of their clinical effectiveness in slowing CKD progression provides further evidence supporting the importance of these mechanisms. Each category of drugs will be reviewed here only briefly. For further details regarding the trials and recommendations for clinical use, please see [Chapter 54](#).

RENIN-ANGIOTENSIN- ALDOSTERONE SYSTEM INHIBITORS

Experimental evidence showing a central role for Ang II in mechanisms of CKD progression through hemodynamic and nonhemodynamic effects has been borne out in randomized clinical trials of ACE inhibitor and ARB treatment in persons with all forms of CKD. ACE inhibitor treatment has been shown to be kidney protective in individual trials of persons with type 1 diabetes and overt nephropathy,²⁹¹ microalbuminuria and type 2 diabetes mellitus,⁵⁵⁰ and nondiabetic CKD.^{292,295,551,552} Further trials have found that ARB treatment affords kidney protection in persons with type 2 diabetes and microalbuminuria⁵⁵³ or overt nephropathy.^{293,294} These findings have been confirmed in several meta-analyses.⁵⁵⁴⁻⁵⁵⁶ However, more complete inhibition of the RAAS with combination ACE inhibitor and ARB^{557,558} or combination ARB and direct renin inhibitor treatment⁵⁵⁹ was associated with an excess of adverse effects including acute kidney injury (AKI), hypotension, and hyperkalemia, implying that near-complete blockade of the RAAS may be undesirable in many patient groups. In summary, the large body of clinical evidence showing kidney-protective effects of RAAS inhibitor treatment confirms that Ang II is a critical mediator of mechanisms of CKD progression in humans and supports the consensus that RAAS inhibition should be central to treatment

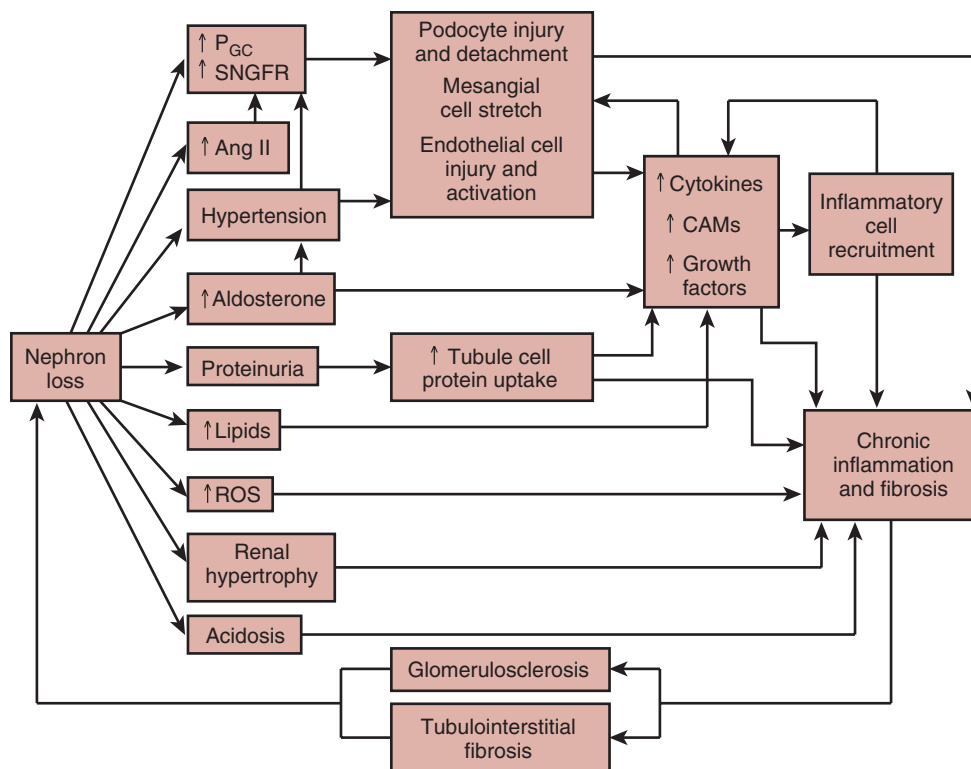


Fig. 50.10 Schematic illustrating the hypothesized interaction of multiple hemodynamic and nonhemodynamic factors in the pathogenesis of progressive nephron injury in chronic kidney disease. CAM, Cell adhesion molecules; GFR, Glomerular filtration rate; P_{GC} , glomerular capillary hydraulic pressure; ROS, reactive oxygen species; SNGFR, single-nephron GFR.

strategies for slowing CKD progression.⁴⁹⁸ The role of RAAS inhibitor treatment in achieving optimal kidney protection is discussed further in [Chapter 54](#).

SODIUM GLUCOSE COTRANSPORTER 2 INHIBITORS

Evidence of the kidney-protective effects of a class of drug developed for the treatment of diabetes, the sodium-glucose cotransporter 2 (SGLT2) inhibitors, has provided further support for importance of glomerular hemodynamic factors in CKD progression. These drugs act by inhibiting the reabsorption of filtered glucose and sodium in the proximal tubule, resulting in glycosuria that reduces hyperglycemia, and modest natriuresis. Importantly, it has been proposed that increased delivery of sodium chloride to the macula densa results in increased tubuloglomerular feedback, which reduces glomerular hyperfiltration by provoking the constriction of afferent arterioles.²⁹⁷⁻²⁹⁹ In rats with streptozotocin-induced diabetes, treatment with an SGLT2 inhibitor resulted in a marked reduction in SNGFR that was inversely proportional to the distal tubular chloride concentration, indicating activation of tubuloglomerular feedback.³⁰⁰ Similarly, treatment with the SGLT2 inhibitor after 5/6 nephrectomy afforded kidney protection similar to treatment with an ARB and provoked an increase in urinary adenosine excretion, a marker of tubuloglomerular feedback activity, that was inversely related to the severity of tubulointerstitial fibrosis.⁹⁷ In persons with type 1 diabetes, treatment with an SGLT2 inhibitor reduced glomerular hyperfiltration to normal levels in conjunction with a decrease in effective renal plasma flow and an increase in renal vascular resistance, observations consistent with preglomerular (afferent arteriole) vasoconstriction.⁵⁶⁰ Thus treatment with SGLT2 inhibitors ameliorates glomerular hyperfiltration and (by implication) glomerular

hypertension. Landmark clinical trials reported substantial kidney protection when SGLT2 inhibitors were added to RAAS inhibitor treatment in persons with CKD associated with type 2 diabetes⁵⁶¹ or without diabetes⁵⁶² and with or without albuminuria.⁵⁶³ In a meta-analysis of large clinical trials in persons with and without type 2 diabetes, treatment with SGLT2 inhibitors was associated with a significant reduction in risk of the composite primary endpoint of CKD progression (50% or greater decrease in eGFR, a sustained low eGFR, ESKD, or death from kidney failure) (relative risk 0.63; 95% CI, 0.58–0.69),²⁹⁶ confirming the importance of glomerular hyperfiltration (and glomerular hypertension) in human CKD progression. Nevertheless, several other potential kidney-protective effects of SGLT2 inhibitors have been identified including lowering of systemic BP, reduction of albuminuria, weight loss, amelioration of arterial stiffness, reduction of renal oxygen demand, and induction of hypoxia-inducible factor 1 (HIF1).²⁹⁷⁻²⁹⁹ The kidney-protective effects of SGLT2 inhibitors are reviewed in more detail in [Chapters 41 and 54](#).

MINERALOCORTICOID INHIBITORS

Several small clinical trials reported an additional reduction of proteinuria by 15% to 54%, BP by approximately 40%, and GFR by approximately 25% when aldosterone receptor blockers were added to ACE inhibitor or ARB treatment.⁵⁶⁴ More novel nonsteroidal aldosterone antagonists are also being evaluated. In one randomized trial that enrolled 336 subjects with hypertension, urine ACR of 30 to 599 mg/g, and an eGFR of 50 mL/min/1.73 m² or higher, treatment with eplerenone for 52 weeks, added to ACE inhibitor or ARB therapy, resulted in a 17.3% decrease in urine ACR, whereas a 10.3% increase was observed in those who received placebo ($P = 0.02$).⁵⁶⁵ The most compelling evidence of

kidney protection has come from trials of a nonsteroidal MRA, finerenone. In the Finerenone in Reducing Kidney Failure and Disease Progression in Diabetic Kidney Disease (FIDELIO-DKD) trial, 5734 persons with type 2 diabetes, GFR 25 to 75 mL/min/1.73 m² and urine albumin-to-creatinine ratio (ACR) 30 to 5000 mg/g were randomized to treatment with finerenone or placebo, added to RAAS inhibition. Finerenone treatment resulted in a significant reduction in the primary composite endpoint of kidney failure (GFR <15 mL/min/1.73 m² or KRT), 40% GFR decline, or renal death (HR 0.82; 95% CI, 0.73–0.93).⁵⁶⁶ Similarly, The Finerenone in Reducing Cardiovascular Mortality and Morbidity in Diabetic Kidney Disease (FIGARO-DKD) trial enrolled 7437 persons with type 2 diabetes and albuminuria but with a wider range of GFR (25–90 mL/min/1.73 m²). In this trial the primary composite endpoint was a major cardiovascular event and the secondary composite outcome was CKD progression (same as the primary endpoint in FIDELIO-DKD). Finerenone treatment significantly reduced the risk of the primary cardiovascular endpoint (HR 0.87; 95% CI, 0.76–0.98) and tended to reduce the risk of the secondary endpoint (HR 0.87; 95% CI, 0.76–1.01).⁵⁶⁷ Data from FIDELIO-DKD and FIGARO-DKD were combined in a prespecified pooled analysis showing a significant 23% reduction in the primary composite endpoint of kidney failure, 57% reduction in GFR, or death due to renal causes (HR 0.77, 95% CI 0.67–0.88).⁵⁶⁸

Together, these findings provide strong support for an important role of aldosterone and mineralocorticoid receptor activation, separate from Ang II, in CKD progression mechanisms. Single-cell transcriptomic studies in animal models have provided new insights into the molecular mechanisms of MRA.⁵⁶⁹

GLUCAGON-LIKE PEPTIDE-1 RECEPTOR AGONISTS

Glucagon-like peptide-1 (GLP-1) is a hormone secreted by gastrointestinal tract cells in response to a meal that stimulates glucose-dependent insulin secretion. GLP-1 also slows gastric emptying and acts on the hypothalamus to reduce appetite. GLP-1 receptor agonists (GLP-1RAs) were developed as drugs for the treatment of type 2 diabetes. Subsequent trials showed efficacy in improving glycemic control but also observed substantial weight loss. Following evidence of kidney protection with GLP-1RA treatment from secondary analyses of earlier trials, the “Evaluate Renal Function with Semaglutide Once Weekly” (FLOW) trial was the first to investigate kidney endpoints in persons with CKD. Persons ($n = 3533$) with CKD and type 2 diabetes were randomized to treatment with semaglutide or placebo in addition to RAAS inhibitor treatment. The primary endpoint was a composite of kidney failure (KF), 50% decline in estimated glomerular filtration rate (eGFR), or death due to kidney disease or CVD. After a median follow-up period of 3.4 years, the primary composite endpoint of KF, 50% decline in estimated glomerular filtration rate (eGFR), or death due to kidney disease or CVD was reduced by 24% in the semaglutide versus placebo group (HR 0.76; 95% CI, 0.66–0.88).⁵⁷⁰ Secondary analysis of data from a trial that observed a 20% reduction in cardiovascular events with semaglutide treatment in persons with obesity but not diabetes reported 22% reduction in the composite kidney endpoint of eGFR <15 mL/min/1.73 m², initiation

of KRT, death due to kidney disease, eGFR decline of $\geq 50\%$, or progression to macroalbuminuria (HR 0.78; 95% CI, 0.63–0.96).⁵⁷¹ In a phase 2 trial of semaglutide treatment in persons with BMI ≥ 27 kg/m², semaglutide treatment reduced albuminuria by 52.1% (95% CI, –65.5 to –33.4; $P < 0.0001$) compared with placebo.⁵⁷²

Kidney protection with GLP-1RA treatment has been attributed to a reduction in risk factors for CKD progression due to weight loss and BP lowering, as well as improvements in dyslipidemia and glycemic control. In addition, GLP-1 receptors are expressed in the kidneys and exert antiinflammatory effects by inhibiting signaling by receptors for advanced glycation end products (RAGE).⁵⁷³

Currently recommended kidney-protective drugs therefore impact multiple different common pathway mechanisms of CKD progression. The benefits of SGLT2 inhibitors, MRAs, and GLP-1RAs have been demonstrated to be additive to those of RAAS inhibitors. Further trials are required to investigate whether MRAs and GLP-1RAs afford additional kidney protection when added to combination RAAS inhibitor and SGLT2 inhibitor treatment, but their differing mechanisms of action suggest that this will be the case. For a figure summarizing the impact of kidney-protective drugs on different mechanisms of CKD progression, please see fig. 2 in Chapter 54.

ARTERIAL HYPERTENSION

Several population-based studies have observed an increased risk of developing progressive kidney damage with higher levels of BP.^{574–577} In a population of 332,544 men, there was a strong graded relationship between BP and the risk of developing or dying with ESKD over 15 to 17 years, but kidney function was not assessed at screening or during follow-up.⁵⁷⁸ However, in the Atherosclerosis Risk in Communities (ARIC) Study, GFR was evaluated at 5 study visits over 30 years in 14,854 participants. Hypertension was associated with a more rapid rate of GFR decline and a higher risk of incident CKD, and this association was attenuated by antihypertensive treatment.⁵⁷⁹ Even small increases in BP, below the threshold usually used to define hypertension, are associated with an increased risk of ESKD.^{574,576,580} Interestingly, in normotensive potential living kidney donors, urine biomarkers of podocyte detachment and podocyte stress were positively associated with higher MAP.⁵⁸¹ In another community-based study, a rising trend in systolic BP was associated with a 1.57-fold higher risk (95% CI, 1.20–2.06) of incident eGFR <60 mL/min/1.73 m² and a higher risk of new albuminuria than stable BP.⁵⁸² Similarly, in a study of young adults, an increase in BP during the first 10 years of observation was associated with a more rapid subsequent decline in GFR. Hypertension has also been identified as a risk factor for developing albuminuria or kidney impairment among persons with type 2 diabetes mellitus.⁵⁸³

Whereas the role of hypertension in initiating kidney disease requires further clarification, there is clear evidence that hypertension accelerates the rate of progression of preexisting kidney disease, most likely through the transmission of raised hydraulic BP to the glomerulus, resulting in the exacerbation of glomerular capillary hypertension associated with nephron loss.³ Among

persons with diabetic nephropathy and nondiabetic CKD, the initiation of antihypertensive therapy results in significant reductions in rates of GFR decline, implying that hypertension, an almost universal consequence of impaired kidney function, also contributes to the progression of CKD.⁵⁸⁴ The potential impact of hypertension on the kidney has been exemplified by case reports of persons with unilateral renal artery stenosis who manifested diabetic nephropathy or FSGS only in the nonstenotic kidney and not in the stenotic side that was shielded from hypertension.^{585,586} Analysis of data from the Chronic Renal Insufficiency Cohort (CRIC) study has emphasized the importance of BP control over time. Time-updated SBP higher than 130 mm Hg was more strongly associated with an increased risk of CKD progression than a single baseline measurement.⁵⁸⁷

Uncertainty remains, however, as to the level of BP lowering that is required to achieve optimal kidney protection.⁵⁸⁸ Several randomized trials have sought to resolve this issue and are reviewed in detail in [Chapter 54](#).⁵⁸⁹⁻⁵⁹¹ The overall picture is one of lower BP targets being associated with more effective kidney protection among those with more severe proteinuria, and the data supported a previous recommendation that BP should be lowered to <130/80 mm Hg in all persons with diabetic or proteinuric CKD and <140/90 mm Hg in persons without these risk factors.⁵⁹²

A subsequent large study suggested that an even lower BP target would improve outcomes further. In The Systolic Blood Pressure Intervention Trial (SPRINT), 9361 persons aged 50 years or older with risk factors for cardiovascular disease (including 2646 with CKD) were randomized to a SBP target of <120 mm Hg versus <140 mm Hg. The trial was stopped early due to evidence of significant benefit in the primary outcome of cardiovascular events (HR 0.75; 95% CI, 0.64–0.89) and all-cause mortality (HR 0.73; 95% CI, 0.60–0.90) in the lower BP target group, and there was no evidence of effect modification by CKD status.⁵⁹³ A prespecified subgroup analysis of participants with CKD at baseline found no between groups in the kidney outcome of 50% decline in the GFR or ESKD after a median of 3.3 years. The lower BP target was associated with some initial decline in the eGFR, likely due to increased use of RAASi, whereas a small initial increase in the GFR was observed with the higher BP. After excluding the first 6 months of observation, the low BP target was associated with a slightly higher rate of GFR decline (0.47 vs. 0.32 mL/min/1.73 m² per year; *P* < 0.03). The lower BP target was also associated with a lower urine ACR at all time points to 48 months. There was no difference in serious adverse events between the groups, but the low BP target was associated with a higher incidence of hyperkalemia, hypokalemia, and AKI.⁵⁹⁴ Posttrial follow-up of a subgroup of 3041 participants found no difference in GFR decline between groups in the postintervention period.⁵⁹⁵ The results of SPRINT therefore indicate survival and cardiovascular benefits but no kidney-protective benefit with lower BP targets, and they question the notion that the risk of lower BP targets outweighs the benefits in all older adults. A similar trial of 11,255 persons aged 50 years or older with high cardiovascular risk (4359 with diabetes and 3022 with previous stroke) also reported a significant reduction in cardiovascular events (HR 0.88; 95% CI,

0.78–0.99) associated with a low SBP target of <120 mm Hg (vs. <140 mm Hg) though only 6% of participants had a baseline eGFR below 60 mL/min/1.73 m² and albuminuria was not reported.⁵⁹⁶

In contrast, several other studies have reported that aggressive BP lowering may be associated with adverse effects in some patient groups. In persons with diabetes, the IDNT study reported that achieved SBP <120 mm Hg was associated with increased all-cause mortality and no further improvement in kidney outcomes⁵⁹⁷; the ACCORD study reported no additional benefit with respect to cardiovascular endpoints in participants randomized to an SBP target of <120 mm Hg (vs. conventional control to <130/80 mm Hg), but the lower BP target was associated with more treatment-related adverse events and a greater decline in the GFR.⁵⁹⁸ In a meta-analysis of trials including persons with nondiabetic CKD, achieved SBP <110 mm Hg was associated with a higher risk of CKD progression.⁵⁹⁹ Furthermore, secondary analysis of the Ongoing Telmisartan Alone and in combination with Ramipril Global EndpointT (ONTARGET) trial found that participants who achieved SBP of <120 mm Hg had significantly higher cardiovascular mortality than those who achieved SBP of 120 to 129 mm Hg.⁶⁰⁰

On the basis of the results of SPRINT, the 2024 Kidney Disease Improving Global Outcomes (KDIGO) guidelines suggested a lower target systolic BP of <120 mm Hg for all persons with CKD and hypertension when tolerated but also advised that less intensive BP lowering should be considered in persons with symptomatic postural hypotension, high risk of falls, frailty, or limited life expectancy.⁴⁹⁸

DIETARY PROTEIN INTAKE

Increased dietary protein intake and intravenous protein loading in animals or humans with intact kidneys are associated with increases in kidney mass, RBF, and GFR, as well as a decrease in kidney vascular resistance. The magnitude of the increases in the GFR and RBF in response to a protein load is a function of kidney reserve. In persons with reduced GFR, some studies have shown that the percentage increase in the GFR in response to a protein meal is reduced in those with a lower baseline GFR.⁶⁰¹ In contrast, a study comparing the renal response to an oral protein load in persons with moderate and advanced CKD found a similar percentage increase in the GFR over baseline in both groups, demonstrating that even with advanced kidney disease, some kidney reserve is still present, and that elevated intake of dietary protein may have undesirable effects on glomerular hemodynamics at all levels of kidney function.⁶⁰²

Micropuncture experiments have demonstrated that amino acid infusion results in increases in glomerular plasma flow and transcapillary hydraulic pressure difference, thereby raising SNGFR without affecting the ultrafiltration coefficient.⁶⁰³ Interestingly, however, perfusion of the isolated kidney with an amino acid mixture resulted in only a modest increase in the GFR,⁶⁰⁴ suggesting that amino acids themselves do not have a major direct effect on renal hemodynamics. Rather, their effects appear to be mediated by glucagon, the secretion of which is stimulated by protein feeding.^{603,605,606} Other mechanisms that may contribute to glomerular hyperfiltration after a protein meal include

increased sodium chloride reabsorption in the proximal tubule and loop of Henle,⁶⁰⁷ resulting in decreased tubuloglomerular feedback and increased nitric oxide and prostaglandin synthesis.^{607,608} In addition, Ang II and ET have been proposed as mediators of protein-induced kidney injury because low-protein diets have been shown to reduce renal ET-1, ET receptors A and B, and AT₁ receptor mRNA expression in PAN-injected and normal rats.^{609,610}

It has been proposed that the augmented kidney function induced by dietary protein may be an evolutionary adaptation of the kidney to the intermittent heavy protein intake of the hunter-gatherer.⁴ Kidney hyperfunction following a protein load would serve to facilitate excretion of the waste products of protein catabolism and other dietary components, thereby achieving homeostasis in the presence of an abrupt increase in consumption in times of nutritional plenty; the subsequent decline of the GFR to baseline during the intervals between meals would then favor mechanisms suited to conservation of fluid and electrolytes in times of scarcity. Persistent kidney hyperfunction due to continuous excessive protein intake, however, leads to kidney injury in experimental models. Laboratory animals with intact kidneys and ingesting food ad libitum become proteinuric and develop glomerulosclerosis with age.^{4,204,611} This progression is significantly attenuated by feeding animals on alternate days only.²⁰⁴

In animal models, a high-protein diet accelerates kidney damage, whereas a low-protein diet slows progression in aging rats,⁶¹¹ diabetic rats,²⁶¹ and after 5/6 nephrectomy.⁹ Despite unambiguous evidence from experimental studies, confirmation of a beneficial effect of dietary protein restriction in clinical trials has proved elusive. The Modification of Diet in Renal Disease (MDRD) study found no difference in the rate of GFR decline in participants randomized to lower versus higher protein diet,⁵⁸⁹ though secondary analyses that accounted for an initial reduction in the GFR that likely resulted from the functional effects of decreased protein intake did suggest some benefit.⁶¹² Long-term follow-up of 255 participants in study 2 of the MDRD trial found no kidney-protective benefit associated with randomization to a very-low-protein diet in the original study but did report a higher risk of death in this group (HR, 1.92; 95% CI, 1.15–3.20).⁶¹³ Several meta-analyses have each concluded that dietary protein restriction is associated with a modest reduction in risk of ESKD.^{614,615,616,617} A further meta-analysis assessed the effect of low-protein diet supplemented with keto-analogs versus low-protein diet alone and found that the former was associated with better preservation of GFR and, in persons without diabetes, a marginally lower risk of ESKD.⁶¹⁸ An important caveat to these findings is that the proportion of the participants treated with RAAS inhibitors was variable. Though evidence suggests that low-protein diet and RAAS inhibitors may have synergistic kidney-protective effects, this has not yet been adequately evaluated in RCTs. Concerns about the risks of malnutrition associated with dietary protein restriction, particularly in older persons with CKD, have to some extent been addressed by results from an observational study in persons aged >65 years with eGFR <20 mL/min/1.73 m,² which found no nutritional status decline or excess mortality in those prescribed low-protein diets.⁶¹⁹

In summary, there is evidence from some small trials and several meta-analyses that dietary protein restriction may slow the progression of human CKD, particularly in more advanced and proteinuric diseases. Whereas the kidney-protective benefit appears modest, such dietary restriction is associated with other benefits including improvement in acidosis and a reduction in the phosphorus and potassium load. Current guidelines do not recommend dietary protein restriction for all persons with CKD but advise that high-protein intake should be avoided and that low-protein diet with amino acid or essential keto-acid supplementation can be considered with careful supervision in persons at high risk of progression.⁴⁹⁸ The role of dietary intervention in the management of CKD is discussed further in [Chapter 55](#).

SEX DIFFERENCES

Laboratory studies have indicated that male animals appear to be at greater risk of developing kidney disease and disease progression than females. Age-associated glomerulosclerosis is much more pronounced in male than female rats, and it is notable that the male propensity for age-related glomerulosclerosis can be prevented by castration.⁶²⁰ This sex difference was found to be independent of P_{GC} or glomerular hypertrophy, suggesting a role for sex hormones as modulators of kidney injury. Ovariectomy had no effect on the development of glomerular injury in some animal models^{620,621} but did accelerate progression in glomerulosclerosis-prone mice.⁶²²

In contrast, in the hypercholesterolemic Imai rat, the development of spontaneous glomerulosclerosis in males can be significantly reduced by castration or administration of exogenous estrogens.^{623,624} These data suggest an important role for androgens in the development of kidney injury and raise the possibility that estrogens may, to some extent, counteract the adverse effects of androgens. In an apparently conflicting observation, female Nagase albuminemic rats developed more severe kidney injury than males, a characteristic that is ameliorated by ovariectomy.⁶²⁵ These rats may be unique, however, in that triglyceride levels, which are higher in females, may have an independent and overriding effect on kidney disease propensity. Glomerulosclerosis also develops to a significantly greater extent in male versus female rats subjected to extensive kidney ablation.⁶²⁶ This difference was independent of BP and glomerular hypertrophy, but the degree of glomerulosclerosis and the extent of mesangial expansion were each found to correlate significantly with an increased expression of glomerular procollagen α_1 (IV) mRNA in males. Similarly, in aging Munich-Wistar rats, glomerular metalloproteinase activity was found to decrease with age in males but not in females or castrated rats, suggesting that suppression of metalloproteinase activity by androgens could account for the sex difference in disease susceptibility.⁶²⁷ Finally, estrogens, but not androgens, possess antioxidant activity and have been shown to inhibit mesangial cell low-density lipoprotein (LDL) oxidation,⁶²⁸ a property that may contribute to kidney protection.

Clinical studies confirm that sex differences with respect to CKD progression also exist in humans. Even in healthy

adults, age-related decline in measured GFR was noted to be significantly greater in men (mean -1.20 ; 95% CI, 1.12 – 1.28 mL/min/ 1.73 m 2) than in women (mean -0.96 ; 95% CI, -0.88 to -1.04 mL/min/ 1.73 m 2).⁶²⁹ In most studies, the prevalence of CKD stage 3 to 5 is higher in females but the risk of progression and ESKD is higher in males. In a Japanese community-based mass screening program, the risk of developing ESKD (if baseline serum creatinine level was >1.2 mg/dL for males or >1 mg/dL for females) was almost 50% higher in men than in women.⁶³⁰ In a large population-based study in the United States, male sex was associated with a significantly increased risk of ESKD or death associated with CKD.⁵⁷⁶ Similarly, in France, studies of factors influencing the development of ESKD in persons with moderate and severe kidney disease found that disease progression was accelerated in males versus females, especially in those with chronic glomerulonephritis or autosomal dominant polycystic kidney disease (ADPKD). Furthermore, the effect of hypertension as a risk factor for CKD progression appeared to be greater in males.^{631,632} In Italy, a pooled analysis of observational studies of older people with eGFR <45 mL/min/ 1.73 m 2 similarly observed a higher risk of ESKD in men in a multivariable analysis (HR 1.50; 95% CI, 1.28–1.77) and the slope of GFR decline was steeper in men than women.⁶³³ In the CRIC study, multivariable analysis found that women evidenced a lower risk of ESKD, 50% decline in GFR, progression to CKD stage 5, and death.⁶³⁴ Other studies have also reported a lower risk of ESKD among females with CKD stage 3⁶³⁵ and a shorter time to RRT among males with CKD stages 4 and 5.⁶³⁶

One meta-analysis of 68 studies that included 11,345 persons with CKD has reported a higher rate of decline of kidney function in men,⁶³⁷ but another meta-analysis of individual-participant data from 11 randomized trials evaluating the efficacy of ACE inhibitor treatment in CKD did not show an increased risk of doubling the serum creatinine level or ESKD or ESKD alone in men.⁶³⁸ On the contrary, after adjustment for baseline variables including BP and urinary protein excretion, women had a significantly higher risk of these endpoints than men.⁶³⁸ One limitation of these studies is that the menopausal status of the women was often not documented. Interestingly, in one study, bilateral oophorectomy in 1653 premenopausal women was associated with a higher risk of incident CKD over a median of 14 years of observation when compared with age-matched women who did not undergo oophorectomy (adjusted HR, 1.42; 95% CI, 1.14–1.77; absolute risk increase, 6.6%).⁶³⁹

In general, the prevalence of hypertension and uncontrolled hypertension is higher among men. Men tend to consume more protein than women; the prevalence of dyslipidemias is greater in men than in premenopausal women. All these factors may contribute to the increased severity of kidney disease observed in men but do not explain all the differences.^{640,641} That hormonal effects are also important is indicated by Mendelian randomization studies that reported a higher risk of CKD associated with higher genetically predicted testosterone levels in men but not women⁶⁴² and a lower risk of CKD associated with higher genetically predicted sex hormone binding globulin (which reduces bioactivity of sex hormones) in men but not women.⁶⁴³ Reviews have also highlighted that disparities in access to health care between men and women in

many regions may contribute to the observed differences in outcomes.^{644,645} The impact of sex differences on the epidemiology of kidney disease is reviewed in more detail in Chapter 22.

NEPHRON ENDOWMENT

Experimental and clinical studies have shown that the number of nephrons per kidney is variable and may be influenced by several factors during development in utero. Furthermore, low nephron endowment predisposes individuals to hypertension and CKD. This has been confirmed in studies using a rat model of spontaneous kidney agenesis. Rats born with a single kidney had 19% fewer nephrons per kidney than their two-kidney littermates, resulting in a 60.2% reduction in nephron endowment that was associated with subsequent kidney and glomerular hypertrophy, proteinuria, glomerular sclerosis, and TIF.³⁵⁰ It has been proposed that reduced nephron endowment results in an increase in the single-nephron GFR and therefore a reduction in kidney reserve.⁶⁴⁶ Whereas the glomerular hemodynamic changes associated with mild to moderate congenital nephron deficiencies may not in themselves be sufficient to provoke kidney injury, they could be predicted to compound the effects of an acquired nephron loss and predispose an individual to progressive kidney damage. Thus CKD should be viewed as a multihit process, in which the first hit may be reduced nephron endowment.⁶⁴⁷ The impact of developmental programming on nephron endowment, BP, and kidney function is discussed in detail in Chapter 20.

RACE AND ETHNICITY

Data from the USRDS have consistently shown a substantially higher incidence of ESKF among Black Americans, Hispanics, and Indigenous Americans versus White Americans. In 2022, the adjusted ESKF incidence was 994 per million population (pmp) in Black Americans, 572 pmp in Indigenous Americans, 523 pmp in Hispanic Americans, 364 pmp in Asian Americans, and 256 in White Americans (Fig. 50.11).⁶⁴⁸ Analysis of long-term trends showed that between 2002 and 2019 there was no change in the white population but there was a 20.6% decrease in incidence observed in the Black population. In 2020–2022 the disparity between Black and White populations tended to increase again, a trend attributed to the impact of the COVID-19 pandemic. A similar trend was observed when comparing the trends for Indigenous versus White Americans.⁶⁴⁸ Analysis of longer-term trends in ESKD incidence from 1980 to 2019 observed that in the period 1980 to 1993 the growth in incidence was higher in Black than White Americans but that subsequently the incidence declined in Black Americans while it continued to increase in White Americans.⁶⁴⁹

The reasons for these obvious discrepancies are complex and include both social and biological factors.^{650,651} Interestingly, data from the Reasons for Geographic and Racial Differences in Stroke (REGARDS) Cohort Study have shown a lower prevalence of eGFR 50 to 59 mL/min/ 1.73 m 2 , among Black American versus White subjects but a higher prevalence of eGFR, 10 to 19 mL/min/ 1.73 m 2 , suggesting

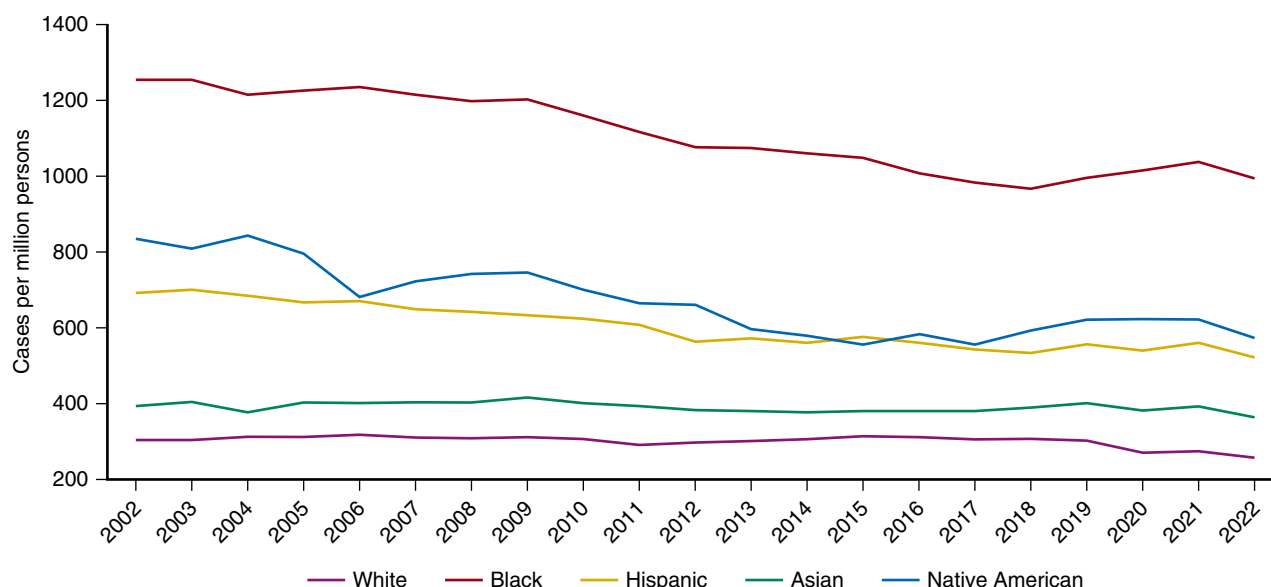


Fig. 50.11 Trends in adjusted incidence rate of end-stage kidney disease (ESKD), by race, in the U.S. population, 2002–2022. (From the U.S. Renal Data System. Annual Data Report 2024, Chapter 1: Incidence, Prevalence, Patient Characteristics, and Treatment Modalities.)

that Black Americans have a lower risk of developing CKD but a higher risk of progression of CKD to ESKF.⁶⁵² Analysis of data from 547,188 U.S. veterans has found that Black participants developed CKD on average 7.8 years younger than White participants. Consistent with other data, Black Americans evidenced a 2.5-fold higher cumulative incidence of ESKF but importantly had a 17% to 48% lower hazard of the competing risk of death. The authors conclude that these discrepancies were largely explained by the younger age at onset of CKD in Black Americans, emphasizing the need for early detection and treatment.⁶⁵³

There are substantial racial discrepancies in the incidence of ESKF due to hypertensive and diabetic nephropathies. One longitudinal study that examined data from 1,306,825 Medicare beneficiaries has reported substantially increased risks of developing ESKF in Black American versus Caucasian participants in all categories—among persons with diabetes, a 2.4- to 2.7-fold increase; among hypertensive persons, a 2.5- to 2.9-fold increase; and among persons with neither hypertension nor diabetes, a 3.5-fold increase.⁶⁵⁴ MDRD study data showed the prevalence of hypertension to be higher in Black Americans versus Caucasians among persons with CKD, despite a higher mean GFR in the Black American participants.⁶⁴¹ Hypertensive persons were found to have had more rapid progression of kidney disease before entry into the study, suggesting that the higher prevalence of hypertension in Black Americans is likely to be a significant contributor to the accelerated progression of CKD. On the other hand, both higher MAP and Black American race/ethnicity were independent predictors of a faster decline in the GFR in the MDRD study.⁶⁵⁵

In a large community-based epidemiologic study, Black Americans were found to have a 5.6 times higher unadjusted incidence of hypertensive ESKF.⁶⁵⁶ This increased incidence was directly related to the prevalence of hypertension, severe hypertension, and diabetes in the study population and was inversely related to age at diagnosis of hypertension and socioeconomic status. After adjustment for these factors,

the risk of hypertensive ESKF remained 4.5 times greater among Black Americans compared with White Americans, providing further evidence that Black Americans have an increased susceptibility to kidney disease beyond that attributable to their increased prevalence of hypertension and diabetes. Salt-sensitive hypertension, in particular, is more prevalent in the Black American population than in the White population.⁶⁵⁷ Comparing kidney responses to a high sodium intake in salt-sensitive versus salt-resistant persons, RBF was found to decrease in the presence of an increased filtration fraction (implying an increased P_{GC}) in salt-sensitive persons, whereas the converse occurred in salt-resistant persons.⁶⁵⁸

These observations are consistent with the notion that salt loading injures the glomerulus through glomerular capillary hypertension and that salt-sensitive individuals, and Black American subjects in particular, are at added risk of this form of injury. The incidence of ESKF due to diabetic nephropathy is fourfold higher in Black Americans than among White Americans.⁶⁵⁹ It is notable that after controlling for the higher prevalence of diabetes and hypertension, as well as age, socioeconomic status, and access to health care, the excess incidence of ESKF due to diabetes in Black versus White Americans was confined to type 2 diabetes.⁶⁶⁰ Among persons with type 1 diabetes, Black Americans were not found to be at higher risk than White Americans. Indeed, most Black Americans with diabetic ESKF (77%) had type 2 diabetes, whereas most White Americans with diabetic ESKF (58%) had type 1 diabetes.⁶⁶¹ Black American race/ethnicity was also found to be associated with a threefold higher risk of early kidney function decline (increase in serum creatinine of ≥ 0.4 mg/dL) among adults with diabetes.⁶⁶²

Several potential factors contributing to the different prevalence and severity of kidney disease among population groups have been analyzed. Adjustment for socioeconomic factors reduces but does not eliminate, the increased risk of Black Americans to develop

ESKF.^{651,659,662} Black Americans generally have lower birth weights than their White counterparts and may therefore have programmed or genetically determined deficits in nephron number, rendering them more susceptible to hypertension and subsequent ESKF.^{663,664} Finally, 40% of Black Americans with hypertensive ESKF and 35% with type 2 diabetes-associated ESKF have a first-, second-, or third-degree relative with ESKF, implying a strong familial susceptibility to ESKF and therefore a genetic predisposition.⁶⁶⁵

Evidence of a genetic explanation for the high incidence of ESKF observed in Black Americans was provided by research that identified a strong association between ESKF and two coding variants of the gene for apolipoprotein L1 (*APOL1*).⁶⁶⁶ These gene variants confer resistance to infection with *Trypanosoma brucei rhodesiense*, which causes sleeping sickness, providing an explanation for how selection likely resulted in a high prevalence of these variants in the population.⁶⁶⁷ Subsequent studies have identified associations between *APOL1* risk variants and several kidney pathologies including FSGS, HIV-associated nephropathy (HIVAN), sickle cell kidney disease, and severe lupus nephritis. *APOL1* risk variants have also been associated with HIVAN in black South Africans.⁶⁶⁸ Moreover, cohort studies have reported associations between *APOL1* risk variants and the risk of progression to ESKD. The risk of progression was the lowest in European Americans (with no risk variants), intermediate in Black Americans with no or one risk variant, and highest in Black Americans with two risk variants.⁶⁶⁹ It is estimated that *APOL1* variants account for 40% of disease burden due to CKD in Black Americans.

The biological role of *APOL1* in the progression of CKD remains to be fully elucidated. Expression of the *APOL1* risk variants in divergent species, *Drosophila* and *Saccharomyces* (a yeast), has identified impairment of conserved core intracellular endosomal trafficking processes as a key mechanism of cellular injury.⁶⁷⁰ The *APOL1* protein is expressed in the kidney but is also secreted and bound to circulating HDL particles. Current evidence suggests that it is the locally expressed form of *APOL1* that is involved in CKD pathogenesis.⁶⁷¹ In a mouse model, expression of inducible *APOL1* gene risk variants in podocytes provokes kidney disease similar to that observed in humans, suggesting a possible mechanism.⁶⁷² Further studies have demonstrated that the GI *APOL1* variant initiates podocyte damage by causing abnormal sodium influx and potassium efflux in podocytes, an effect that was prevented by an *APOL1* protein blocker.⁶⁷³ Despite the strong association between the inheritance of two *APOL1* risk variants and ESKD, only a minority of people with this genotype actually develop kidney disease, suggesting that the action of a second factor is required to cause disease in genetically susceptible individuals.⁶⁷⁴ HIV is one example of such a second hit, but it has been proposed that other viruses and other gene variants may also be important.⁶⁶⁷ Exposure to air pollution may also serve as a second hit.⁶⁷⁵

Other ethnic and racial minority groups including Asians,^{583,676} Hispanics,⁶⁷⁷ Indigenous Americans,⁶⁷⁸ Mexican Americans,⁶⁷⁹ and Indigenous Australians⁶⁸⁰ have also been found to be at increased risk of developing

CKD and ESKF. See [Chapter 18](#) for a further discussion of ethnicity and the epidemiology of CKD and [Chapter 44](#) for a detailed discussion of genetic factors in the pathogenesis of CKD.

OBESEITY AND METABOLIC SYNDROME

Obesity may directly cause a glomerulopathy characterized by proteinuria and histologic features of focal and segmental glomerulosclerosis,^{681,682} but it is likely that it also exacerbates the progression of other forms of CKD. Micropuncture studies have confirmed that obesity is another cause of glomerular hypertension and hyperfiltration that may contribute to the progression of CKD.^{290,683} Griffin and colleagues⁶⁸⁴ have pointed out that whereas obesity is widespread, only a minority of obese individuals develop obesity-related glomerulopathy. They have proposed that low nephron endowment (associated with low birth weight, which is also associated with an increased risk of later life obesity) or acquired nephron loss constitute a necessary additional factor that increases glomerular hypertrophy and preglomerular vasodilation and transmission of elevated systemic BP to the glomerulus, leading to glomerulosclerosis. Additional obesity-related mechanisms that may exacerbate CKD progression include proinflammatory cytokine production by adipocytes,^{685,686} increased production of aldosterone, and expanded extracellular volume.⁶⁸⁷

In humans, severe obesity is associated with increased renal plasma flow, glomerular hyperfiltration, and albuminuria, abnormalities that are reversed by weight loss.⁶⁸⁸ Several large population-based studies have identified obesity as an independent risk factor for developing CKD.^{577,689,690} Change in body weight has also been identified as a risk factor for incident CKD. In one study of 8792 previously healthy men, an increase in body weight of 0.75 or more kg/year (and a decrease of <0.75 kg/year) was associated with an increased risk of developing CKD in previously obese and nonobese subjects.⁶⁹¹

The metabolic syndrome (insulin resistance), defined by the presence of abdominal obesity, dyslipidemia, hypertension, and fasting hyperglycemia, is frequently associated with obesity and is also associated with an increased risk of developing CKD.⁶⁹²⁻⁶⁹⁵ The waist-hip ratio, a marker of central fat distribution and insulin resistance, was also independently associated with impaired kidney function, even in lean individuals (BMI <25 kg/m²) among a population-based cohort of 7676 subjects.^{696,697}

Whether obesity confers increased risk of developing CKD in the absence of metabolic syndrome or other comorbid conditions was previously uncertain. In the Framingham Heart Study, an increased risk of developing CKD stage 3 associated with obesity was no longer significant after adjustment for known cardiovascular risk factors, though the increased risk of incident proteinuria persisted in multivariable models.⁶⁹⁸ Analysis of data from 281,228 participants in the U.K. Biobank study using conventional logistic regression analysis and Mendelian randomization identified that increasing BMI and waist-hip ratio were each associated with a higher risk of CKD but that this association was largely attributable to the associated increase in hypertension and diabetes.⁶⁹⁹ On the other hand, a meta-analysis that included 4,965,285 participants from 16

studies reported that in “healthy” (defined as fewer than 3 components of the metabolic syndrome) persons with obesity and overweight evidenced an increased risk of CKD (RR 1.47; 95% CI, 1.31–1.65 for obesity; RR 1.29; 95% CI, 1.27–1.32 vs. controls with healthy weight), suggesting that obesity per se does confer an increased risk of developing CKD.⁷⁰⁰

The association of obesity with progression of established CKD has been less well established. In community-based studies, higher BMI has been reported to be an independent risk factor for ESKF,^{690,701} but longitudinal studies of CKD have reported less consistent findings. Higher BMI was an independent predictor of CKD progression among 162 persons with IgA nephropathy⁷⁰² and in a cohort of persons with predominantly CKD stage 3,⁷⁰³ but not in a cohort with CKD stages 4 and 5.⁶³⁶ In the CRIC study, higher BMI was paradoxically associated with a lower risk of CKD progression,⁷⁰⁴ and in a combined analysis of data from seven large CKD cohort studies, BMI was significantly associated with CKD progression in only three studies and with ESKD in only two.⁷⁰⁵

Nevertheless, studies of the impact of weight loss interventions on CKD progression support that weight loss may be an important intervention for achieving kidney protection. Sustained weight loss is notoriously difficult to achieve, but a meta-analysis of several small, short-duration studies has found that weight loss achieved through diet or medication is associated with a reduction in proteinuria and BP. Surgical procedures to achieve weight loss in the morbidly obese were associated with normalization of glomerular hyperfiltration and reductions in microalbuminuria and BP.⁷⁰⁶ One systematic review has analyzed the effects of weight loss achieved by bariatric surgery, medication, or diet in 31 studies and found that in most studies, weight loss is associated with reductions in proteinuria. In people with glomerular hyperfiltration, the GFR tended to decrease with weight loss and, in those with a reduced GFR, it tended to increase.⁷⁰⁷

The increasing use of bariatric surgery to manage obesity has allowed investigation of the impact of surgically induced weight loss on CKD. In a long-term study of 2144 persons who underwent bariatric surgery, change in CKD risk category, as defined by the KDIGO classification, was assessed after 7 years. Among those with moderate risk at baseline, the risk category improved in 53% and deteriorated in 5% to 8%. In the high-risk group, improvement was observed in 56% and deterioration in 3% to 10%, and, in the very-high-risk group, 23% improved. The eGFR initially improved, with a peak at 2 years, and then gradually declined. Albuminuria showed large and sustained decreases in the moderate- (median urine ACR decreased from 48 to 14 mg/g) and high-risk groups (median urine ACR decreased from 326 to 26 mg/g).⁷⁰⁸

In another study comparing outcomes in persons with CKD stages 3 and 4 undergoing bariatric surgery with similarly obese propensity-matched persons who did not undergo surgery, the eGFR was significantly higher in the surgery group after 3 years by a mean of 9.84 mL/min/1.73 m².⁷⁰⁹ In a meta-analysis that included 23 cohort studies and 3015 persons who underwent all forms of bariatric surgery, small but statistically significant improvements were observed in serum creatinine levels (mean decrease, 0.08 mg/dL) and

proteinuria (mean decrease, 0.04 g/day).⁷¹⁰ Treatment with GLP-1RAs offers a novel intervention for achieving weight loss and kidney protection, though not all the benefit is attributable to weight loss (see earlier discussion).

DYSLIPIDEMIA

Moorhead and colleagues⁷¹¹ advanced the hypothesis that abnormalities in lipid metabolism may contribute to the progression of CKD. Glomerular injury, accompanied by an alteration in basement membrane permeability, was envisaged as the initiator of a vicious cycle of hyperlipidemia and progressive glomerular injury. They proposed that urinary losses of albumin and lipoprotein lipase activators result in an increase in circulating LDLs, which in turn bind to the glomerular basement membrane, further impairing its permselectivity. Filtered lipoproteins accumulate in the mesangium, stimulating extracellular matrix synthesis and mesangial cell proliferation; filtered LDL is taken up and metabolized by the tubules, leading to cell injury and interstitial disease. Notably, this hypothesis did not propose hyperlipidemia as an initiating factor in kidney injury but rather as a participant in a self-sustaining mechanism of disease progression.

Several lines of experimental evidence have confirmed the association between dyslipidemia and kidney damage. Both intact and uninephrectomized rats with dietary-induced hypercholesterolemia developed more extensive glomerulosclerosis than normocholesterolemic controls, and the severity of glomerulosclerosis correlated with serum cholesterol levels⁷¹²; aging female Nagase albuminemic rats have endogenous hypertriglyceridemia and hypercholesterolemia and develop proteinuria and glomerulosclerosis by 9 and 18 months of age, respectively, whereas male Nagase albuminemic rats have lower lipid levels and no glomerulosclerosis by 22 months of age.⁶²⁵ Interestingly, ovariectomy in female Nagase albuminemic rats lowers triglyceride levels and reduces their kidney injury. In seeming contradiction, however, young and aging male Sprague-Dawley rats developed more extensive glomerulosclerosis than age- and sex-matched Nagase albuminemic rats, despite higher cholesterol levels in the NARs.⁷¹³ Triglyceride levels, however, were lower in the Nagase albuminemic rats, again suggesting an independent role for triglycerides in lipid-mediated kidney injury.

Whereas data regarding the role of lipids in initiating kidney disease are conflicting, several experiments have supported the notion that dyslipidemia may promote kidney damage. Cholesterol feeding has been shown to exacerbate glomerulosclerosis in uninephrectomized rats, prediabetic rabbits, rats with puromycin aminonucleoside nephropathy, and in the unclipped kidney of rats with two-kidney, one-clip hypertension. When hypertension and dyslipidemia are superimposed, a synergistic effect that dramatically accelerates kidney functional deterioration is observed.^{714,715} In the 5/6 nephrectomy model, progressive kidney damage is associated with kidney tissue accumulation of lipids, as well as upregulation of pathways involved in the tubular reabsorption of protein-bound lipids and downregulation of pathways involved in lipid catabolism.^{716,717}

In humans, the role of lipids in the initiation and progression of CKD remains unclear. At autopsy, a highly significant correlation was found between the presence of systemic atherosclerosis and the percentage of sclerotic glomeruli in normal individuals, fostering speculation that the development of glomerulosclerosis may be analogous to that of atherosclerosis.⁷¹⁸ Furthermore, dyslipidemia has been identified in several large studies as a risk factor for the subsequent development of CKD in apparently healthy individuals.^{577,719,720} The common forms of primary hypercholesterolemia are not associated with an increased incidence of CKD in the general population, but kidney damage has been described in association with rare inherited disorders of lipoprotein metabolism.^{721,722} In addition, in a study of 631 routine kidney biopsies, lipid deposits were detected in nonsclerotic glomeruli in 8.4% of kidneys, and staining for apolipoprotein B was positive in approximately 25% of biopsies, suggesting that lipid deposition may occur in diverse kidney diseases.⁷²³

Some epidemiologic studies have found a strong association between CKD progression and dyslipidemia,^{655,724-728} but others have not.⁷²⁹⁻⁷³² Interpretation of these data is complicated by the fact that in persons with kidney insufficiency, dyslipidemias do not occur in isolation and are associated with other factors that also affect kidney disease progression including hypertension, hyperglycemia, and proteinuria.

The mechanisms whereby hyperlipidemia may contribute to kidney injury have not been fully elucidated. Cholesterol feeding has been associated with an increase in mesangial lipid content,⁷¹² glomerular macrophages, and TGF- β , as well as fibronectin mRNA levels.^{733,734} Furthermore, reduction of glomerular macrophages by whole-body x-irradiation in the setting of nephrotic syndrome significantly reduced albuminuria without affecting serum lipid levels, indicating that macrophages play a central role in hyperlipidemic glomerular damage.⁷³⁴ Mesangial cells express receptors for LDL, and uptake is stimulated by vasoconstrictor and mitogenic peptides such as ET-1 and PDGF.⁷³⁵ Metabolism of LDL by mesangial cells leads to increased synthesis of fibronectin and MCP-1, which may contribute to mesangial matrix expansion and the recruitment of circulating macrophage/monocytes into the glomerulus.⁷³⁶ Moreover, triglyceride-rich lipoproteins (very-low-density lipoprotein [VLDL] and intermediate-density lipoprotein [IDL]) induce mesangial cell proliferation and elaboration of IL-6, PDGF, and TGF- β in vitro.⁷³⁷ Mesangial cells, macrophages, and renal tubule cells all have the capacity to oxidize LDL via the formation of ROS, a step that may be inhibited by antioxidants and HDL.⁷³⁸⁻⁷⁴⁰ Oxidized LDLs may induce dose-dependent mesangial cell proliferation or mesangial cell death, as well as production of TNF- α , eicosanoids, monocyte chemotaxins, and glomerular vasoconstriction. These pathways, together with free radicals generated during LDL oxidation, may each contribute to kidney inflammation and injury.^{737,739}

Hyperlipidemia is also associated with elevated P_{GC} , raising the possibility of a further pathway to glomerulosclerosis via hemodynamic injury.⁷¹² The elevated P_{GC} appears to be mediated, in part, by an increase in renal vascular resistance that occurs in the context of increased plasma viscosity. In diabetic persons, circulating Ang II levels have been found

to correlate with serum cholesterol,⁷⁴¹ and both oxidized LDLs and lipoprotein(a) have been shown to stimulate renin production by juxtaglomerular cells in vitro.⁷⁴⁰ Moreover, oxidized LDL has been found to reduce nitric oxide synthesis by endothelial cells,⁷⁴⁰ raising the possibility that alterations in the activity of the RAAS and nitric oxide metabolism could also contribute to the increase in P_{GC} observed with hyperlipidemia.

It would follow that if hyperlipidemia exacerbates kidney damage, interventions that lower serum lipid levels should ameliorate CKD progression. Treatment with a 3-hydroxyl-3-methylglutaryl coenzyme A (HMG-CoA) reductase inhibitor (statin) or clofibrate in the obese Zucker rat, a strain with endogenous hyperlipidemia and spontaneous glomerulosclerosis, and 5/6 nephrectomized rats, which develop hyperlipidemia secondary to kidney insufficiency, resulted in lowering of serum lipid levels, reduction in albuminuria, reduction in mesangial cell DNA synthesis, and attenuation of glomerulosclerosis, despite a lack of effect on either systemic BP or P_{GC} .^{13,742} Indeed, statin treatment resulted in additional lowering of proteinuria, regression of glomerulosclerosis, normalization of podocyte number, and abrogation of tubulointerstitial injury when added to combination of ACE inhibitor and ARB treatment.⁷⁴³ In rats in the nephrotic phase of PAN, statin treatment resulted in reduction of albuminuria and serum cholesterol levels, reduction of MCP-1 mRNA expression, and a 77% reduction in glomerular macrophage accumulation.⁷⁴⁴ The statins may therefore exert beneficial effects on kidney disease progression by not only reducing serum lipid levels but also inhibiting mesangial cell proliferation and mechanisms for the recruitment of macrophages due to the decreased expression of chemotactic factors and cell adhesion molecules.⁷⁴⁵

Niacin treatment after 5/6 nephrectomy resulted in lower BP, less proteinuria, less kidney tissue accumulation of lipids, and attenuation of tubulointerstitial injury. This indicates that lipid lowering through strategies other than statins may also be kidney protective.⁷⁴⁶

Despite evidence of potential benefit in animal experiments, evidence of kidney protection from dietary or pharmacologic lowering of serum lipids levels in humans is limited. A meta-analysis of 13 small studies that included both diabetic and nondiabetic kidney disease found that lipid-lowering therapy significantly reduced the rate of GFR decline (mean reduction of 1.9 mL/min/1.73 m² per year).⁷⁴⁷ In a further meta-analysis of randomized statin studies in persons with CKD, a relatively small subgroup with kidney endpoints available evidenced a reduction in proteinuria with statin treatment but no improvement in creatinine clearance.⁷⁴⁸ Whereas these limited kidney-protective effects were associated with cholesterol lowering, it is possible that they may also be due to the direct pleiotropic effects of statins, a notion supported by the observation that lipid lowering with fibrates was not associated with preservation of kidney function.^{749,750}

The Study of Heart and Renal Protection (SHARP) investigated the cardiovascular- and kidney-protective effects of lipid lowering with simvastatin and ezetimibe in 9438 subjects with CKD and ESKD.⁷⁵¹ Whereas the treatment arm showed a mean reduction of 43 mg/dL in LDL cholesterol and a 17% reduction in major atherosclerotic events, no

significant effect was observed on the incidence of the kidney endpoints ESKF (risk ratio, 0.97; 95% CI, 0.89–1.05) and ESKF or creatinine doubling (risk ratio, 0.93; 95% CI, 0.86–1.01). It should be noted, however, that the subjects with CKD had relatively advanced disease (mean eGFR, 27 ± 13 mL/min/1.73 m²), and these observations therefore do not exclude the possibility that lipid lowering may have kidney-protective effects in less advanced CKD. However, a further meta-analysis of 38 studies that included 37,274 participants with CKD has found that statin therapy is associated with a reduction in mortality and cardiovascular events but no clear effect on CKD progression.⁷⁵²

Inhibitors of the enzyme, proprotein convertase subtilisin/kexin type 9 (PCSK9), a key regulator of hepatic LDL receptor recycling, have been developed as novel potent treatments to lower cholesterol levels in persons with familial hyperlipidemias or those resistant to treatment with statins and represent new tools to further investigate the impact of lipid lowering on CKD progression. However, current indirect evidence does not indicate kidney-protective effects. A pooled analysis of participants with CKD (eGFR, 30–59 mL/min/1.73 m²; mean eGFR, 51 mL/min/1.73 m²) in eight trials evaluating the monoclonal PCSK9 antibody, alirocumab, reported substantial reductions in LDL cholesterol and triglyceride levels, as well as other lipids versus placebo or ezetimibe but no difference in eGFR at 24 or 104 weeks.⁷⁵³ Subgroup analysis in a large trial of evolocumab similarly confirmed that PCSK9 antibody treatment was effective in lowering cholesterol and reducing cardiovascular events in persons with CKD, but there was no difference in GFR decline between randomized groups.⁷⁵⁴ Clinical trials with primary kidney endpoints in persons with CKD are needed.

At present there is insufficient evidence to recommend statin treatment specifically for kidney protection, but statin treatment is recommended for all persons with CKD aged 50 years or older to reduce cardiovascular risk.⁴⁹⁸

CALCIUM AND PHOSPHATE METABOLISM

Alterations in calcium and phosphate metabolism that result from a declining kidney function may also contribute to progressive kidney damage.

Uninephrectomized rodents fed a high-phosphate diet developed renal calcium and phosphate deposition and tubulointerstitial injury.^{246,755} In a similar experiment, treatment with 3-phosphocitrate, an inhibitor of calcium phosphate deposition, led to a significant reduction in kidney damage compared with controls, indicating that calcium phosphate deposition in the kidney occurs during the evolution of kidney damage and may exacerbate nephron loss.⁷⁵⁶ In clinical studies, serum phosphate has been identified as an independent risk factor for CKD progression⁷⁵⁷ that may also attenuate the efficacy of ACE inhibitor treatment in the context of proteinuric nephropathies.⁷⁵⁸ Phosphate excess therefore does appear to have some intrinsic nephrotoxicity that is enhanced in the setting of reduced nephron number. In animal experiments and small human studies, dietary phosphate restriction or treatment with oral phosphate binders has been associated with reductions in proteinuria and glomerulosclerosis and attenuation of disease progression, as well as prevention of

hyperparathyroidism.⁷⁵⁹⁻⁷⁶² Dietary phosphate restriction, however, almost inevitably also imposes dietary protein restriction. It is therefore not clear whether the benefit was derived directly from the reduced phosphate intake or indirectly from protein restriction. One study in humans has reported additional kidney protection when phosphate restriction is superimposed on protein restriction.⁷⁶³

Elevated PTH levels may also impact kidney damage. Podocytes express a unique transcript of PTH receptor, and PTH has been shown to have several effects on the kidney including decreasing SNGFR (without change in Q_A , P_{GC} , or ΔP), lowering K_F and stimulating renin production.⁷⁶² Furthermore, increased PTH levels may exacerbate kidney damage through effects on BP,⁷⁶⁴ glucose intolerance, and lipid metabolism.^{765,766} Two experimental studies have provided evidence that elevated PTH may contribute to CKD progression. In rats fed a high-protein diet after 5/6 nephrectomy, parathyroidectomy improved survival, limited kidney hypertrophy, and calcium content and attenuated the rise in serum creatinine.⁷⁶⁷ In the other, calcimimetic treatment and parathyroidectomy after 5/6 nephrectomy each abrogated TIF and glomerulosclerosis.⁷⁶⁸

Vitamin D, normally 1-hydroxylated in the kidney and therefore reduced in CKD, has several potentially beneficial effects on the kidney. Several experiments have reported amelioration of kidney damage in rats treated with 1,25(OH)₂D₃ or vitamin D analog after 5/6 nephrectomy.⁷⁶⁹⁻⁷⁷¹ A further study found that 1,25(OH)₂D₃ treatment also preserved podocyte number, volume, and structure after 5/6 nephrectomy.³³⁴ In other experimental models, vitamin D or vitamin D analogs have been shown to abrogate interstitial inflammation by promoting the sequestration of NF-κB signaling,⁷⁷² inhibit kidney hypertrophy after uninephrectomy,⁷⁷³ reduce renin^{762,774} and TGF-β expression,⁷⁷¹ and restore glomerular filtration barrier structure, as well as slit diaphragm protein expression.⁷⁷⁴ Several small trials reported reductions in proteinuria following treatment with vitamin D analog and paricalcitol in diabetic⁷⁷⁵ and nondiabetic CKD,^{776,777} but larger trials have not been published.

Fibroblast growth factor 23 (FGF23) has been identified as a key regulator of the bone mineral and vitamin D changes observed in CKD and may also contribute to CKD progression, as well as mediate some of the adverse cardiovascular consequences associated with CKD. It is produced by osteoblasts and osteocytes; levels rise early during CKD. FGF23 production is stimulated chiefly by 1,25(OH)₂D₃ and dietary phosphate intake.⁷⁷⁸ Its chief actions are to reduce phosphate reabsorption in the proximal tubule by downregulating sodium phosphate cotransporters and reduce intestinal phosphate absorption by reducing 1,25(OH)₂D₃ levels by inhibiting renal 25(OH)D₃ 1α-hydroxylase and stimulating the catabolic 25(OH)D₃ 24-hydroxylase.⁷⁷⁸ Thus decreased phosphate excretion early in the course of CKD stimulates FGF23 production, which increases phosphate excretion to prevent hyperphosphatemia. This response is achieved at the expense of low 1,25(OH)₂D₃ levels, which in turn facilitate the development of secondary hyperparathyroidism.^{254,779} In addition to its role in bone mineral metabolism, longitudinal studies have identified FGF23 as an independent risk factor for mortality in persons receiving hemodialysis^{780,781} and in

persons with earlier stage CKD,⁷⁸² although one group has suggested that the relationship may not be causal.⁷⁸³ Several studies have also identified FGF23 as an independent risk factor for CKD progression in persons with diabetes⁷⁸⁴ and without diabetes,⁷⁸⁵ including Black Americans⁷⁸⁶ and children with CKD.⁷⁸⁷ Overexpression of mutant FGF23 in the Thy1 model of glomerulonephritis resulted in lowering of serum phosphate levels and amelioration of glomerulosclerosis,⁷⁸⁸ but whether FGF23 contributes directly to CKD progression is unclear. Unfortunately, a randomized trial of the phosphate binder, lanthanum carbonate, and nicotinamide, an inhibitor of intestinal phosphate absorption, in persons with CKD stages 3b and 4, designed to investigate the benefits of lowering phosphate and FGF23, failed to achieve a reduction in either.⁷⁸⁹ CKD Mineral Bone Disorder is discussed in more detail in Chapter 52.

ANEMIA

Anemia is a frequent consequence of CKD but may also influence its progression. Both acute and chronic anemia are associated with reversible increases in renal vascular resistance and a normal or reduced filtration fraction in animals and humans. Conversely, an increase in hematocrit is associated with an increase in filtration fraction. Thus hematocrit may influence renal hemodynamics and thereby affect the rate of progression of CKD. The effects of anemia on glomerular hemodynamics have been studied in rats subjected to 5/6 nephrectomy, DOCA-salt hypertension, and diabetes.⁷⁹⁰⁻⁷⁹² Irrespective of the model, anemia was associated with significant amelioration of glomerulosclerosis and a reduction in P_{GC} .

Reduced P_{GC} resulted predominantly from reductions in efferent arteriolar resistance in rats with kidney ablation, lowered SBP in DOCA-salt rats, and increased afferent arteriolar resistance in diabetic rats. Similarly, in the MWF/Ztm rat, which develops spontaneous glomerulosclerosis with age, anemia induced by dietary iron deficiency was associated with lower BP, reduced urinary protein excretion, and less extensive glomerulosclerosis compared with controls fed diets of normal iron content.⁷⁹³ In contrast, prevention of anemia by the administration of erythropoietin to rats with remnant kidneys to maintain a normal hematocrit resulted in increased systemic and glomerular BPs, as well as markedly increased glomerulosclerosis.⁷⁹⁰ In another apparently contradictory study, treatment with epoetin delta after 5/6 nephrectomy was associated with slower rates of decline in kidney function, decreased kidney fibrosis, and less interstitial macrophage accumulation.⁷⁹⁴ Interestingly, these effects were observed at subhemopoietic doses, indicating that they may have resulted from direct actions of the epoetin rather than anemia correction.

Despite the apparently favorable hemodynamic effects of anemia in experimental models of CKD, human studies have suggested that anemia may accelerate CKD progression. In persons with inherited hemoglobinopathies, chronic anemia is associated with glomerular hyperfiltration that eventuates in proteinuria, hypertension, and ESKD.^{795,796} Furthermore, reduced hemoglobin was an independent predictor of increased risk of developing ESKD in several longitudinal studies.⁷⁹⁷⁻⁸⁰⁰

Two small randomized studies have reported a kidney-protective benefit when anemia was corrected with erythropoietin.^{801,802} On the other hand, two other studies that had effects on left ventricular mass as their primary endpoint,^{803,804} as well as the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT),⁸⁰⁵ found no effect of high versus low hemoglobin target on rate of decline in the GFR. In the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, randomization to a higher hemoglobin target (13–15 mg/dL) was associated with a shorter time to initiation of dialysis than the lower target (10.5–11.5 mg/dL).⁸⁰⁶

The reasons for the apparent contradiction between the beneficial hemodynamic effects of anemia in experimental models and the identification of anemia as a risk factor for CKD progression in clinical studies are unknown. It is possible that the benefit of the hemodynamic effects is outweighed by other factors, such as increased renal hypoxia and ROS formation, which may contribute to progressive kidney damage.⁸⁰⁷ Nevertheless, several studies have indicated that normalization of hemoglobin in CKD may be associated with serious adverse effects including increased risk of stroke⁸⁰⁵ and death.⁸⁰⁸ Issues related to the treatment of anemia in CKD are discussed further in Chapter 53.

TOBACCO SMOKING

Smoking produces acute sympathetic nervous system activation resulting in tachycardia and an increase in SBP of up to 21 mm Hg.⁸⁰⁹ Vasoconstriction occurs in several vascular beds including the kidneys. Among healthy nonsmoking volunteers, acute exposure to cigarette smoke caused an 11% increase in renovascular resistance accompanied by a 15% reduction in the GFR and an 18% decrease in filtration fraction. These effects appear to be mediated, at least in part, by nicotine, because similar responses were observed after chewing nicotine gum.⁸¹⁰ The renal hemodynamic effects of smoking can be blocked by pretreatment with a β -blocker, indicating that β -adrenergic stimulation is also involved.⁸¹¹

The effects of chronic smoking on the normal kidney are less well defined. Renal plasma flow but not the GFR is reduced in chronic smokers, and plasma endothelin levels are elevated. In one population-based study, chronic smoking was associated with a small increase in creatinine clearance, implying that smoking may cause glomerular hyperfiltration.⁸¹² That these functional abnormalities may result in structural changes to blood vessels is suggested by the observation of abnormal intrarenal vasculature in smokers.^{813,814}

Epidemiologic studies have found smoking to be an important predictor of proteinuria in the general population.^{812,815-818} Furthermore, smoking has been identified as a significant risk factor for CKD^{577,819-821} and the development of ESKF.⁵⁷⁶

Several studies have reported that smoking is a risk factor for progression in a variety of forms of CKD including diabetic kidney disease,⁸²²⁻⁸²⁹ ADPKD, IgA nephropathy,⁸³⁰ lupus nephritis,⁸³¹ primary glomerulonephritis,⁸³² and general population cohorts.⁸³³⁻⁸³⁵ On the other hand, two large cohort studies of persons with CKD have

reported no association between smoking status and CKD progression.^{836,837}

Mechanisms whereby cigarette smoking may result in kidney damage include glomerular capillary hypertension, endothelial cell injury, direct tubulotoxicity,⁸³⁸ mesangial cell proliferation,⁸³⁹ glomerular hypertrophy, increased ROS generation,⁸⁴⁰ and sympathetic nervous system activation.⁸⁴¹ The kidney is therefore another organ that is adversely affected by smoking and smoking cessation may contribute to slowing the rate of progression of CKD.^{786,842}

ACUTE KIDNEY INJURY

A growing body of evidence has indicated that recovery from acute kidney injury (AKI) is associated with multiple adverse outcomes⁸⁴³ including a substantially increased risk of CKD. AKI superimposed on CKD has been proposed as a previously underappreciated mechanism for CKD progression. A meta-analysis of 82 studies that included 2,017,437 participants who survived an episode of AKI found that they were at substantially increased risk of new or progressive CKD (HR 2.67; 95% CI, 1.99–3.58), ESKD (HR 4.81; 95% CI, 3.04–7.62; 0.47), and death (HR 1.80; 95% CI, 1.61–2.02).⁸⁴⁴

Studies in animal models of AKI have identified failure normal healing characterized by persistent dedifferentiation of tubule cells as a key mechanism associated with progressive tubulointerstitial inflammation and fibrosis after AKI. After acute tubular necrosis, regeneration of tubules is achieved by dedifferentiation of remaining tubule cells, followed by proliferation to replace lost cells and redifferentiation. If this process fails, tubule cells experience cell-cycle arrest in the dedifferentiated state and continue to produce proinflammatory and profibrotic cytokines that drive progressive interstitial fibrosis. Activation of pericytes results in differentiation into myofibroblasts that contribute to fibrosis. The loss of pericytes contributes to loss of endothelial integrity and capillary rarefaction that exacerbates tissue hypoxia and fibrosis.^{845,846} The specific factors that provoke progressive kidney damage after AKI remain to be fully elucidated. Ferroptosis and necroptosis,

two forms of cell death that provoke inflammation, have been proposed to play an important role.⁸⁴⁷ It has been suggested that a single episode of AKI normally heals without progressive kidney damage but that repeated episodes of AKI, a single severe episode of AKI, or AKI superimposed on preexisting CKD may provoke the above mechanisms.⁸⁴⁸ The interaction between AKI and CKD progression has been demonstrated by a study of 39,805 persons with eGFR <45 mL/min/1.73 m² before hospitalization.⁸⁴⁹ Those who survived an episode of dialysis-requiring AKI had a high risk of developing ESKF within 30 days of hospital discharge—that is, nonrecovery of AKI that was related to preadmission eGFR. After adjustment for multiple risk factors, AKI was associated with a 30% increase in long-term risk for death or ESKF (adjusted HR, 1.30; 95% CI 1.04–1.64).

The interaction between AKI and CKD has been explored in multiple animal models.⁸⁵⁰ In rats subjected to kidney ischemia 2 weeks after 3/4 nephrectomy, uninephrectomy, or sham operation, ischemia after 3/4 nephrectomy was associated with a sustained increase in serum creatinine levels and more tubules that failed to redifferentiate, associated with more severe capillary rarefaction and TIF. Furthermore, rats that were initially normotensive after 3/4 nephrectomy developed hypertension and proteinuria at 2 to 4 weeks after ischemia.⁸⁵¹ The investigators proposed that loss of autoregulation results in greater transmission of elevated systemic BP to the glomerulus that exacerbates glomerular damage and contributes to CKD progression.⁸⁴⁸ In another animal model, investigators observed that moderate ischemia-reperfusion injury resulted in AKI with transient activation of Wnt/ β -catenin signaling and kidney recovery, but severe ischemia-reperfusion injury caused a sustained and exaggerated activation of Wnt/ β -catenin signaling associated with progressive kidney fibrosis.⁸⁵² Moreover, overexpression of Wnt1 accelerated the progression of AKI to CKD, and blockade of Wnt/ β -catenin ameliorated the progression of AKI to CKD. Proposed interactions between mechanisms of kidney injury post-AKI and mechanisms of CKD progression are illustrated in Fig. 50.12. AKI is discussed in more detail in Chapters 27 and 28.

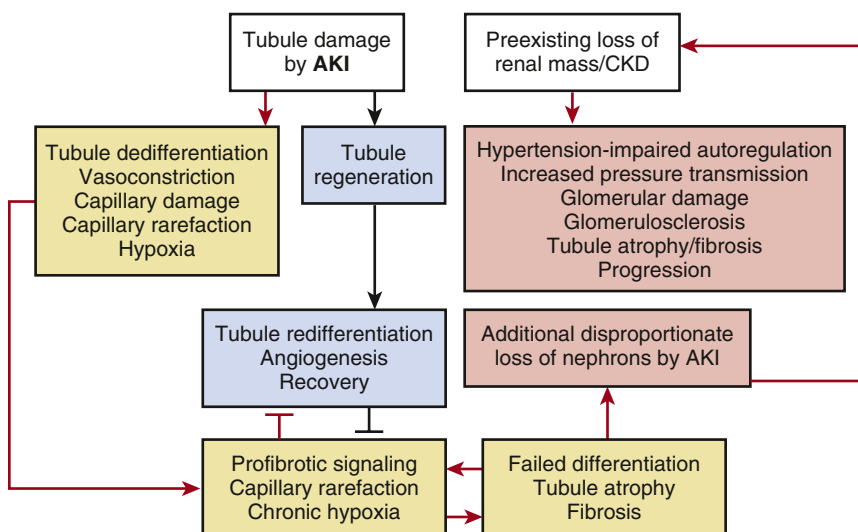


Fig. 50.12 Failed tubule differentiation and loss of renal mass after acute kidney injury (AKI) lead to hemodynamic abnormalities that cause chronic kidney disease (CKD) progression. This schematic diagram illustrates the effects of AKI that lead to tubulointerstitial fibrosis, the renal mass reduction that retards recovery of tubules regenerating after AKI, and the resulting disproportionate further reduction of renal mass that triggers hemodynamic mechanisms of renal disease progression. (From Venkatachalam MA, Weinberg JM, Kriz W, Bidani AK. Failed tubule recovery, AKI-CKD transition, and kidney disease progression. *J Am Soc Nephrol*. 2015;26(8):1765–1776.)

FUTURE DIRECTIONS

The development of pharmacologic inhibitors of the RAAS provided powerful and incisive tools to explore kidney hemodynamic and other associated adaptations in the setting of progressive kidney injury. These insights have paved the way for clinical studies that provided clear evidence for the use of RAAS inhibitor treatment as the mainstay of kidney-protective strategies. More recent trials have provided robust evidence of improved kidney protection when novel therapies are added to RAAS inhibition. SGLT2 inhibitors improved kidney protection across a wide spectrum of persons with CKD, an effect attributable at least in part to amelioration of glomerular hyperfiltration. Finerenone, a nonsteroidal MRA, augmented kidney protection in persons with diabetes and CKD through antiinflammatory and antifibrotic actions. The GLP-1RA, semaglutide slowed CKD progression in persons with diabetes likely by ameliorating risk factors for progression and antiinflammatory effects. We have thus entered an exciting era in which multiple interventions are available that impact different progression mechanisms to optimize kidney protection. Importantly, all novel therapies impact previously identified common pathways and mechanisms of CKD progression, demonstrating the importance of research in this area. Nevertheless, even in these studies a substantial proportion of participants evidenced progression of their CKD and we have not yet identified treatments that improve kidney function. Ongoing research involving cell biology and molecular cloning, as well as genomics and proteomics, continues to yield novel insights into the mechanisms of CKD progression that promise to direct researchers to potential new molecular targets for kidney-protective interventions. The development of the means to inhibit molecular targets specifically may provide new forms of therapy for those with CKD and enable physicians to realize the goal of achieving remission of progressive kidney injury in the majority and to look beyond this to aim for regression of kidney damage.⁸⁵³

In Memoriam

This chapter is dedicated to the memory of Barry M. Brenner, M.D. (October 4, 1937–August 6, 2024), who made a singular contribution to understanding the mechanisms of progression in chronic kidney disease, enabling the development of therapies now in clinical use to slow progression and reduce the global burden kidney failure.

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QUESTIONS

1. Micropuncture studies showed that after 5/6 nephrectomy in rats, glomerular capillary hydraulic pressure increased due to:
 - a. An increase in efferent arteriolar resistance.
 - b. An increase in afferent arteriolar resistance.
 - c. A greater increase in afferent arteriolar resistance than efferent arteriolar resistance.
 - d. A greater decrease in afferent arteriolar resistance than efferent arteriolar resistance.
 - e. A greater decrease in efferent arteriolar resistance than afferent arteriolar resistance.

Answer: d

Rationale: After 5/6 nephrectomy, the glomerular hemodynamic changes observed are the result of the combined effects of multiple vasodilator and vasoconstrictor molecules. Overall, there is vasodilation of both afferent and efferent arterioles. However, because of the differential vasoconstrictor effects of angiotensin II, the efferent arteriole dilates relatively less than the afferent arteriole, resulting in an increase in glomerular capillary hydraulic pressure and an increase in the single-nephron glomerular filtration rate (see [Table 50.1](#)).

2. On the basis of experimental evidence, which of the following statements about renal hypertrophy is false?
 - a. Biochemical evidence of a hypertrophic response is evident within hours of uninephrectomy.
 - b. Renal hypertrophy is accompanied by an increase in nephron number.
 - c. The extent of hypertrophy increases with the amount of renal tissue removed.
 - d. The extent of hypertrophy decreases with increasing age.
 - e. Kidney weight is increased at 48 to 72 hours after uninephrectomy and achieves a 30% to 40% gain at 2 to 3 weeks.

Answer: b

Rationale: In humans and other mammals, no new nephrons are formed after birth (or shortly thereafter). Therefore renal hypertrophy after birth is achieved by growth of existing nephrons, not the development of new nephrons.

3. In rats subjected to 5/6 nephrectomy, treatment with an angiotensin-converting enzyme (ACE) inhibitor resulted in which of the following responses?
 - a. Normalization of both glomerular capillary hydraulic pressure and single-nephron glomerular filtration rate (SNGFR).
 - b. Normalization of glomerular capillary hydraulic pressure but no change in SNGFR.
 - c. No change in glomerular capillary hydraulic pressure and normalization of SNGFR.
 - d. No change in either glomerular capillary hydraulic pressure or SNGFR.
 - e. Normalization of glomerular capillary hydraulic pressure and a further increase in SNGFR.

Answer: b

Rationale: Treatment with an ACE inhibitor reduces efferent arteriolar resistance to normalize glomerular capillary hydraulic pressure, but the SNGFR remains elevated. Because treatment with an ACE inhibitor affords renoprotection, this implies that glomerular capillary hypertension rather than glomerular hyperfiltration is the most important hemodynamic factor that drives chronic kidney disease progression.

4. As the glomerular filtration rate (GFR) decreases, decreased tubular reabsorption is the dominant mechanism that allows the nephron to maintain equilibrium for which of the following solutes?
 - a. Sodium
 - b. Potassium
 - c. Phosphate
 - d. Calcium
 - e. Sodium and phosphate

Answer: c

Rationale: As the GFR decreases, sodium and phosphate balance is maintained largely through decreased reabsorption of filtered solute. Maintaining potassium balance requires increased secretion of potassium in the distal nephron of fewer nephrons. Fractional excretion of calcium remains unchanged until the GFR falls below 25 mL/min/1.73 m², when fractional excretion increases due to the obligatory solute diuresis.

5. On the basis of observations in experimental models of chronic kidney disease (CKD), which of the following statements is false?
 - a. Glomerular capillary hypertension is a key factor driving a vicious cycle of progressive nephron loss.
 - b. Hemodynamic and nonhemodynamic factors contribute to progressive kidney damage.
 - c. Kidney function decreases due to a similar decline in the function of all nephrons.
 - d. Angiotensin II is a key molecular mediator of mechanisms that contribute to CKD progression.
 - e. Transforming growth factor- β is an important mediator of glomerulosclerosis and interstitial fibrosis.

Answer: c

Rationale: There is strong evidence from animal and clinical studies to support the Bricker hypothesis, which states that in CKD, kidney function declines due to a decreasing number of functioning (or hyperfunctioning) nephrons. This concept is central to our understanding of the mechanisms of CKD progression.